CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU), MEERUT

BCA 5th SEMESTER
MASTER DEDUPLICATED QUESTION BANK

Complete University Past Year Exam Questions (2013 – 2025) • Deduplicated • Year Stamped

📚 4 Core Subjects | 📑 37 Past Exam Papers Analyzed | ✨ 211 Unique Master Questions | 🎯 100% Fully Solved

📖 About This Master Publication

- **Zero Clutter / Deduplicated:** Identical questions repeated over 12 years are consolidated into single high-yield master entries.
- **Exam Year Stamps:** Every single question shows the exact university exam sessions in which it appeared (e.g. `CCSU 2025, 2024, 2021, 2018`).
- **Probability Badges:** Most frequent questions are highlighted with 🔥 `Highly Repeated (3x to 10x)` tags for guaranteed scoring.
- **Comprehensive Model Answers:** Complete, verified, step-by-step answers including relational schemas, Java code, network layer diagrams, and numerical calculations.
- **Covers All 4 Subjects:** BCA-501 (DBMS), BCA-502 (Java), BCA-503 (Networks), and BCA-504 (Numerical Methods).

Published for CCSU BCA 5th Semester Students • Examination Preparation • 2026 Edition

🔥 MASTER HIGH-YIELD "GOLDEN QUESTIONS" MATRIX

Top 49 Most Repeated Questions Across All 4 Subjects • Must-Study for 90%+ Exam Guarantee

Subject	Frequency	Exam Years Appeared	Marks	Core Exam Question Statement
BCA-503 Networks	🔥 9 Times	2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016	**15M / 3M / 7.5M**	Explain the ISO/OSI Reference Model in detail. Describe the primary functions of each of the seven layers.
BCA-502 Java	🔥 9 Times	2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016	**15M / 3M / 7.5M**	Explain Applet Life Cycle. Write a program to demonstrate a simple Java applet to display an image. Compare Applets over HTML.
BCA-502 Java	🔥 9 Times	2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016	**15M / 3M / 7.5M**	Explain Servlet Life Cycle and compare it with JSP Life Cycle. Write a complete HttpServlet code example.
BCA-501 DBMS	🔥 9 Times	2025, 2024, 2020, 2019, 2017, 2016, 2015, 2014, 2013	**15M / 3M**	What is Relational Algebra? Discuss its fundamental operations (Select, Project, Union, Set Difference, Cartesian Product, Rename) and derived Join operation.
BCA-503 Networks	🔥 8 Times	2025, 2024, 2023, 2021, 2020, 2019, 2017, 2016	**15M / 7.5M**	Explain Congestion Control Mechanisms in Computer Networks: Leaky Bucket and Token Bucket Algorithms with comparative analysis.
BCA-504 NM	🔥 7 Times	2024, 2023, 2022, 2021, 2020, 2019, 2018	**15M / 22.5M / 3M**	Using Runge-Kutta 4th order method, find $y(0.2)$ if $dy/dx = x + y^2$, given $y(0) = 1$ and step size $h = 0.1$.
BCA-504 NM	🔥 7 Times	2024, 2023, 2022, 2021, 2020, 2019, 2018	**15M / 3M / 7.5M**	Obtain a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$ using Bisection method correct to three decimal places.
BCA-504 NM	🔥 7 Times	2024, 2023, 2022, 2021, 2020, 2019, 2018	**15M / 3M / 7.5M**	Solve the differential equation $dy/dx = x + y^2$, $y(0) = 1$ using Picard's method (two approximations).
BCA-503 Networks	🔥 7 Times	2025, 2021, 2020, 2019, 2018, 2017, 2016	**15M / 22.5M / 7.5M**	Explain Sliding Window Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ with efficiency analysis.
BCA-503 Networks	🔥 7 Times	2025, 2024, 2020, 2019, 2018, 2017, 2016	**15M / 3M / 7.5M**	Compare Circuit Switching, Packet Switching, and Message Switching.
BCA-502 Java	🔥 7 Times	2025, 2023, 2020, 2019, 2018, 2017, 2016	**15M / 3M / 7.5M**	Explain Exception Handling mechanism in Java with keywords: try, catch, finally, throw, and throws.
BCA-501 DBMS	🔥 7 Times	2025, 2023, 2019, 2018, 2016, 2014, 2013	**3M / 7.5M**	Explain Primary Key, Foreign Key, and Candidate Key with suitable SQL examples.
BCA-504 NM	🔥 6 Times	2024, 2023, 2022, 2021, 2019, 2018	**15M / 22.5M / 3M / 7.5M**	Evaluate $\int_0^1 dx/(1 + x^2)$ using Simpson's 3/8 rule. Hence obtain the approximate value of π.

Subject	Frequency	Exam Years Appeared	Marks	Core Exam Question Statement
BCA-504 NM	🔥 6 Times	2024, 2023, 2022, 2020, 2019, 2018	**15M / 3M**	Solve the system of linear equations by Gauss-Seidel Iteration method: $6x + y + z = 20$, $x + 4y - z = 6$, $x - y + 5z = 7$.
BCA-503 Networks	🔥 6 Times	2025, 2021, 2020, 2019, 2018, 2016	**15M / 22.5M / 7.5M**	Explain Routing Algorithms: (a) Distance Vector Routing (Bellman-Ford) and (b) Link State Routing (Dijkstra). Discuss Count-to-Infinity problem.
BCA-503 Networks	🔥 6 Times	2025, 2020, 2019, 2018, 2017, 2016	**15M / 7.5M**	Explain Error Detection Techniques: Parity Check, Checksum, and CRC (Cyclic Redundancy Check).
BCA-503 Networks	🔥 6 Times	2023, 2020, 2019, 2018, 2017, 2016	**15M**	Explain Guided Media: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.
BCA-502 Java	🔥 6 Times	2025, 2020, 2019, 2018, 2017, 2016	**15M / 3M / 7.5M**	What is the difference between Process-based and Thread-based multitasking in Java?
BCA-502 Java	🔥 6 Times	2025, 2023, 2020, 2018, 2017, 2016	**15M / 3M / 7.5M**	Define Thread in Java. Explain how to create threads using (a) Thread class and (b) Runnable interface. Write a complete multithreaded program.
BCA-501 DBMS	🔥 6 Times	2025, 2024, 2021, 2017, 2015, 2014	**15M / 22.5M / 7.5M**	What is Normalization? Explain 1NF, 2NF, 3NF, and BCNF with suitable relational examples and anomaly reduction.
BCA-504 NM	🔥 5 Times	2024, 2023, 2022, 2021, 2020	**22.5M / 3M / 7.5M**	Using method of False Position (Regula-Falsi), find a positive root of the equation $x^3 - 2x - 5 = 0$.
BCA-501 DBMS	🔥 5 Times	2025, 2024, 2020, 2019, 2015	**15M / 3M**	Differentiate between Schema and Instance in DBMS.
BCA-501 DBMS	🔥 5 Times	2025, 2020, 2015, 2014, 2013	**15M / 7.5M**	Draw an ER Diagram for a Hospital Management System specifying entities, attributes, primary keys, relationships, and cardinalities. Show conversion to Relational Schema.
BCA-503 Networks	★ 4 Times	2023, 2019, 2018, 2016	**3M**	What is Network Topology? List advantages and disadvantages of Mesh Topology.
BCA-502 Java	★ 4 Times	2024, 2023, 2018, 2016	**15M / 3M / 7.5M**	What are Packages in Java? How is a user-defined package created and imported? Explain with complete code example.
BCA-502 Java	★ 4 Times	2025, 2019, 2018, 2016	**3M**	Explain Garbage Collection in Java.
BCA-501 DBMS	★ 4 Times	2025, 2016, 2015, 2013	**15M / 3M / 7.5M**	Explain the ACID properties of a transaction with practical examples. Why are they essential?
BCA-501 DBMS	★ 4 Times	2021, 2018, 2014, 2013	**15M / 7.5M**	What are the pitfalls of Lock-Based Protocols in concurrency control?

Subject	Frequency	Exam Years Appeared	Marks	Core Exam Question Statement
BCA-501 DBMS	★ 4 Times	2018, 2016, 2014, 2013	**15M / 7.5M**	Explain Indexed Sequential, B-Tree, and B+ Tree file organizations with examples.
BCA-501 DBMS	★ 4 Times	2025, 2023, 2019, 2018	**3M / 7.5M**	State the key advantages of DBMS over File Processing Systems.
BCA-501 DBMS	★ 4 Times	2023, 2016, 2015, 2013	**3M**	Define Database Management System (DBMS).
BCA-504 NM	★ 3 Times	2024, 2022, 2021	**15M / 3M**	Find the value of dy/dx at x = 1.2 for the following ordered pairs: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).
BCA-504 NM	★ 3 Times	2023, 2022, 2019	**15M**	Using Lagrange's interpolation formula, find the value of y corresponding to x = 2 from: (1, 1), (3, 27), (5, 125), (6, 216).
BCA-504 NM	★ 3 Times	2021, 2020, 2019	**15M / 7.5M**	Write the conditions for applicability of: (a) Trapezoidal rule, (b) Simpson's 1/3 rule. Evaluate $\int_0^1 dx/(1+x)$ using both rules with h = 0.25.
BCA-504 NM	★ 3 Times	2021, 2019, 2018	**3M / 7.5M**	Solve by Gauss elimination: $8x - 3y + 2z = 23$, $4x + 11y - z = 33$, $6x + 3y + 12z = 36$.
BCA-503 Networks	★ 3 Times	2025, 2023, 2016	**15M / 3M / 7.5M**	What is Subnetting? Give an example.
BCA-503 Networks	★ 3 Times	2019, 2018, 2016	**3M**	Which device is used for Modulation and Demodulation? Explain its function.
BCA-502 Java	★ 3 Times	2025, 2017, 2016	**15M**	What is JDBC? Explain the Four Types of JDBC Drivers with architectural diagrams and characteristics.
BCA-502 Java	★ 3 Times	2025, 2024, 2017	**15M**	What are Layout Managers in AWT/Swing? Explain BorderLayout, FlowLayout, and GridLayout with examples.
BCA-502 Java	★ 3 Times	2024, 2019, 2018	**15M**	What is JDBC? Write a complete JDBC program to insert and display records of employees using PreparedStatement.
BCA-502 Java	★ 3 Times	2024, 2023, 2019	**7.5M**	Explain the concepts of Inheritance in Java and its various types.
BCA-502 Java	★ 3 Times	2025, 2019, 2017	**3M**	What is Bytecode in Java? Why does it enable platform independence?
BCA-501 DBMS	★ 3 Times	2025, 2024, 2023	**15M / 3M**	Who are Data Administrators (DA)? What are the functions of a Database Administrator (DBA)? Differentiate between DA and DBA.
BCA-501 DBMS	★ 3 Times	2025, 2023, 2017	**15M / 3M / 7.5M**	Differentiate between DDL and DML commands with syntax and examples.

Subject	Frequency	Exam Years Appeared	Marks	Core Exam Question Statement
BCA-501 DBMS	★ 3 Times	2023, 2018, 2015	**15M / 3M**	Explain Data Independence with its types: Physical and Logical Data Independence. Why is it significant?
BCA-501 DBMS	★ 3 Times	2017, 2016, 2015	**15M / 7.5M**	Explain BCNF (Boyce-Codd Normal Form) with example.
BCA-501 DBMS	★ 3 Times	2021, 2015, 2014	**7.5M**	Explain the Three-Level Architecture of DBMS with a neat block diagram.
BCA-501 DBMS	★ 3 Times	2021, 2018, 2013	**3M / 7.5M**	Explain Hashing in brief with its collision handling techniques.
BCA-501 DBMS	★ 3 Times	2020, 2019, 2018	**3M / 7.5M**	What is a Referential Integrity Constraint? Why do we need it? Explain with a suitable example.

SUBJECT: BCA-501 • INTRODUCTION TO DBMS

SECTION C • LONG ANSWER QUESTIONS (15 MARKS / DETAILED SOLUTIONS) (38 UNIQUE

QUESTIONS)

BCA-501 | Marks: 15M / 3M | 🔥 Highly Repeated (9×)

CCSU Exam Years: 2025, 2024, 2020, 2019, 2017, 2016, 2015, 2014, 2013

What is Relational Algebra? Discuss its fundamental operations (Select, Project, Union, Set Difference, Cartesian Product, Rename) and derived Join operation.

Definition: Relational Algebra is a formal procedural query language that takes one or two relations as input and produces a new relation as output without modifying the underlying stored tables.

Fundamental Operators:

1. **Select (σ):** Unary operator that filters rows satisfying a predicate condition `P`.

```
σ_{Salary > 50000 AND Dept = 'IT'} (Employee)
```

2. **Project (π):** Unary operator that selects specified columns and eliminates duplicate tuples.

```
π_{EmpName, Salary} (Employee)
```

3. **Union (∪):** Binary operator combining tuples from two union-compatible relations.

```
Student_CS ∪ Student_IT
```

4. **Set Difference (−):** Returns tuples present in relation R but not in S (union compatible).

```
All_Students - Placed_Students
```

5. **Cartesian Product (×):** Cross product pairing every tuple of R with every tuple of S.

```
Employee × Department
```

6. **Rename (ρ):** Renames relation or attributes: `ρ_{S(A_1, A_2)} (R)`.

Derived Operation — Natural Join (⋈): Combines tuples of two relations matching on identical common attribute names:

```
Employee ⋈ Department = π_{...} (σ_{Employee.DeptID = Department.DeptID} (Employee × Department))
```

BCA-501 | Marks: 15M / 22.5M / 7.5M | 🔥 Highly Repeated (6×)

CCSU Exam Years: 2025, 2024, 2021, 2017, 2015, 2014

What is Normalization? Explain 1NF, 2NF, 3NF, and BCNF with suitable relational examples and anomaly reduction.

Definition: Normalization is a formal process of decomposing relations with anomalies into structured relations to eliminate insertion, deletion, and update anomalies while avoiding redundant data storage.

1. First Normal Form (1NF):

- A relation is in 1NF if and only if all attribute values are atomic (no multivalued, composite, or nested attributes).
- *Violation:* `Employee(EmpID, Skills)` where `Skills = "Java, Python, C++"`.
- *Decomposition:* Split into atomic rows: `(101, 'Java')`, `(101, 'Python')`, `(101, 'C++')`.

2. Second Normal Form (2NF):

- Relation must be in 1NF and have **no partial dependency** (no non-prime attribute should depend on a proper subset of any candidate key).
- *Example:* `Enrollment(StudentID, CourseID, CourseFee, Grade)`. Here `CourseFee` depends only on `CourseID` (partial).
- *Decomposition:* `StudentCourses(StudentID, CourseID, Grade)` and `CourseDetails(CourseID, CourseFee)`.

3. Third Normal Form (3NF):

- Relation must be in 2NF and have **no transitive dependency** (for every non-trivial FD `X \to Y`, either `X` is a super key or `Y` is a prime attribute).
- *Example:* `Employee(EmpID, EmpName, DeptNo, DeptHead)`. Here `EmpID → DeptNo` and `DeptNo → DeptHead`, creating transitive dependency.
- *Decomposition:* `Emp(EmpID, EmpName, DeptNo)` and `Dept(DeptNo, DeptHead)`.

4. Boyce-Codd Normal Form (BCNF):

- A stricter version of 3NF where for every non-trivial FD `X \to Y`, determinant `X` **must strictly be a Super Key**.

BCA-501 | Marks: 15M / 3M | 🔥 Highly Repeated (5×) | CCSU Exam Years: 2025, 2024, 2020, 2019, 2015

Differentiate between Schema and Instance in DBMS.

Comparison:

- **Database Schema:** The overall logical blueprint, structure, and design of the database (e.g., table names, column data types, constraints). It is specified during system design and changes very infrequently.
- **Database Instance (State):** The actual data content populated within the database at a specific moment in time. It changes dynamically whenever records are inserted, updated, or deleted.

Analogy: In programming languages, a Schema corresponds to a Class/Data Type, whereas an Instance corresponds to an Object/Value of that type.

BCA-501 | Marks: 15M / 7.5M | 🔥 Highly Repeated (5×) | CCSU Exam Years: 2025, 2020, 2015, 2014, 2013

Draw an ER Diagram for a Hospital Management System specifying entities, attributes, primary keys, relationships, and cardinalities. Show conversion to Relational Schema.

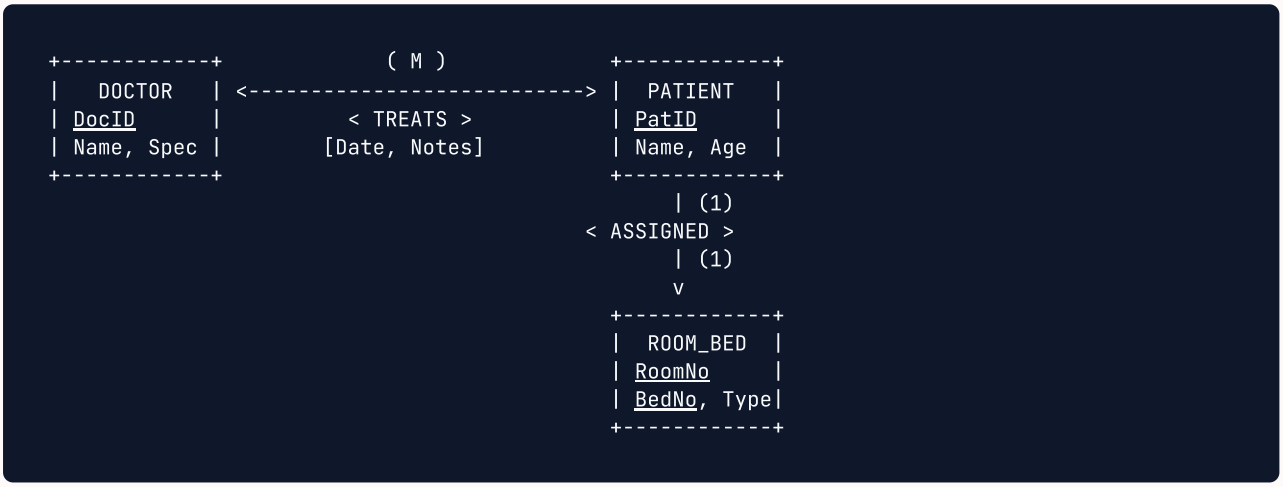
1. Entity Sets & Attributes:

- **DOCTOR**: DocID, Name, Specialty, Phone
- **PATIENT**: PatID, Name, Age, Gender, Address
- **ROOM / BED**: RoomNo, BedNo, RoomType, DailyCharge
- **BILL / ACCOUNT**: BillID, BillDate, TotalAmount, PaymentStatus

2. Relationships & Cardinalities:

- `TREATS` : Many-to-Many between DOCTOR and PATIENT with attribute `DateOfTreatment` .
- `ASSIGNED_TO` : Many-to-One between PATIENT and BED (A patient is assigned to one bed; a bed holds one patient at a time).
- `GENERATES` : One-to-One between PATIENT and BILL.

3. ASCII ER Diagram Representation:



4. Relational Schema Reduction:

```
1. Doctor(DocID, Name, Specialty, Phone)
2. Patient(PatID, Name, Age, Gender, Address, RoomNo, BedNo)
   FOREIGN KEY (RoomNo, BedNo) REFERENCES Room_Bed(RoomNo, BedNo)
3. Room_Bed(RoomNo, BedNo, RoomType, DailyCharge)
4. Treats(DocID, PatID, TreatmentDate, Prescription)
   FOREIGN KEY (DocID) REFERENCES Doctor(DocID),
   FOREIGN KEY (PatID) REFERENCES Patient(PatID)
5. Bill(BillID, PatID, BillDate, TotalAmount, Status)
   FOREIGN KEY (PatID) REFERENCES Patient(PatID)
```

BCA-501 | Marks: 15M / 3M / 7.5M | Highly Repeated (4×) | CCSU Exam Years: 2025, 2016, 2015, 2013

Explain the ACID properties of a transaction with practical examples. Why are they essential?

Definition: A **Transaction** is a logical unit of database processing that includes one or more database access operations. To preserve database consistency and reliability, every DBMS must guarantee the four **ACID Properties**:

1. Atomicity ("All or Nothing"):

- Either all operations of the transaction are committed successfully, or none take effect. If failure occurs mid-transaction, all changes are rolled back.
- *Example:* Bank transfer of `\500 from Account A to Account B. If \` 500 is debited from A and power fails before crediting B, the recovery subsystem rolls back A's debit.

2. Consistency (Preservation of Invariants):

- Execution of a transaction in isolation preserves database integrity constraints (e.g., total balance of A + B must remain unchanged before and after the transfer).

3. Isolation (Independence of Concurrent Executions):

- Even if multiple transactions execute concurrently, the intermediate uncommitted state of one transaction is hidden from others, producing a serializable execution schedule. Enforced via locking (2PL) and timestamp ordering.

4. Durability (Survivability of Committed Data):

- Once a transaction commits, its updates persist permanently in secondary storage and will survive subsequent power failures, crashes, or OS reboots via Write-Ahead Logging (WAL) and checkpointing.

BCA-501 | Marks: 15M / 7.5M | Highly Repeated (4×) | CCSU Exam Years: 2021, 2018, 2014, 2013

What are the pitfalls of Lock-Based Protocols in concurrency control?

Pitfalls:

1. **Deadlock:** Occurs when two or more transactions are waiting indefinitely for locks held by each other (e.g., `T_1` holds Lock on A and waits for B, while `T_2` holds Lock on B and waits for A). Requires deadlock detection (Wait-For Graph) and victim rollback.
2. **Starvation:** A transaction waits indefinitely because other higher-priority transactions repeatedly acquire locks on the required data item.
3. **Cascading Rollback:** In non-strict 2PL, if a transaction aborts, all dependent uncommitted transactions that read its dirty data must also be rolled back in a cascade.
4. **Concurrency Bottleneck:** Excessive long-duration exclusive locks reduce system throughput and degrade response times.

BCA-501 | Marks: 15M / 7.5M | Highly Repeated (4×) | CCSU Exam Years: 2018, 2016, 2014, 2013

Explain Indexed Sequential, B-Tree, and B+ Tree file organizations with examples.

Comparison of File Organizations:

- **Indexed Sequential (ISAM):** Records stored in sequential key order with a multilevel static index. Good for sequential and direct access, but degrades with many overflow records.
- **B-Tree:** Self-balancing dynamic multi-way search tree. Keys and data record pointers reside in both internal nodes and leaf nodes.
- **B+ Tree:** Modern standard for relational DBMS storage engines. Internal nodes store only search keys to guide navigation, while **all actual data records/pointers reside in leaf nodes** connected via a doubly linked list, enabling superior range scans.

BCA-501 | Marks: 15M / 3M | ★ Repeated (3×) | CCSU Exam Years: 2025, 2024, 2023

Who are Data Administrators (DA)? What are the functions of a Database Administrator (DBA)? Differentiate between DA and DBA.

Data Administrator (DA) vs Database Administrator (DBA):

Aspect	Data Administrator (DA)	Database Administrator (DBA)
Role Level	Managerial and business level	Technical and operational level
Focus	Business data policies, enterprise data dictionary, data governance.	Physical database design, database server installation, query tuning, security.
Scope	Decides *what data* needs to be collected across the enterprise.	Decides *how data is stored, indexed, secured, and recovered.*

Key Functions of DBA:

1. Hardware/software installation and database upgrades.
2. Creating database users, granting roles, and monitoring access privileges.
3. Performance tuning: Analyzing execution plans, optimizing SQL queries, re-indexing.
4. Automating database backups and executing disaster recovery drills.

BCA-501 | Marks: 15M / 3M / 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2025, 2023, 2017

Differentiate between DDL and DML commands with syntax and examples.

Comparison:

Feature	Data Definition Language (DDL)	Data Manipulation Language (DML)
Purpose	Defines and modifies the schema structure of the database.	Manages, accesses, and manipulates the data within the tables.
Commit Nature	Auto-committed permanently (cannot be rolled back in most RDBMS).	Transactional (can be committed or rolled back using TCL).
Commands	`CREATE`, `ALTER`, `DROP`, `TRUNCATE`, `RENAME`	`INSERT`, `UPDATE`, `DELETE`, `SELECT`

SQL Examples:

```
-- DDL Commands:
CREATE TABLE Student (RollNo INT PRIMARY KEY, Name VARCHAR(50));
ALTER TABLE Student ADD COLUMN City VARCHAR(30);

-- DML Commands:
INSERT INTO Student (RollNo, Name, City) VALUES (101, 'Aman', 'Meerut');
UPDATE Student SET City = 'Delhi' WHERE RollNo = 101;
```

BCA-501 | Marks: 15M / 3M | ★ Repeated (3×) | CCSU Exam Years: 2023, 2018, 2015

Explain Data Independence with its types: Physical and Logical Data Independence. Why is it significant?

Definition: Data Independence is the ability to modify schema definitions at one level of the 3-level ANSI/SPARC architecture without altering schemas at the higher level.

1. Logical Data Independence:

- The ability to modify the **Conceptual Schema** (e.g., adding/deleting attributes, splitting tables) without having to rewrite external schemas or existing application programs.
- *How achieved:* Using SQL Views that present the original logical schema to old applications.
- *Difficulty:* Harder to achieve because applications are closely tied to logical data models.

2. Physical Data Independence:

- The ability to modify the **Internal/Physical Schema** (e.g., changing file organizations, building B+ Tree indexes, moving to new storage media) without changing the Conceptual Schema.
- *Ease:* Easier to achieve because logical structure remains completely decoupled from disk-level storage.

BCA-501 | Marks: 15M / 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2017, 2016, 2015

Explain BCNF (Boyce-Codd Normal Form) with example.

Answer:

Boyce-Codd Normal Form (BCNF) is an advanced, stricter version of Third Normal Form (3NF), specifically developed by Raymond Boyce and Edgar Codd to resolve anomalies arising from overlapping composite candidate keys.

> **Formal Definition:** A relation *R* is in BCNF if and only if for every non-trivial functional dependency `X → Y` holding on *R*: **X is a Super Key of R**.

Comparative Analysis: 3NF vs BCNF

Criteria	Third Normal Form (3NF)	Boyce-Codd Normal Form (BCNF)
Condition for X → Y	Valid if X is Super Key **OR** Y is a prime attribute.	Valid **ONLY** if X is a Super Key (no prime relaxation).
Redundancy	May permit slight redundancy if overlapping candidate keys exist.	Completely eliminates all redundancy resulting from functional dependencies.
Dependency Preservation	**Always preserves** all functional dependencies.	May occasionally **not preserve** all FDs upon decomposition.

Classic Example Violating BCNF & Resolution:

> Consider relation: `StudentAdvisor(StudentID, Subject, Advisor)` with constraints:
>
> - A student can have multiple advisors for different subjects: `{StudentID, Subject} → Advisor`
> - Each advisor specializes in only one subject: `Advisor → Subject`
> - Candidate Keys: `{StudentID, Subject}` and `{StudentID, Advisor}`. Prime Attributes: all 3.
>
> **Check 3NF:** For `Advisor → Subject`, Advisor is not super key, but Subject is prime attribute. Hence table is in **3NF** ✓.
>
> **Check BCNF:** Advisor is **NOT** a super key. Hence violates **BCNF** ✗.
>
> **BCNF Decomposition:** Split into two relations:
>
> 1. `AdvisorSubject(Advisor, Subject)` — Advisor is primary key ✓
> 2. `StudentAdvisor(StudentID, Advisor)` — Both form composite primary key ✓

BCA-501 | Marks: 15M / 22.5M | ★ Repeated (2×) | CCSU Exam Years: 2021, 2020

Explain Two-Phase Locking (2PL) Protocol. Compare Strict 2PL and Rigorous 2PL with serializability guarantees.

Two-Phase Locking (2PL): A concurrency control protocol guaranteeing conflict serializability by dividing a transaction into two distinct phases:

1. **Growing Phase:** Transaction may obtain locks (Shared or Exclusive) but cannot release any lock.
2. **Shrinking Phase:** Transaction may release locks but cannot acquire any new lock.

Variants of 2PL:

- **Basic 2PL:** Guaranteed conflict serializability, but vulnerable to cascading aborts and deadlocks.
- **Strict 2PL:** All Exclusive (X) locks held by a transaction must be retained until the transaction commits or aborts. *Guarantees strict schedules and eliminates cascading aborts!*
- **Rigorous 2PL:** Both Shared (S) and Exclusive (X) locks must be retained until transaction commit/abort. Ensures serializability in commit order.

BCA-501 | Marks: 15M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2023, 2013

What is Normalization? Explain the importance of Normalization in database design.

Definition: Normalization is the systematic process of organizing relation schemas based on functional dependencies to eliminate anomalies and avoid redundant storage.
Importance:

- Eliminates Insertion, Deletion, and Modification anomalies.
- Reduces storage space by ensuring each fact is recorded in only one place.
- Prevents data corruption by enforcing referential integrity constraints across smaller, well-structured tables.

BCA-501 | Marks: 15M / 3M | ★ Repeated (2×) | CCSU Exam Years: 2020, 2017

Explain Views in SQL. What are the advantages of Views? When is a View updatable?

Views in SQL: A **View** is a virtual table defined by an underlying stored SQL query.
Advantages: Data security (restricts column access), simplified queries, and logical data independence.
Updatable View Criteria:

- Query contains only one base table in `FROM` clause.
- No aggregate functions (`SUM` , `AVG` , `COUNT`).
- No `GROUP BY` , `HAVING` , `DISTINCT` , or `UNION` .
- Primary key of base table must be included in the view columns.

BCA-501 | Marks: 15M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2017, 2015

Explain Log-Based Recovery with WAL (Write-Ahead Logging) and Checkpoints.

Answer:

Log-Based Recovery guarantees the Atomicity and Durability of transactions by recording every database modification into an append-only log file residing on stable storage before writing changes to disk.

> **Write-Ahead Logging (WAL) Protocol:**
>
> - The log record corresponding to an update must be written to stable storage **before** the actual database disk buffer is flushed.
> - A transaction enters the *Committed* state only after its `<commit T>` log record is securely written to stable disk.
>
> **Checkpoint Mechanism:** Periodic checkpoints flush all dirty in-memory buffers to disk and append `<checkpoint L>` to the log. During system crash recovery, the recovery manager only needs to scan back to the most recent checkpoint rather than reading from the start of time.

Recovery Algorithms (Undo / Redo):

Transaction State at Crash	Action Taken	Recovery Phase
Both `<start T>` and `<commit T>` present in log.	**REDO(T):** Reapplies all new values to ensure permanence (Durability).	Redo Pass (Roll-forward)
`<start T>` present, but **NO** `<commit T>` or `<abort T>`.	**UNDO(T):** Restores all updated attributes back to their old values (Atomicity).	Undo Pass (Roll-back)

BCA-501 | Marks: 15M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2017, 2013

Explain the Three-Level ANSI/SPARC Database Architecture with detailed diagram.

Answer:

The **ANSI/SPARC Three-Schema Architecture** separates user views from physical database storage to achieve complete Data Independence.

1. **1. External Level (View Level):** The highest level closest to users. Consists of multiple customized external schemas (views) displaying only relevant data subsets and hiding complex business structures and unauthorized fields.
2. **2. Conceptual Level (Logical Level):** The global logical blueprint of the entire enterprise database. Describes entities, attributes, relationships, constraints, and business rules without exposing physical disk details.
3. **3. Internal Level (Physical Level):** The lowest level describing physical disk storage, block sizes, file organization (B+ trees, hashing), compression, and encryption.

Data Independence Across Levels:

Level of Independence	Definition	Example Scenario
Physical Data Independence	Capacity to alter the Internal Schema (re-indexing, RAID striping, hashing) without altering the Conceptual Schema.	Adding a B+ Tree index on `Student(Email)` without altering table definitions.
Logical Data Independence	Capacity to alter the Conceptual Schema (adding tables or attributes) without modifying existing External Views or application queries.	Adding an `AadharNo` column to the Student table without breaking existing report generation scripts.

BCA-501 | Marks: 15M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2016, 2015

Compare Dense Index and Sparse Index in database indexing.

Answer:

An **Index** is an auxiliary data structure that speeds up tuple retrieval in database tables without full table scans.

Feature	Dense Index	Sparse Index
Index Entries	Contains an index record for **every single search key value** in the data file.	Contains index records for **only a subset** of search key values (usually one per disk block).
File Ordering	Can be constructed on either sorted (primary) or unsorted (secondary) data files.	Requires data file to be **strictly sorted** on the search key.
Storage Overhead	High storage space requirement (index size scales linearly with number of rows).	Very low storage space requirement; index easily fits into main memory RAM.
Search Speed	Faster: Locates record pointer directly in index without inspecting adjacent disk rows.	Slightly slower: Finds matching block pointer, then scans within the block to find exact record.
Maintenance (Insert/Delete)	High maintenance overhead on inserts and deletes.	Lower maintenance; changes only when block boundaries shift.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2025

Explain SQL Aggregate Functions (COUNT, SUM, AVG, MIN, MAX) with GROUP BY and HAVING clauses. Provide full query demonstrations.

Aggregate Functions: Functions that perform calculations on a set of column values and return a single scalar summary value.

- `COUNT(column)` : Returns total count of non-null values. `COUNT(*)` counts all rows.
- `SUM(column)` : Calculates total numeric sum of values.
- `AVG(column)` : Calculates arithmetic mean of non-null values.
- `MIN(column)` / `MAX(column)` : Returns the minimum / maximum value.

Role of GROUP BY and HAVING:

- `GROUP BY` divides records into distinct groups based on identical column values.
- `HAVING` filters grouped rows after aggregation (whereas `WHERE` filters individual tuples prior to grouping).

Comprehensive SQL Example:

```
-- Schema: Employee (EmpID, EmpName, Salary, DeptID, City)
-- Query: Find DeptID, total employees, average salary, and max salary
-- for departments having more than 3 employees and average salary > 45000:

SELECT
    DeptID,
    COUNT(EmpID) AS TotalEmployees,
    AVG(Salary) AS AverageSalary,
    MAX(Salary) AS HighestSalary,
    MIN(Salary) AS LowestSalary
FROM Employee
WHERE City IN ('Delhi', 'Noida', 'Gurugram')
GROUP BY DeptID
HAVING COUNT(EmpID) > 3 AND AVG(Salary) > 45000
ORDER BY AverageSalary DESC;
```

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2024

Explain Normalization with its advantages. What anomalies arise in unnormalized databases?

Anomalies in Unnormalized Tables:

1. **Insertion Anomaly:** Cannot record information about an entity without having to insert another unrelated entity (e.g., cannot register a new department until at least one employee is hired).
2. **Deletion Anomaly:** Deleting a record of one entity unintentionally erases information about another entity (e.g., deleting the last employee in a department deletes all department details).
3. **Update / Modification Anomaly:** Changing an entity's attribute requires updating hundreds of duplicate rows; if one row is missed, data becomes contradictory.

Advantages of Normalization:

- Eliminates data redundancy and conserves disk storage.
- Maintains referential integrity automatically across tables.
- Speeds up row updates, inserts, and index lookups.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2024

Explain SQL with its various characteristics and features.

Structured Query Language (SQL): The ANSI/ISO standard declarative language for relational databases.
Key Characteristics:

- **Declarative / Non-Procedural:** Users specify *what data* they want, leaving the query optimizer to determine *how* to retrieve it.
- **Comprehensive Capabilities:** Encompasses DDL (structure), DML (manipulation), DCL (security/privileges), and TCL (transactions).
- **Portability & Standards:** Works uniformly across major vendors (Oracle, MySQL, SQL Server, PostgreSQL).
- **Integrity Constraints Support:** Native enforcement of `PRIMARY KEY`, `FOREIGN KEY`, `CHECK`, `UNIQUE`, and `DEFAULT`.
- **View & Subquery Support:** Enables nested queries, common table expressions (CTEs), and dynamic virtual tables (Views).

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2024

Explain the reduction of ER Schema to Relational Tables with a complete case study.

Case Study: Company Database:
Entities: `EMPLOYEE`, `DEPARTMENT`, `PROJECT`.
Relationships:

- `WORKS_FOR` (N:1 from Employee to Department)
- `MANAGES` (1:1 from Employee to Department with `StartDate`)
- `WORKS_ON` (M:N between Employee and Project with `Hours`)

Relational Reduction:

```
1. DEPARTMENT(DeptNo, DeptName, MgrEmpID, MgrStartDate)
   -- 1:1 MANAGES folded into DEPARTMENT with MgrEmpID as FK
2. EMPLOYEE(EmpID, EmpName, Salary, DeptNo)
   -- N:1 WORKS_FOR represented via DeptNo as FK referencing DEPARTMENT
3. PROJECT(ProjNo, ProjName, DeptNo)
   -- N:1 CONTROLS represented via DeptNo as FK
4. WORKS_ON(EmpID, ProjNo, Hours)
   -- M:N relationship becomes a separate junction table with composite PK
   -- FK EmpID → EMPLOYEE, FK ProjNo → PROJECT
```

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2023

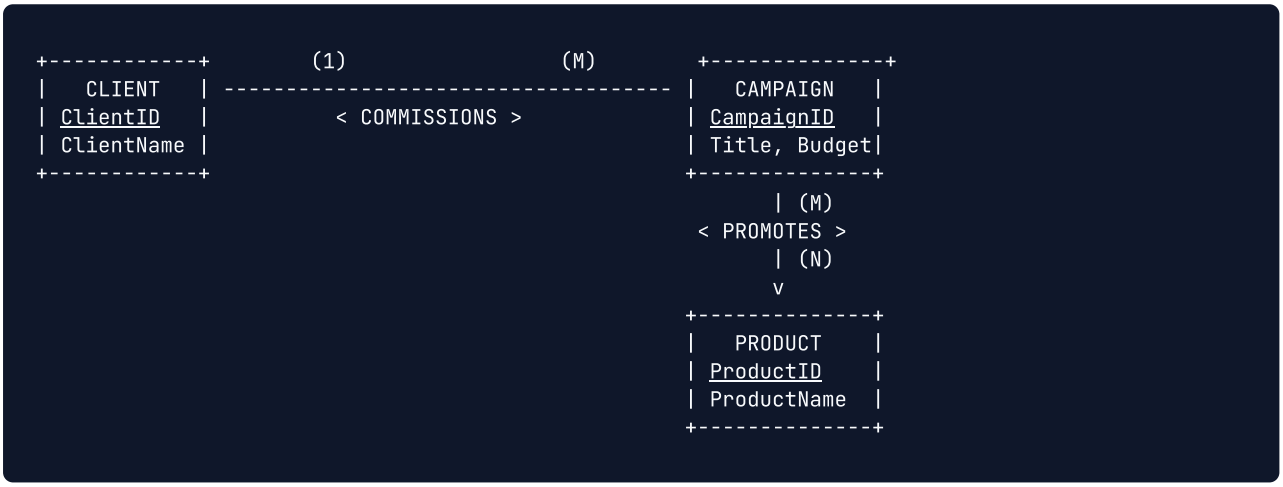
Draw an ER Diagram for a small Marketing Company database, assuming your own data requirements.

Enterprise Requirements:

- **CAMPAIGN:** CampaignID, Title, Budget, StartDate, EndDate.
- **CLIENT:** ClientID, ClientName, ContactPerson, Email, Phone.
- **PRODUCT:** ProductID, ProductName, Category.
- **MARKETING_AGENT:** AgentID, AgentName, CommissionRate.

Relationships:

- A Client commissions one or more Campaigns (1:N).
- Each Campaign promotes one or more Products (M:N).
- Agents are assigned to Campaigns (M:N).



BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2023

Describe the classification of Database Languages. Which type of language is SQL?

Classification of Database Languages:

1. **Data Definition Language (DDL):** Specifies the database schema, storage structure, and constraints (`CREATE`, `ALTER`, `DROP`, `TRUNCATE`).
2. **Data Manipulation Language (DML):** Retrieves, inserts, updates, and deletes stored data. Can be procedural (relational algebra) or non-procedural/declarative (relational calculus, SQL).
3. **Data Control Language (DCL):** Controls permissions and access authorization (`GRANT`, `REVOKE`).
4. **Transaction Control Language (TCL):** Manages the logical unit of work (`COMMIT`, `ROLLBACK`, `SAVEPOINT`).

Classification of SQL: SQL is a **comprehensive declarative (non-procedural) language** that integrates DDL, DML, DCL, and TCL in a single standardized unified language.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2021

Given the Relational Schema:
EMPLOYEE(EmpID, EmpName, Age, Gender, Salary, City, DeptID, Designation, HireDate)
DEPARTMENT(DeptID, DeptName, Location)
Write SQL Queries for:
1. Display the age of 'Male' employees.
2. Display the name of the highest salary employee.
3. Display the name of department of employee 'Aman'.
4. Display all employees belonging to the same address (city).
5. Which employee is the oldest manager in the company?

SQL Queries:

```
-- Query 1: Age of Male employees
SELECT EmpName, Age
FROM EMPLOYEE
WHERE Gender = 'Male';

-- Query 2: Name of the highest salary employee
SELECT EmpName, Salary
FROM EMPLOYEE
WHERE Salary = (SELECT MAX(Salary) FROM EMPLOYEE);

-- Query 3: Department name of employee 'Aman'
SELECT D.DeptName
FROM DEPARTMENT D
JOIN EMPLOYEE E ON D.DeptID = E.DeptID
WHERE E.EmpName = 'Aman';

-- Query 4: Display all employees grouped by identical city
SELECT City, COUNT(EmpID) AS TotalEmps,
       GROUP_CONCAT(EmpName) AS EmployeeNames
FROM EMPLOYEE
GROUP BY City
HAVING COUNT(EmpID) > 1;

-- Query 5: Oldest manager in the company
SELECT EmpName, Age, Designation, HireDate
FROM EMPLOYEE
WHERE Designation = 'Manager'
ORDER BY Age DESC
LIMIT 1;
```

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2020

Explain File Organization Techniques: Sequential, Indexed Sequential (ISAM), and Hashed File Organization.

Answer:

File Organization refers to how records are mapped onto physical storage blocks on disk drives:

1. **1. Sequential File Organization:**
   - Records are stored in continuous sequential order sorted by a designated search key.
   - *Pros:* Highly efficient for batch processing applications (e.g., end-of-month payroll processing).
   - *Cons:* Extremely slow for single-record random access (*O(N)* scan); costly overflow blocks for new inserts.
2. **2. Indexed Sequential Access Method (ISAM):**
   - Combines sequential file storage with a hierarchical multi-level index tree (cylinder and track indexes).
   - *Pros:* Supports both fast sequential scans and fast random access via index lookup (*O(log N)*).
   - *Cons:* Periodic reorganization required when overflow chains become deep.
3. **3. Hashed (Direct) File Organization:**
   - Uses a mathematical hash function `h(Key) → Block Address` to directly compute disk block addresses.
   - *Pros:* Near-instantaneous constant time access (*O(1)*) for equality lookups.
   - *Cons:* Ineffective for range queries (`WHERE Age BETWEEN 20 AND 30`); prone to hash collisions.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2020

Explain Database Security and Authorization. Discuss GRANT and REVOKE commands in SQL.

Answer:

Database Security protects the system against unauthorized access, malicious data corruption, and privacy breaches using authentication and role-based authorization.

SQL Authorization Commands (DCL):

- `GRANT` : Bestows specific privileges (SELECT, INSERT, UPDATE, DELETE) to database users or roles.
- `REVOKE` : Withdraws previously granted privileges.
- `WITH GRANT OPTION` : Allows the recipient user to delegate the granted privilege to other third-party users.

SQL Query Demonstrations:

```
-- 1. Grant SELECT and UPDATE permissions to user 'finance_officer'
GRANT SELECT, UPDATE(Salary) ON Employee TO finance_officer;

-- 2. Grant all privileges on Student table with permission to delegate
GRANT ALL PRIVILEGES ON Student TO registrar WITH GRANT OPTION;

-- 3. Revoke UPDATE privilege from user 'finance_officer'
REVOKE UPDATE ON Employee FROM finance_officer;

-- 4. Cascade revocation (removes delegated permissions as well)
REVOKE SELECT ON Student FROM registrar CASCADE;
```

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2019

Explain the Three-Tier Client/Server Architecture of DBMS with Data Independence.

Three-Tier Architecture:

- **Presentation Tier (User Interface):** Web browser or desktop client GUI (React, HTML).
- **Application Tier (Business Logic):** Web/Application server executing business validation and query processing (Java, Node.js, Python).
- **Database Tier (Data Storage):** Enterprise RDBMS managing database engine, query processing, transactions, and tables.

Provides enhanced security, scalability, and clean modular separation.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2019

Discuss SQL Characteristics and explain five Aggregate Functions with complete query examples.

SQL Queries for Aggregate Functions:

```
-- 1. COUNT: Total employees in IT
SELECT COUNT(*) FROM Employee WHERE Dept = 'IT';

-- 2. SUM: Total payroll for Sales
SELECT SUM(Salary) FROM Employee WHERE Dept = 'Sales';

-- 3. AVG: Average salary in company
SELECT AVG(Salary) FROM Employee;

-- 4. MIN and MAX: Salary bounds
SELECT MIN(Salary) AS MinSalary, MAX(Salary) AS MaxSalary FROM Employee;
```

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2019

Explain Database Recovery Techniques: Log-Based Recovery, Immediate Update, Deferred Update, and Checkpoints.

Log-Based Recovery:

- **Deferred Update (NO-UNDO / REDO):** Updates are recorded only in the log until commit. During recovery, only committed transactions are REDONE; uncommitted transactions require NO UNDO.
- **Immediate Update (UNDO / REDO):** Updates can be written to disk before commit. Active uncommitted transactions at crash must be UNDONE; committed ones are REDONE.
- **Checkpoints:** Periodically flushes dirty log buffers to disk. Recovery scans backward only to the most recent checkpoint, dramatically shortening crash recovery time.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2018

Explain Super Class, Sub Class, Inheritance, and Generalization with examples.

Enhanced ER (EER) Concepts:

- **Superclass:** A generic entity set that includes distinct subgroups (e.g., `PERSON`).
- **Subclass:** A specialized sub-entity set that inherits all attributes and relationships from its superclass, while adding specialized attributes (e.g., `STUDENT` with GPA, `TEACHER` with Rank).
- **Attribute Inheritance:** Subclass entities automatically inherit the primary key and all general attributes of their superclass.
- **Generalization:** Bottom-up synthesis where lower-level entity sets are combined into a generalized superclass entity.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2018

Write short notes on: (i) Strong and Weak Entity, (ii) Different Levels of Data Abstraction, (iii) Database Recovery Techniques.

(i) Strong vs Weak Entity: A Strong entity has a primary key of its own. A Weak entity does not have a primary key and depends on the existence of an identifying strong entity set (identified by discriminator + strong entity PK).
(ii) Data Abstraction Levels: Physical level (low-level block storage), Logical level (community tables and relations), and View level (user-specific abstractions).
(iii) Database Recovery Techniques: Log-based recovery (Write-Ahead Logging), Deferred/Immediate update schemes, and Checkpoints to restore consistency following system failures.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2018

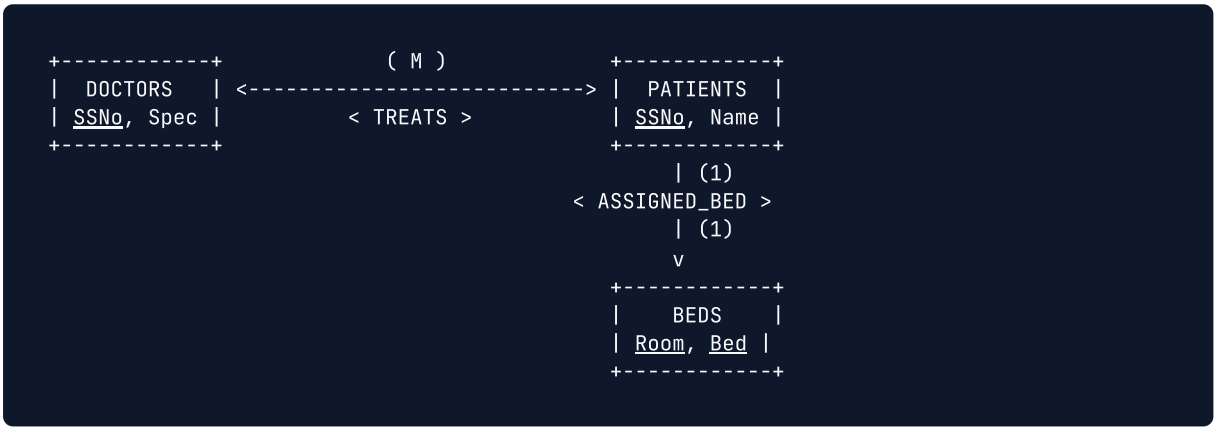
A hospital maintains data about entities: PATIENTS, DOCTORS, BEDS, ACCOUNTS. Construct an E-R diagram specifying keys, mapping cardinalities, and participation constraints. Show table mapping.

Entities & Attributes:

- `PATIENTS(SSNo, LastName, FirstName, HomePhone, Sex, DOB, Age, Street, City, State, Zip)`
- `DOCTORS(SSNo, LastName, FirstName, OfficePhone, Pager, Specialty)`
- `BEDS(RoomNumber, BedNumber, Type, Status, PricePerHour)`
- `ACCOUNTS(AccountNo, DateIn, DateOut, TotalAmount, Status)`

Relationships:

- `TREATS` (M:N between DOCTORS and PATIENTS)
- `ASSIGNED_BED` (1:1 between PATIENTS and BEDS)
- `HAS_ACCOUNT` (1:1 between PATIENTS and ACCOUNTS)



BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2017

Given tables VENDOR(VCode, Name, Address, Phone) and PRODUCT(PCode, PName, Price, VCode). Write SQL queries for:
(i) Display all vendors from 'Delhi'.
(ii) Display products with price greater than 500.
(iii) Display vendor name who supplies product 'Laptop'.
(iv) Count total products supplied by each vendor.

SQL Solutions:

```
-- (i) Vendors from Delhi
SELECT * FROM VENDOR WHERE Address LIKE '%Delhi%';

-- (ii) Products with price > 500
SELECT PCode, PName, Price FROM PRODUCT WHERE Price > 500;

-- (iii) Vendor supplying 'Laptop'
SELECT V.Name
FROM VENDOR V
JOIN PRODUCT P ON V.VCode = P.VCode
WHERE P.PName = 'Laptop';

-- (iv) Count total products per vendor
SELECT V.VCode, V.Name, COUNT(P.PCode) AS TotalProducts
FROM VENDOR V
LEFT JOIN PRODUCT P ON V.VCode = P.VCode
GROUP BY V.VCode, V.Name;
```

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2017

Given tables: PLAY(PlayCode, PlayName, DirNum), DIRECTOR(DirNum, DirName, DirDOB). Write SQL queries to find director names and plays directed.

SQL Solutions:

```
-- Director of play 'Hamlet'
SELECT D.DirName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
WHERE P.PlayName = 'Hamlet';

-- All plays directed by each director
SELECT D.DirName, P.PlayName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
ORDER BY D.DirName;
```

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2016

Draw a detailed ER Diagram for a Banking Enterprise.

Answer:

An **Entity-Relationship (ER) Diagram** visually models the conceptual data architecture of an enterprise.

Core Entities & Attributes for Banking System:

- **CUSTOMER:** CustID, Name (First, Last), Address, Phone, PAN.
- **BRANCH:** BranchID, BranchName, City, Assets.
- **ACCOUNT:** AccountNo, AccountType (Savings, Current), Balance.
- **LOAN:** LoanNo, Amount, InterestRate, LoanType.
- **TRANSACTION:** TxnID, Date, Amount, TxnType (Deposit, Withdrawal).

Relationships & Cardinalities:

- **Cust_Account (Many-to-Many M:N):** A Customer can hold multiple accounts; a Joint account can have multiple customers.
- **Branch_Account (One-to-Many 1:N):** Each Account is maintained at exactly one Branch; a Branch maintains many accounts.
- **Cust_Loan (Many-to-Many M:N):** A Customer can take multiple loans; a joint loan belongs to multiple customers.
- **Account_Txn (One-to-Many 1:N):** An Account records many Transactions; each Transaction belongs to one Account.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2016

Explain Nested Subqueries and Correlated Subqueries in SQL with examples.

Answer:

Feature	Nested (Non-Correlated) Subquery	Correlated Subquery
Execution Mechanism	The inner query executes **once independently**, and its result set is passed to the outer query.	The inner query executes **repeatedly, once for each candidate row** processed by the outer query.
Dependency	Inner query has no reference to outer query columns.	Inner query explicitly references columns from the outer query table (correlation).
Performance	Fast; executed a single time in memory.	Slower on large tables; *O(N × M)* complexity unless optimized by engine.
Code Example	`SELECT Name, Salary FROM Employee WHERE Salary > (SELECT AVG(Salary) FROM Employee);`	`SELECT E1.Name, E1.Salary FROM Employee E1 WHERE E1.Salary > (SELECT AVG(E2.Salary) FROM Employee E2 WHERE E2.DeptID = E1.DeptID);`

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2014

Given tables: DEPT(Deptno, Dname, Loc), PROJECT(Projno, Deptno, Projname). Write SQL queries for project allocations and department summaries.

SQL Solutions: Complete SQL join queries, aggregate counts per department, and filter criteria.

BCA-501 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2014

Explain Log-Based Recovery and Checkpoint Mechanisms in DBMS.

Database Recovery: Deferred update, immediate update, write-ahead logging (WAL), and checkpoint-based fast recovery.

SECTION B • MEDIUM ANSWER QUESTIONS (7.5 MARKS / CORE CONCEPTS) (19 UNIQUE QUESTIONS)

BCA-501 | Marks: 3M / 7.5M | 🔥 Highly Repeated (7×)

CCSU Exam Years: 2025, 2023, 2019, 2018, 2016, 2014, 2013

Explain Primary Key, Foreign Key, and Candidate Key with suitable SQL examples.

1. Candidate Key: A minimal super key — an attribute or minimal set of attributes that uniquely identifies each tuple in a relation without any redundant attributes.
2. Primary Key: The specific candidate key chosen by the database designer to uniquely identify tuples in a table. It cannot accept duplicate values and cannot contain `NULL`.
3. Foreign Key: An attribute in a child table whose values must match the primary key of a referenced parent table (or be `NULL`), enforcing referential integrity.

SQL Implementation Example:

```
-- Parent Table: DEPARTMENT
CREATE TABLE Department (
    DeptID INT PRIMARY KEY,              -- Primary Key
    DeptName VARCHAR(50) NOT NULL UNIQUE -- Candidate Key
);

-- Child Table: EMPLOYEE
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,               -- Primary Key
    EmpName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE,           -- Candidate Key
    DeptID INT,                          -- Foreign Key
    CONSTRAINT fk_dept FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

BCA-501 | Marks: 3M / 7.5M | 🔥 Highly Repeated (4×) | CCSU Exam Years: 2025, 2023, 2019, 2018

State the key advantages of DBMS over File Processing Systems.

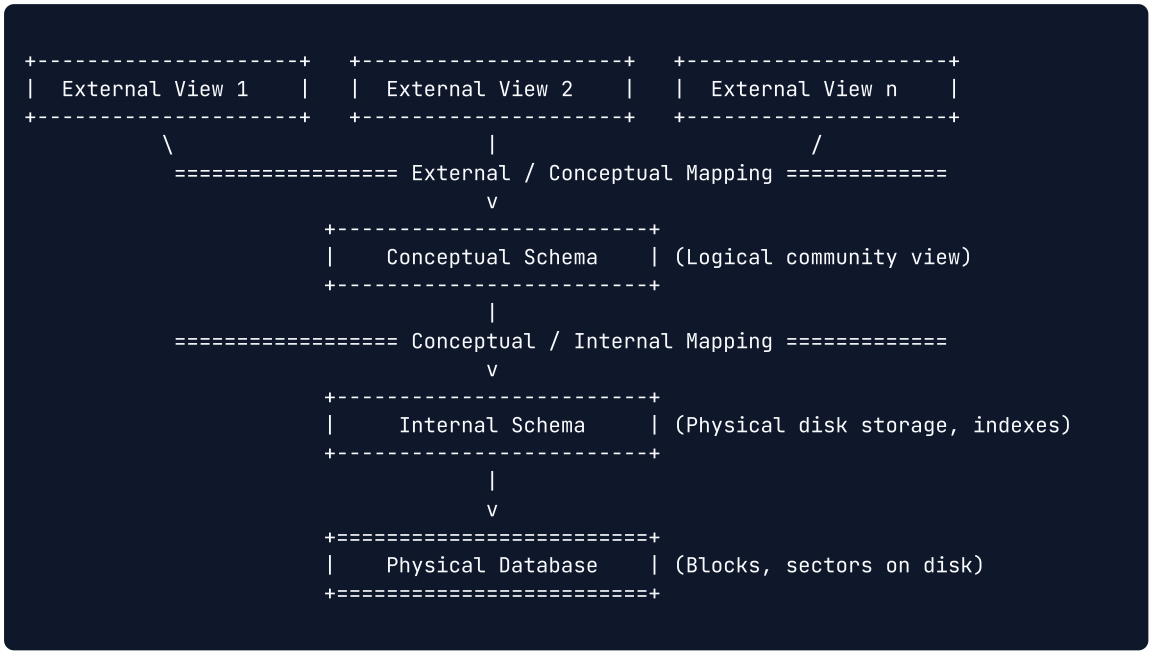
Advantages:

Aspect	Traditional File System	Database Management System (DBMS)
Data Redundancy	High redundancy across duplicate files	Controlled and minimized via normalization
Data Inconsistency	High risk due to isolated updates	Eliminated via centralized repository
Data Independence	None; changes in files break code	Physical and Logical Data Independence achieved
Concurrent Access	Prone to race conditions and lost updates	ACID compliant concurrency control (2PL, timestamp)
Security & Integrity	OS-level passwords, hard to enforce	Fine-grained role-based access & constraints

BCA-501 | Marks: 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2021, 2015, 2014

Explain the Three-Level Architecture of DBMS with a neat block diagram.

Three-Level ANSI/SPARC Architecture:



1. External Level (View Level): User-specific customized views presenting only relevant subsets of data while hiding confidential fields.
2. Conceptual Level (Logical Level): Describes *what data* is stored in the entire database and the relationships among data without physical storage details.
3. Internal Level (Physical Level): Describes *how data* is physically stored on storage media (record formats, B+ trees, hashing).

BCA-501 | Marks: 3M / 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2021, 2018, 2013

Explain Hashing in brief with its collision handling techniques.

Definition: Hashing is an efficient direct search technique where a mathematical hash function `h(k)` maps a search key `k` directly to the disk address/bucket number where the record resides, providing average `O(1)` time complexity.
Collision: Occurs when two distinct search keys hash to the same bucket address: `h(k_1) = h(k_2)`.
Collision Handling Techniques:

- **Open Addressing (Linear Probing):** Searches the next contiguous sequential bucket until an empty slot is located.
- **Chaining / Separate Overflow Buckets:** Maintains a linked list of overflow buckets chained to the primary bucket.
- **Extendible Hashing (Dynamic Hashing):** Uses a directory of pointers with local and global depths to double directories and split buckets dynamically without rehashing all records.

BCA-501 | Marks: 3M / 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2020, 2019, 2018

What is a Referential Integrity Constraint? Why do we need it? Explain with a suitable example.

Explanation: Referential integrity requires that a foreign key value in a referencing table must match an existing primary key value in the referenced table, or be `NULL`.
Need: Prevents orphan records and inconsistent states.
Example:

```
-- Parent Table
CREATE TABLE Department (DeptNo INT PRIMARY KEY, DName VARCHAR(30));

-- Child Table with Foreign Key
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    DeptNo INT,
    CONSTRAINT fk_emp_dept FOREIGN KEY (DeptNo) REFERENCES Department(DeptNo)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

If department 10 is deleted, `ON DELETE CASCADE` automatically removes all employees assigned to department 10, preventing dangling references.

BCA-501 | Marks: 3M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2024, 2023

Define the following terms: (a) Data Model, (b) Database Schema.

(a) Data Model: A conceptual collection of tools, constructs, and notations used to describe data, data relationships, data semantics, and consistency constraints. Major models include:

- **Relational Model:** Tables of tuples and attributes (SQL).
- **Entity-Relationship (ER) Model:** High-level conceptual modeling.
- **Object-Oriented & Document (NoSQL) Models:** JSON, key-value, graph.

(b) Database Schema: The overall structural skeleton of the database defined during the design phase, distinct from the dynamic instances/states stored inside it.

BCA-501 | Marks: 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2019, 2018

Explain Third Normal Form (3NF) with example.

Definition: A relation is in **3NF** if it is in 2NF and has no transitive functional dependencies for non-prime attributes. Formally, for every non-trivial FD `X \to Y`, either `X` is a super key or `Y` is a prime attribute.
Example: `Student(RollNo, Name, DeptID, DeptHead)`.
`RollNo \to DeptID` and `DeptID \to DeptHead` (Transitive!).
Decompose into: `Student(RollNo, Name, DeptID)` and `Department(DeptID, DeptHead)`.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2025

Explain ER Diagram. Describe all standard ER notations and symbols with diagrams.

Definition: An **Entity-Relationship (ER) Diagram** is a graphical semantic modeling tool that represents the logical structure of a database in terms of entities, their attributes, and relationships.

Standard Symbols & Notations:

```
[ Rectangle ]           → Entity Set (e.g., STUDENT, COURSE)
[[ Double Rectangle ]]  → Weak Entity Set (e.g., DEPENDENT, INSTALLMENT)
( Ellipse / Oval )      → Attribute (e.g., Age, Name)
( Underlined Text )     → Primary Key Attribute (e.g., RollNo)
(( Double Ellipse ))    → Multivalued Attribute (e.g., PhoneNumbers, Skills)
( Dashed Ellipse )      → Derived Attribute (e.g., Age calculated from DOB)
{ Diamond }             → Relationship Set (e.g., ENROLLED_IN)
{{ Double Diamond }}    → Identifying Relationship for Weak Entity Set
======== Double Line    → Total Participation (Every entity must participate)
-------- Single Line    → Partial Participation
```

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2024

Find the minimum number of tables required for the given ER diagram in the relational model. Explain table conversion rules.

Rules for ER to Relational Reduction:

1. **Strong Entity Sets:** Each strong entity converts into a separate independent table.
2. **Weak Entity Sets:** Converts into a table containing its discriminator (partial key) combined with the primary key of the identifying strong entity as composite primary key.
3. **1:1 Relationships:** Combine both entities into 1 table, or place the primary key of one table as a foreign key in the other table.
4. **1:N Relationships:** Place the primary key of the '1' side entity into the table of the 'N' side entity as a foreign key (no new table needed).
5. **M:N Relationships:** Requires a new separate relationship table containing primary keys of both participating entities as foreign keys.

Result: For an ER model with two strong entities A and B in 1:N relationship, exactly **2 tables** are needed. If the relationship is M:N, exactly **3 tables** are required.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2024

What are UNION, MINUS, and INTERSECT commands in DBMS? Explain with syntax and examples.

Conditions: Both queries must be *Union-Compatible* (same number of columns with compatible data types).

- `UNION` : Combines results of two SELECT queries and eliminates duplicate tuples.
- `INTERSECT` : Returns only the common tuples present in both result sets.
- `MINUS / EXCEPT` : Returns tuples from the first query that are not present in the second query.

SQL Examples:

```
-- UNION: All students from CS or IT
SELECT RollNo, Name FROM Student_CS
UNION
SELECT RollNo, Name FROM Student_IT;

-- INTERSECT: Students enrolled in both Math and Physics clubs
SELECT StudentID FROM MathClub
INTERSECT
SELECT StudentID FROM PhysicsClub;

-- MINUS: Registered students who have NOT paid exam fees
SELECT StudentID FROM RegisteredStudents
EXCEPT
SELECT StudentID FROM PaidStudents;
```

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2023

Compare Generalization and Specialization with examples.

Comparison:

Criteria	Generalization	Specialization
Approach	**Bottom-up** design process	**Top-down** design process
Mechanism	Combines common features of multiple lower-level entities into a single higher-level entity.	Splits a higher-level entity into specialized sub-entities based on distinct features.
Goal	Minimizes redundancy by factoring out common attributes.	Identifies distinguishing characteristics of specific sub-categories.
Example	Entities `CAR` and `TRUCK` generalized into higher entity `VEHICLE`.	Entity `EMPLOYEE` specialized into `SALARIED_EMP` and `HOURLY_EMP`.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2021

Define Primary Key, NOT NULL Key, and UNIQUE Key with differences.

Definitions & Differences:

Feature	PRIMARY KEY	UNIQUE KEY	NOT NULL
Purpose	Uniquely identifies each record.	Ensures distinct non-duplicate values.	Prevents missing or null values.
NULL Allowed	**Never** allowed (strictly 0 nulls).	Allowed (usually 1 or more nulls per column).	Does not enforce uniqueness, only disallows null.
Count per table	Only **1** per table.	Multiple unique constraints allowed.	Can be applied to every column.
Index created	Automatically creates Primary Clustered Index.	Creates Secondary Unique Non-Clustered Index.	No index created by default.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2020

Explain Lossless-Join Decomposition and Dependency Preservation in relational design.

1. Lossless-Join Decomposition:
A decomposition of relation `R` into `R_1` and `R_2` is lossless if and only if natural join of the projections equals the original relation without generating spurious tuples:

```
R_1 ⋈ R_2 = R ⇔ (R_1 ∩ R_2 → R_1) OR (R_1 ∩ R_2 → R_2)
```

2. Dependency Preservation:
A decomposition preserves dependencies if the union of functional dependencies enforceable within individual decomposed relations covers all functional dependencies of the original relation: `(F_1 ∪ F_2)^+ = F^+`.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2020

Compare 3NF and BCNF with an illustrative example.

Key Differences:

- **3NF:** For every non-trivial FD `X \to Y`, either `X` is a super key *OR* `Y` is a prime attribute (part of any candidate key). Allows dependency preservation always.
- **BCNF:** For every non-trivial FD `X \to Y`, `X` **must strictly be a super key**. Eliminates all FD redundancy, but may not always preserve all functional dependencies.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2019

Compare B-Tree and B+ Tree Index Structures.

Comparison:

Feature	B-Tree	B+ Tree
Data Storage	Keys and data records/pointers stored in both internal and leaf nodes.	Data pointers stored **only in leaf nodes**; internal nodes store only routing keys.
Leaf Node Linking	Leaf nodes are not linked.	Leaf nodes are connected via a **doubly linked list**.
Range Queries	Slow (requires in-order tree traversal).	Extremely fast (scan linked list across leaf nodes).
Node Capacity	Fewer keys per node due to data pointers.	High fan-out, fewer levels, reduced disk I/O.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2019

Explain Conflict Serializability and Testing for Serializability using Precedence Graph.

Answer:

A schedule S is **Conflict Serializable** if it is conflict equivalent to some serial schedule of the same transactions by swapping non-conflicting operations.

> **Conflict Operations:** Two operations conflict if: (1) they belong to different transactions, (2) they access the same data item, and (3) at least one of them is a `write()` operation.
>
> **Precedence Graph (Serialization Graph) Testing:**
>
> 1. Construct a directed graph `G = (V, E)` where each vertex represents an active transaction T_i.
> 2. Draw a directed edge $T_i \rightarrow T_j$ if T_i executes an operation that conflicts with an operation executed later by T_j.
> 3. **Theorem:** A schedule S is Conflict Serializable **if and only if its Precedence Graph contains NO directed cycles**.
> 4. If acyclic, Topological Sorting yields the equivalent serial execution order.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2017

What is Query Optimization? Explain heuristic optimization.

Answer:

Query Optimization is the process of selecting the most efficient physical execution plan (evaluating lowest disk I/O, CPU, and memory cost) among equivalent relational algebra strategies.

Heuristic Optimization Rules (Pushing Down Operations):

1. **Perform Selections (σ) as early as possible:** Pushing selections down reduces the number of tuples before costly joins.
2. **Perform Projections (π) as early as possible:** Discarding unneeded columns reduces tuple byte width and memory consumption.
3. **Combine Selection with Cartesian Product into Join:** Replace $\sigma_{R.A = S.A}(R \times S)$ with Equi-Join $R \bowtie S$ to eliminate full cross products.
4. **Evaluate Restrictive Joins First:** Order joins so intermediate relation sizes are minimized.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2016

Explain Deadlock Detection and Prevention methods in DBMS.

Answer:

A **Deadlock** is a condition where two or more transactions are in a simultaneous circular wait state, each holding locks that the other requires to proceed.

1. Deadlock Prevention Strategies (Timestamp-Based):

- **Wait-Die Scheme (Non-Preemptive):** If older transaction T_i requests a lock held by younger T_j, T_i is allowed to **wait**. If younger requests from older, younger **dies** (aborts and restarts with original timestamp).
- **Wound-Wait Scheme (Preemptive):** If older requests from younger, older **wounds** (preempts/aborts) younger. If younger requests from older, younger is allowed to **wait**.

2. Deadlock Detection & Recovery:

- **Wait-For Graph (WFG):** Directed graph where nodes represent active transactions. Directed edge $T_1 \rightarrow T_2$ indicates T_1 is waiting for T_2. A **cycle in the WFG signals a deadlock**.
- **Deadlock Recovery:** Pick a victim transaction based on lowest rollback cost, abort it, release its locks, and restart it.

BCA-501 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2014

Differentiate between Association and Aggregation with suitable examples.

Answer:

Feature	Association	Aggregation
Concept	A generic structural relationship between two independent classes ("knows-a" relationship).	A specialized form of association representing a whole-part relationship ("has-a" relationship).
Ownership & Lifetime	Both objects have independent lifecycles; neither owns the other.	The parent ("whole") contains the child ("part"), but the child can exist independently of the parent.
Tightness of Coupling	Loose coupling.	Moderate coupling (weaker than Composition where parent death destroys children).
Real-World Example	**Teacher and Student:** A Teacher associates with a Student, but both exist independently.	**Department and Teacher:** A Department aggregates Teachers. If Department closes, Teachers still exist.

SECTION A • SHORT ANSWER QUESTIONS (3 MARKS / HIGH-YIELD DEFINITIONS) (23 UNIQUE QUESTIONS)

BCA-501 | Marks: 3M | 🔥 Highly Repeated (4×) | CCSU Exam Years: 2023, 2016, 2015, 2013

Define Database Management System (DBMS).

Definition: A **Database Management System (DBMS)** is a computerized system software suite that enables users to define, create, maintain, query, and control access to a shared database. It serves as an intelligent interface between end-user applications and physical data storage.

BCA-501 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2025, 2014

What do you mean by Data Inconsistency in File Management Systems? Explain.

Explanation: Data Inconsistency occurs when multiple copies of the exact same data across different files contain conflicting, contradictory values.
Example: In a university file system, a student's address is recorded in the *Library File* and also in the *Fee Accounting File*. If the student changes their residential address and only the Library office updates their local file while the Fee office does not, the university now holds two contradictory addresses for the same student. This occurs due to uncoordinated duplicate storage without centralized integrity control.

BCA-501 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2020, 2017

What is a Data Dictionary (System Catalog)?

Definition: A **Data Dictionary** is a reserved set of read-only system tables maintained by the DBMS that stores metadata (data about data), including schemas, table descriptions, column domains, integrity constraints, indexes, user privileges, and storage allocations.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2024

What are the advantages of using DBMS?

Key Advantages:

1. **Minimal Redundancy:** Normalization ensures data is stored in one primary location.
2. **Data Integrity:** Automatic enforcement of domain and referential integrity constraints.
3. **Concurrency Control:** Multi-user simultaneous access without collisions via locking protocols.
4. **Crash Recovery:** WAL (Write-Ahead Logging) and checkpoints guarantee durability after hardware failure.
5. **Data Independence:** Separates user views from low-level storage modifications.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2024

What is RDBMS?

Answer:

A **Relational Database Management System (RDBMS)** is a database system founded on E.F. Codd's Relational Model (1970) where data is structured into two-dimensional tables (relations) consisting of rows (tuples) and columns (attributes).

> **Core Pillars of RDBMS:**
>
> - **Relational Integrity:** Enforces Entity Integrity (Primary Keys cannot be null) and Referential Integrity (Foreign Keys).
> - **SQL Interface:** Standardized declarative language for DDL, DML, and DCL commands.
> - **Full ACID Compliance:** Guarantees Atomicity, Consistency, Isolation, and Durability across concurrent transactions.
> - **Commercial Examples:** Oracle Database, Microsoft SQL Server, MySQL, PostgreSQL, SQLite.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

List any five applications of DBMS.

Applications:

1. **Banking:** Customer accounts, loans, ATM transactions, fund transfers.
2. **Airlines & Railways:** Real-time reservation systems, flight schedules, ticketing.
3. **Universities & Colleges:** Student admissions, course registrations, grades, fee accounts.
4. **E-Commerce & Retail:** Online product catalogs, shopping carts, billing, inventory control.
5. **Telecommunications:** Call records, monthly billing, subscriber information, network routing.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

What are Attributes? Give examples.

Definition: An **Attribute** is a descriptive property or characteristic that describes an entity in an entity set.

Examples for Entity STUDENT:

- *Simple Attribute:* `RollNo`, `Age`
- *Composite Attribute:* `FullName` (composed of `FirstName`, `LastName`)
- *Multivalued Attribute:* `PhoneNumbers`, `Hobbies`
- *Derived Attribute:* `Age` (computed from `DateOfBirth` and current date)

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

Define DML (Data Manipulation Language).

Definition: Data Manipulation Language (DML) provides constructs for accessing, inserting, updating, and querying stored database records. Classified into *Procedural DML* (user specifies what data and how to retrieve it) and *Non-Procedural / Declarative DML* (SQL, where user specifies only what data is needed).

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

Give examples of Simple and Composite Attributes of an entity.

Examples:

- **Simple (Atomic) Attribute:** Cannot be divided into subparts (e.g., `RollNo = 101`, `Age = 21`, `Salary = 50000`).
- **Composite Attribute:** Can be divided into smaller sub-components with independent meanings (e.g., `Address` divided into `HouseNo`, `Street`, `City`, `Pincode`; or `Name` into `FirstName`, `LastName`).

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2019

Name the three levels of Data Abstraction.

Three Levels:

1. **Physical Level (Lowest):** Describes how data is stored on disk blocks.
2. **Logical / Conceptual Level (Middle):** Describes what data is stored and relationships among data.
3. **View / External Level (Highest):** Describes customized views for specific user groups.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2019

Distinguish between Functional Dependency and Multivalued Dependency.

Comparison:

- **Functional Dependency (X → Y):** For any two tuples with identical `X` values, their `Y` values must be identical (single-valued dependency). Basis of 2NF, 3NF, BCNF.
- **Multivalued Dependency (X ↠ Y):** The presence of `(x, y_1, z_1)` and `(x, y_2, z_2)` implies the presence of `(x, y_1, z_2)` and `(x, y_2, z_1)`, representing independent multi-valued facts. Basis of 4NF.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2018

What do you mean by Relational Database? Describe.

Definition: A **Relational Database** is a collection of data items organized into formal tables (relations) of rows and columns based on first-order predicate logic, where relations are linked together via primary and foreign key constraints.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2018

Briefly explain the terms Security and Authorization.

Definitions:

- **Database Security:** Protection of the database against unauthorized access, malicious attacks, corruption, or destruction.
- **Authorization:** The mechanism that defines and enforces what actions (e.g., SELECT, INSERT, UPDATE, DROP) a validated user or role is permitted to perform on specific database objects.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2017

Define Tuple, Relation, Attribute, and Domain.

Definitions:

- **Relation:** A two-dimensional table of records.
- **Tuple:** A single row in a relation representing an individual entity instance.
- **Attribute:** A named column describing a specific property.
- **Domain:** The set of allowable atomic values for an attribute.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2017

What are Cardinality and Participation Constraints in ER Modeling?

Definitions:

- **Cardinality Ratio:** Specifies the maximum number of relationship instances that an entity can participate in (1:1, 1:N, M:N).
- **Participation Constraint:** Specifies whether existence of an entity depends on its being related to another entity: *Total* (double line) or *Partial* (single line).

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2016

What is a Weak Entity Set? How is it identified?

Answer:

A **Weak Entity Set** is an entity set that does not possess sufficient attributes to form a Primary Key of its own. It is existentially dependent on a parent **Identifying (Owner) Entity Set**.

Identification & Diagrammatic Representation:

- **Discriminator (Partial Key):** The set of attributes that distinguishes weak entities belonging to the same owner entity. Underlined with a **dashed / dotted line**.
- **Primary Key Composition:** Formed by combining the **Primary Key of the Identifying Entity + Discriminator**.
- **ER Diagram Notation:**
  - Weak Entity Box: **Double Rectangle**.
  - Identifying Relationship: **Double Diamond**.
  - Total Participation: **Double Line**.
- *Example:* `DEPENDENT(EmpID, DepName, Age, Relation)` where `EmpID` is borrowed from `EMPLOYEE`.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2016

Differentiate between Natural Join and Theta Join.

Answer:

Feature	Theta Join (θ-Join)	Natural Join (⋈)
Join Condition	Allows arbitrary comparison operators ($=$, $<$, $>$, $\leq$, $\geq$, $\neq$).	Strictly tests equality ($=$) on attributes sharing identical names.
Duplicate Columns	Retains both common columns in the output relation (redundant columns).	**Automatically eliminates duplicate columns**; retains only one instance of shared attributes.
Syntax & Notation	$R \bowtie_{R.A\ \theta\ S.B} S$	$R \bowtie S$ (no explicit condition required).
Relational Equivalence	Equivalent to Cartesian Product followed by Selection: $\sigma_{Predicate}(R \times S)$.	Equivalent to Equi-Join followed by Projection removing duplicate attribute.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2016

Explain COMMIT and ROLLBACK commands in SQL.

Answer:

`COMMIT` and `ROLLBACK` are Transaction Control Language (TCL) commands that preserve the ACID properties of database transactions:

Command	Syntax	Operational Behavior
COMMIT	`COMMIT;`	Permanently saves all changes made by the transaction to disk. Once committed, changes are durable and cannot be undone. Releases all held locks.
ROLLBACK	`ROLLBACK;` `ROLLBACK TO savepoint_name;`	Cancels all uncommitted modifications made by the active transaction, restoring the database back to its last clean committed state.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2015

What do you understand by Hashing? Explain with example.

Answer:

Hashing is an effective file organization and indexing technique that uses a mathematical algorithm (hash function `h(k)`) to compute direct physical block addresses from search key values.

> **Key Concepts:**
>
> - **Hash Function:** `h(Key) = Address`. Desirable properties: uniform distribution and random dispersion to avoid hotspots.
> - **Collision:** Occurs when two distinct keys hash to the same bucket address: $h(k_1) = h(k_2)$.
> - **Collision Resolution Methods:**
>   1. **Open Addressing (Linear Probing):** Searches sequentially for the next available bucket.
>   2. **Chaining (Overflow Buckets):** Maintains a linked list of overflow disk blocks attached to the bucket.
> - **Pros:** Constant time access *O(1)* for equality searches (`WHERE ID = 101`).

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2015

What are Nested SQL Queries? Explain with example.

Definition: A query enclosed inside another SQL query, passed via `WHERE` or `HAVING` clauses using operators like `IN`, `EXISTS`, `ALL`, `ANY`.

```
SELECT EmpName FROM Employee
WHERE Salary > (SELECT AVG(Salary) FROM Employee);
```

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2014

Differentiate between Sequential and Random Access in files.

Differences: Sequential access reads records consecutively in order from the beginning of the file (`O(N)`). Random (direct) access jumps immediately to any record block by its calculated address or key index (`O(1)`).

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2013

What is Data Redundancy?

Answer:

Data Redundancy refers to the unnecessary duplication of the same piece of information across multiple files or tables in a database system.

> **Major Problems Caused by Redundancy:**
>
> 1. **Wasted Storage:** Storing identical department names and manager addresses thousands of times squanders disk space.
> 2. **Data Inconsistency:** If duplicate data is modified in one place but missed in another, conflicting records result.
> 3. **Operational Anomalies:** Causes Insertion, Deletion, and Update anomalies in relational tables.
>
> **Elimination:** Controlled and eliminated through the process of **Relational Normalization (1NF through BCNF)**.

BCA-501 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2013

What is Relational Data Model?

Answer:

Proposed by Dr. Edgar F. Codd in 1970, the **Relational Data Model** represents data in the form of mathematical two-dimensional tables called **Relations**.

> **Fundamental Components:**
>
> - **Relation (Table):** A set of rows representing an entity set.
> - **Tuple (Row):** A single record representing an individual entity instance.
> - **Attribute (Column):** A named field representing a property of the relation.
> - **Domain:** The set of all permissible atomic values for an attribute.
> - **Degree (Arity):** Total count of attributes (columns) in a relation.
> - **Cardinality:** Total count of tuples (rows) in a relation.

SUBJECT: BCA-502 • JAVA PROGRAMMING & DYNAMIC WEBPAGE DESIGN

Total Deduplicated Unique Questions: 50 • All Sessions Fully Consolidated

SECTION C • LONG ANSWER QUESTIONS (15 MARKS / DETAILED SOLUTIONS) (22 UNIQUE QUESTIONS)

BCA-502 | Marks: 15M / 3M / 7.5M | 🔥 Highly Repeated (9×)

CCSU Exam Years: 2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016

Explain Applet Life Cycle. Write a program to demonstrate a simple Java applet to display an image. Compare Applets over HTML.

Applet Image Display Code:

```
import java.applet.*;
import java.awt.*;

/* <applet code="ImageApplet.class" width="400" height="300"></applet> */
public class ImageApplet extends Applet {
    Image img;
    public void init() {
        img = getImage(getDocumentBase(), "college_logo.png");
    }
    public void paint(Graphics g) {
        g.drawImage(img, 20, 20, this);
        g.drawString("CCSU BCA V Semester", 20, 260);
    }
}
```

Comparison over HTML: Applets provide dynamic client-side computation, graphical interactivity, and local state management that static HTML cannot natively provide.

BCA-502 | Marks: 15M / 3M / 7.5M | 🔥 Highly Repeated (9×)

CCSU Exam Years: 2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016

Explain Servlet Life Cycle and compare it with JSP Life Cycle. Write a complete HttpServlet code example.

Servlet Life Cycle Methods:

1. `init(ServletConfig config)` : Loads servlet class, creates instance, and initializes parameters (called once).
2. `service(HttpServletRequest req, HttpServletResponse res)` : Dispatches to `doGet()` or `doPost()` for every incoming HTTP request.
3. `destroy()` : Invoked when container unloads the servlet, releasing database and memory resources.

Servlet Implementation Example:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class WelcomeServlet extends HttpServlet {
    public void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String user = request.getParameter("username");
        out.println("<html><body>");
        out.println("<h2>Welcome, " + user + "!</h2>");
        out.println("<p>Server Response Time: " + new java.util.Date() + "</p>");
        out.println("</body></html>");
    }
}
```

BCA-502 | Marks: 15M / 3M / 7.5M | 🔥 Highly Repeated (7×)

CCSU Exam Years: 2025, 2023, 2020, 2019, 2018, 2017, 2016

Explain Exception Handling mechanism in Java with keywords: try, catch, finally, throw, and throws.

Answer:

Exception Handling in Java is a robust mechanism to handle runtime errors, preventing abnormal program termination and maintaining regular application flow.

> **The Exception Hierarchy:** All exception types inherit from `java.lang.Throwable`. It divides into:
>
> - **Error:** Irrecoverable conditions (e.g., `OutOfMemoryError`, `StackOverflowError`).
> - **Exception:** Conditions that a program should catch and handle. Subdivided into:
> - *Checked Exceptions:* Checked at compile time (e.g., `IOException`, `SQLException`, `ClassNotFoundException`).
> - *Unchecked Exceptions (Runtime):* Inherit from `RuntimeException` (e.g., `NullPointerException`, `ArithmeticException`, `ArrayIndexOutOfBoundsException`).

The Five Core Keywords:

Keyword	Purpose	Syntax Example
`try`	Encloses code that might throw an exception.	`try { int a = 10/0; }`
`catch`	Handles the specific exception thrown by `try` block.	`catch(ArithmeticException e) { ... }`
`finally`	Always executes whether exception occurs or not; used for resource cleanup.	`finally { reader.close(); }`
`throw`	Explicitly throws a single exception instance.	`throw new IllegalArgumentException("Invalid age");`
`throws`	Declares exceptions a method may throw to callers in method signature.	`void readFile() throws IOException { ... }`

Complete Working Java Program Demonstrating All 5 Keywords:

```
public class ExceptionMasterDemo {
    // throws: Declares potential checked exception
    static void checkEligibility(int age) throws ArithmeticException {
        if (age < 18) {
            // throw: Explicitly throws an exception object
            throw new ArithmeticException("Underage: Age must be 18+ to vote.");
        }
        System.out.println("Eligible for voting.");
    }

    public static void main(String[] args) {
        try {
            System.out.println("Opening database resource...");
            checkEligibility(16); // Will trigger exception
        } catch (ArithmeticException ex) {
            System.err.println("Caught Exception: " + ex.getMessage());
        } catch (Exception ex) {
            System.err.println("Generic Handler: " + ex);
        } finally {
            // finally block is GUARANTEED to execute
```

```
            System.out.println("Finally Block: Resources and connections safely
closed.");
        }
    }
}
```

BCA-502 | Marks: 15M / 3M / 7.5M | 🔥 Highly Repeated (6×)

CCSU Exam Years: 2025, 2020, 2019, 2018, 2017, 2016

What is the difference between Process-based and Thread-based multitasking in Java?

Comparison:

Feature	Process-Based Multitasking	Thread-Based Multitasking
Granularity	Heavyweight; each process has its own separate address space.	Lightweight; threads share the same address space in a single process.
Context Switching	High overhead; managed by operating system kernel.	Low overhead; switches quickly within the JVM.
Communication	Requires Inter-Process Communication (IPC, sockets, pipes).	Direct shared memory, wait(), notify(), synchronized blocks.

BCA-502 | Marks: 15M / 3M / 7.5M | 🔥 Highly Repeated (6×)

CCSU Exam Years: 2025, 2023, 2020, 2018, 2017, 2016

Define Thread in Java. Explain how to create threads using (a) Thread class and (b) Runnable interface. Write a complete multithreaded program.

Multithreading in Java:

```
// Approach 1: Extending Thread class
class NumberPrinter extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println("Thread 1: " + i);
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
}

// Approach 2: Implementing Runnable interface
class LetterPrinter implements Runnable {
    public void run() {
        for(char c = 'A'; c ⩽ 'E'; c++) {
            System.out.println("Thread 2: " + c);
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
}

public class MultiThreadDemo {
    public static void main(String[] args) {
        NumberPrinter t1 = new NumberPrinter();
        Thread t2 = new Thread(new LetterPrinter());
        t1.start(); // Spawns new thread
        t2.start();
    }
}
```

BCA-502 | Marks: 15M / 3M / 7.5M | 🔥 Highly Repeated (4×)

CCSU Exam Years: 2024, 2023, 2018, 2016

What are Packages in Java? How is a user-defined package created and imported? Explain with complete code example.

Answer:

A **Package** in Java is a container or namespace that groups related classes, interfaces, and sub-packages together. Packages prevent naming conflicts, control access visibility, and provide modular software architecture.

> **Types of Packages:**
>
> 1. **Built-in Packages:** Provided by Java SE API (e.g., `java.lang`, `java.util`, `java.io`, `java.sql`, `java.net`, `java.awt`).
> 2. **User-Defined Packages:** Created by developers to organize custom application modules.

Creating and Importing a User-Defined Package:

```
// Step 1: File saved as 'Calculator.java' in folder 'com/ccsu/math/'
package com.ccsu.math;

public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
    public int multiply(int a, int b) {
        return a * b;
    }
}
```

```
// Step 2: Main application importing the custom package
package com.ccsu.app;

import com.ccsu.math.Calculator; // Importing specific class

public class MainApp {
    public static void main(String[] args) {
        Calculator calc = new Calculator();
        System.out.println("Addition: " + calc.add(15, 25));       // Output: 40
        System.out.println("Multiplication: " + calc.multiply(6, 7)); // Output: 42
    }
}
```

Compilation and Execution with CLASSPATH:

```
javac -d . Calculator.java    # -d creates the directory structure com/ccsu/math/
javac -d . MainApp.java
java com.ccsu.app.MainApp
```

Package Access Protection Matrix:

Modifier	Same Class	Same Package	Subclass (Different Pkg)	World (Different Pkg)
`public`	Yes	Yes	Yes	Yes
`protected`	Yes	Yes	Yes	No
`default` (package-private)	Yes	Yes	No	No

`private`	Yes	No	No	No

BCA-502 | Marks: 15M | ★ Repeated (3×) | CCSU Exam Years: 2025, 2017, 2016

What is JDBC? Explain the Four Types of JDBC Drivers with architectural diagrams and characteristics.

Definition: Java Database Connectivity (JDBC) is an industry-standard Java API that enables Java programs to interact with relational databases using SQL queries.

Four Types of JDBC Drivers:

1. **Type-1: JDBC-ODBC Bridge Driver:** Translates JDBC calls to ODBC calls. Requires ODBC driver installation on every client machine. Obsolete and slow.
2. **Type-2: Native-API / Partially Java Driver:** Converts JDBC calls directly into native C/C++ vendor database client libraries (e.g., Oracle OCI). Requires client-side installation.
3. **Type-3: Network Protocol / All-Java Middleware Driver:** Translates JDBC calls into an intermediate server middleware protocol, which then talks to multiple databases. Flexible for 3-tier architectures.
4. **Type-4: Thin Driver / Direct-to-Database Pure Java Driver:** Converts JDBC calls directly into vendor-specific database network protocols via TCP/IP sockets. 100% pure Java, zero client installation, highest performance (Modern standard!).

BCA-502 | Marks: 15M | ★ Repeated (3×) | CCSU Exam Years: 2025, 2024, 2017

What are Layout Managers in AWT/Swing? Explain BorderLayout, FlowLayout, and GridLayout with examples.

Layout Managers: Automatically position, size, and rearrange GUI components inside containers regardless of screen resolution or OS window scaling.

- **FlowLayout:** Arranges components sequentially from left to right, wrapping to the next line when space runs out. Default for `Panel` and `Applet`.
- **BorderLayout:** Divides container into 5 regions: `NORTH`, `SOUTH`, `EAST`, `WEST`, and `CENTER`. Default for `Frame` and `JFrame`.
- **GridLayout:** Arranges components into a rectangular grid of equal-sized rows and columns.

BCA-502 | Marks: 15M | ★ Repeated (3×) | CCSU Exam Years: 2024, 2019, 2018

What is JDBC? Write a complete JDBC program to insert and display records of employees using PreparedStatement.

Complete PreparedStatement JDBC Program:

```
import java.sql.*;

public class EmployeeJDBC {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/company_db";
        String user = "root";
        String password = "password";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, user, password);

            // 1. Insert record using PreparedStatement
            String insertSql = "INSERT INTO Employee(EmpID, Name, Salary) VALUES(?,
?, ?)";
            PreparedStatement psInsert = conn.prepareStatement(insertSql);
            psInsert.setInt(1, 101);
            psInsert.setString(2, "Rahul Sharma");
            psInsert.setDouble(3, 55000.00);
            int rows = psInsert.executeUpdate();
            System.out.println("Inserted rows: " + rows);

            // 2. Display all employee records
            String selectSql = "SELECT EmpID, Name, Salary FROM Employee";
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(selectSql);

            System.out.println("\n--- EMPLOYEE RECORDS ---");
            while(rs.next()) {
                System.out.println(rs.getInt("EmpID") + "\t" +
                                   rs.getString("Name") + "\t" +
                                   rs.getDouble("Salary"));
            }
            conn.close();
        } catch(Exception e) {
            e.printStackTrace();
        }
    }
}
```

BCA-502 | Marks: 15M / 22.5M | ★ Repeated (2×) | CCSU Exam Years: 2021, 2018

Explain AWT Event Handling with Delegation Event Model. Write a program to handle Button click events.

Answer:

The **Delegation Event Model** is the standard design pattern in Java AWT and Swing for processing GUI user interactions. It decouples the UI event source from the event listener.

> **Three Core Components of the Delegation Model:**
>
> 1. **Event Source:** The GUI component that generates the event when state changes (e.g., `Button`, `TextField`, `Checkbox`).
> 2. **Event Object:** Encapsulates all event details (e.g., `ActionEvent`, `MouseEvent`, `KeyEvent`) inheriting from `java.util.EventObject`.
> 3. **Event Listener:** An interface containing callback methods (e.g., `ActionListener`, `MouseListener`). The listener registers with the source using registration methods like `addActionListener()`.

Complete Working Java Program (Button Click Event Handling):

```
import java.awt.*;
import java.awt.event.*;

public class ButtonClickEventDemo extends Frame implements ActionListener {
    private Label label;
    private Button btnClick, btnExit;
    private int counter = 0;

    public ButtonClickEventDemo() {
        // Frame Configuration
        setTitle("AWT Delegation Event Model Demo");
        setSize(400, 200);
        setLayout(new FlowLayout());

        // Create Components
        label = new Label("Button clicks: 0");
        btnClick = new Button("Click Me!");
        btnExit = new Button("Exit");

        // Register Event Listeners (Delegation)
        btnClick.addActionListener(this);
        btnExit.addActionListener(this);

        // Add to Frame Container
        add(label);
        add(btnClick);
        add(btnExit);

        // Window closing adapter
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });

        setVisible(true);
    }

    // Callback method implementation for ActionListener
    @Override
    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == btnClick) {
            counter++;
```

```
            label.setText("Button clicks: " + counter);
        } else if (e.getSource() == btnExit) {
            System.exit(0);
        }
    }

    public static void main(String[] args) {
        new ButtonClickEventDemo();
    }
}
```

BCA-502 | Marks: 15M | ★ Repeated (2×) | CCSU Exam Years: 2024, 2020

Discuss JSP in detail. What are JSP Directives? Explain various types of directives with examples.

JSP Directives (`<%@ directive attribute="value" %>`):

1. **page Directive:** Controls page-level settings: `<%@ page import="java.util.*, java.sql.*" contentType="text/html" errorPage="error.jsp" %>`
2. **include Directive:** Inserts static files at translation time: `<%@ include file="header.jsp" %>`
3. **taglib Directive:** Imports custom tag libraries (JSTL): `<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>`

BCA-502 | Marks: 15M | ★ Repeated (2×) | CCSU Exam Years: 2023, 2020

What is JDBC? Explain JDBC Architecture and execute a database SELECT query.

Answer:

Java Database Connectivity (JDBC) is a standard Java API (part of `java.sql` and `javax.sql` packages) that provides uniform, vendor-neutral communication between Java applications and relational database engines.

JDBC Architecture:

- **Application:** Java code issuing SQL queries via JDBC API interfaces.
- **JDBC Driver Manager:** Backbone component that loads database-specific drivers and establishes connections using connection strings.
- **JDBC Drivers:** Translates standard JDBC API calls into vendor-specific database protocols.
  1. **Type-1 (JDBC-ODBC Bridge):** Translates JDBC to ODBC; slow, deprecated.
  2. **Type-2 (Native API):** Converts JDBC to native C/C++ client libraries; client installation needed.
  3. **Type-3 (Network Protocol):** Communicates with middle-tier application server.
  4. **Type-4 (Thin Driver - Pure Java):** Directly speaks database socket protocol; fast, platform-independent, widely used (e.g., MySQL Connector/J).

Complete Working Java Program (Database SELECT & INSERT Query):

```
import java.sql.*;

public class JdbcFullDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/college_db";
        String user = "root";
        String pass = "password";

        // Step 1 & 2: Load Driver & Open Connection
        try (Connection con = DriverManager.getConnection(url, user, pass)) {
            System.out.println("Connected to MySQL Database successfully!");

            // Step 3: Prepared Statement INSERT
            String insertSql = "INSERT INTO students(roll_no, name, branch) VALUES
(?, ?, ?)";
            try (PreparedStatement psInsert = con.prepareStatement(insertSql)) {
                psInsert.setInt(1, 105);
                psInsert.setString(2, "Ananya Verma");
                psInsert.setString(3, "BCA");
                int rowsInserted = psInsert.executeUpdate();
                System.out.println("Rows inserted: " + rowsInserted);
            }

            // Step 4: Execute SELECT Query
            String selectSql = "SELECT roll_no, name, branch FROM students";
            try (Statement stmt = con.createStatement();
                 ResultSet rs = stmt.executeQuery(selectSql)) {
                System.out.println("\n--- Student Records ---");
                while (rs.next()) {
                    int roll = rs.getInt("roll_no");
                    String name = rs.getString("name");
                    String branch = rs.getString("branch");
                    System.out.println("Roll: " + roll + " | Name: " + name + " |
Branch: " + branch);
                }
            }
        } catch (SQLException ex) {
            System.err.println("JDBC Error: " + ex.getMessage());
        }
```

```
    }
}
```

BCA-502 | Marks: 15M | ★ Repeated (2×) | CCSU Exam Years: 2021, 2016

Explain Client-Server Socket Programming in Java with complete code.

Answer:

Socket programming in Java (via `java.net`) enables two computers to communicate over a TCP/IP network using bidirectional stream connections.

> **Key Socket Classes:**
>
> - `ServerSocket` : Used on the server to listen on a designated port and accept incoming connections via `accept()` .
> - `Socket` : Used by the client to initiate connection to the server's IP address and port, and returned by `accept()` on server for data I/O.

1. Server Program (TCPServer.java):

```
import java.io.*;
import java.net.*;

public class TCPServer {
    public static void main(String[] args) {
        int port = 5000;
        try (ServerSocket serverSocket = new ServerSocket(port)) {
            System.out.println("Server listening on port " + port + "... Waiting
for client.");

            Socket socket = serverSocket.accept(); // Blocks until client connects
            System.out.println("Client connected: " + socket.getInetAddress());

            BufferedReader in = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
            PrintWriter out = new PrintWriter(socket.getOutputStream(), true);

            String clientMsg = in.readLine();
            System.out.println("Received from Client: " + clientMsg);

            // Echo back uppercase
            out.println("SERVER ECHO: " + clientMsg.toUpperCase());
            socket.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

2. Client Program (TCPClient.java):

```
import java.io.*;
import java.net.*;

public class TCPClient {
    public static void main(String[] args) {
        String host = "127.0.0.1";
        int port = 5000;
        try (Socket socket = new Socket(host, port)) {
            PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
            BufferedReader in = new BufferedReader(new
InputStreamReader(socket.getInputStream()));

            out.println("Hello CCSU Server from BCA 5th Sem!");
            String response = in.readLine();
            System.out.println("Server Response: " + response);
        } catch (IOException e) {
            e.printStackTrace();
```

```
        }
    }
}
```

BCA-502 | Marks: 22.5M | CCSU Exam Question | CCSU Exam Years: 2021

Explain JDBC Architecture and write a program to update employee salaries based on department.

Program:

```
import java.sql.*;

public class UpdateSalaryDemo {
    public static void main(String[] args) {
        try {
            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/db", "root", "pass");
            PreparedStatement ps = con.prepareStatement("UPDATE Employee SET Salary
= Salary * 1.10 WHERE Dept = ?");
            ps.setString(1, "IT");
            int rows = ps.executeUpdate();
            System.out.println("Salaries updated for " + rows + " employees.");
            con.close();
        } catch(Exception e) { e.printStackTrace(); }
    }
}
```

BCA-502 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2025

Write an HTML code to create a User Registration Web Form containing: Username, Date of Birth, Address, Gender (Radio buttons), Email-ID, Mobile Number, and Submit/Reset buttons.

HTML Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>User Registration Form</title>
</head>
<body>
    <h2>Student / User Registration Form</h2>
    <form action="register" method="POST">
        <table cellpadding="6">
            <tr>
                <td>User Name:</td>
                <td><input type="text" name="username" required /></td>
            </tr>
            <tr>
                <td>Date of Birth:</td>
                <td><input type="date" name="dob" required /></td>
            </tr>
            <tr>
                <td>Address:</td>
                <td><textarea name="address" rows="3" cols="30"></textarea></td>
            </tr>
            <tr>
                <td>Gender:</td>
                <td>
                    <input type="radio" name="gender" value="Male" /> Male
                    <input type="radio" name="gender" value="Female" /> Female
                </td>
            </tr>
            <tr>
                <td>Email-ID:</td>
                <td><input type="email" name="email" required /></td>
            </tr>
            <tr>
                <td>Mobile Number:</td>
                <td><input type="tel" name="mobile" pattern="[0-9]{10}" required />
</td>
            </tr>
            <tr>
                <td colspan="2" align="center">
                    <input type="submit" value="Register Now" />
                    <input type="reset" value="Clear Form" />
                </td>
            </tr>
        </table>
    </form>
</body>
</html>
```

BCA-502 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2024

What are Packages? Explain Java API Packages. List any five methods of the String class with examples.

String Class Methods:

1. `length()` : Returns character count → `"Hello".length() == 5`.
2. `charAt(int index)` : Returns character at index → `"Hello".charAt(1) == 'e'`.
3. `substring(int begin, int end)` : Extracts substring → `"Java".substring(0, 2) == "Ja"`.
4. `toUpperCase()` : Converts to uppercase → `"bca".toUpperCase() == "BCA"`.
5. `equals(Object another)` : Compares string contents → `"Java".equals("Java") == true`.

BCA-502 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2023

Explain HTML form controls and write code for an Admission Form.

Answer:

HTML Forms collect user input on web applications and submit it to web servers (Servlets/JSP) via HTTP `GET` or `POST` methods.

Complete Code for BCA Admission Form:

```
<!DOCTYPE html>
<html>
<head>
    <title>CCSU BCA Admission Form</title>
    <style>
        body { font-family: Arial, sans-serif; background: #f0f2f5; padding: 20px;
}
        .form-card { max-width: 500px; margin: auto; background: white; padding:
25px; border-radius: 8px; box-shadow: 0 2px 8px rgba(0,0,0,0.1); }
        .form-group { margin-bottom: 14px; }
        label { display: block; font-weight: bold; margin-bottom: 4px; font-size:
13px; }
        input, select, textarea { width: 100%; padding: 8px; border: 1px solid
#ccc; border-radius: 4px; box-sizing: border-box; }
        .btn-submit { background: #2563eb; color: white; border: none; padding:
10px 16px; border-radius: 4px; cursor: pointer; font-weight: bold; width: 100%; }
    </style>
</head>
<body>
<div class="form-card">
    <h2 style="text-align:center; color:#1e3a8a;">BCA Admission Registration</h2>
    <form action="AdmissionServlet" method="POST">
        <div class="form-group">
            <label>Full Student Name:</label>
            <input type="text" name="fullName" required placeholder="Enter your
name">
        </div>
        <div class="form-group">
            <label>Email Address:</label>
            <input type="email" name="email" required
placeholder="name@domain.com">
        </div>
        <div class="form-group">
            <label>Date of Birth:</label>
            <input type="date" name="dob" required>
        </div>
        <div class="form-group">
            <label>Gender:</label>
            <input type="radio" name="gender" value="Male" checked
style="width:auto;"> Male
            <input type="radio" name="gender" value="Female" style="width:auto;
margin-left:15px;"> Female
        </div>
        <div class="form-group">
            <label>Course Stream:</label>
            <select name="course">
                <option value="BCA">Bachelor of Computer Applications (BCA)
</option>
                <option value="MCA">Master of Computer Applications (MCA)</option>
            </select>
        </div>
        <div class="form-group">
            <label>Address:</label>
            <textarea name="address" rows="3"></textarea>
        </div>
        <button type="submit" class="btn-submit">Submit Application</button>
    </form>
</div>
```

```
</body>
</html>
```

BCA-502 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2023

Explain JSP implicit objects and session tracking in Web Applications.

Answer:

JavaServer Pages (JSP) provides **9 implicit objects** automatically instantiated by the JSP container and directly accessible inside scriptlets and expressions without explicit declaration:

Implicit Object	Underlying Class / Interface	Scope & Purpose
`request`	`HttpServletRequest`	Encapsulates incoming HTTP request parameters, headers, cookies.
`response`	`HttpServletResponse`	Encapsulates HTTP response to client (cookies, redirects).
`out`	`JspWriter`	Writes buffered HTML output stream back to browser.
`session`	`HttpSession`	Tracks user session across multiple page requests.
`application`	`ServletContext`	Global application context shared by all users.
`config`	`ServletConfig`	Configuration initialization parameters for this JSP page.
`pageContext`	`PageContext`	Provides access to all scopes and attributes.
`page`	`java.lang.Object`	Synonym for `this` (the generated servlet instance).
`exception`	`java.lang.Throwable`	Available only on error pages (`<%@ page isErrorPage="true" %>`).

Session Tracking Techniques in Web Applications:

1. **HttpSession API:** Standard method (`session.setAttribute("user", u);`). Tracks user using `JSESSIONID` cookie.
2. **Cookies:** Small text files stored on client machine sent with HTTP requests.
3. **URL Rewriting:** Appends `;jsessionid=12345` to URLs when cookies are disabled.
4. **Hidden Form Fields:** Embedding hidden HTML inputs (`<input type="hidden" name="sessionId" value="XYZ">`).

BCA-502 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2020

Write HTML code for a complete Web Form with Table formatting and Validation.

Answer:

Web forms collect user input and submit data to server-side scripts. Table-based styling ensures clean alignment of labels and input fields, while JavaScript provides client-side validation before HTTP submission.

```
<!DOCTYPE html>
<html>
<head>
    <title>Validated User Registration Form</title>
    <script>
        function validateForm() {
            var name = document.forms["regForm"]["uname"].value;
            var email = document.forms["regForm"]["email"].value;
            var pass = document.forms["regForm"]["pwd"].value;
            var mobile = document.forms["regForm"]["mob"].value;

            if (name.trim() == "") {
                alert("Validation Error: Full Name is required!");
                return false;
            }
            if (email.indexOf("@") == -1 || email.indexOf(".") == -1) {
                alert("Validation Error: Enter a valid email address!");
                return false;
            }
            if (pass.length < 6) {
                alert("Validation Error: Password must be at least 6 characters
long!");
                return false;
            }
            if (!/^[0-9]{10}$/.test(mobile)) {
                alert("Validation Error: Mobile number must be exactly 10
digits!");
                return false;
            }
            alert("Validation Succeeded: Submitting data to server...");
            return true;
        }
    </script>
</head>
<body style="font-family:Arial, sans-serif; background:#f8fafc; padding:30px;">
    <div style="max-width:550px; margin:auto; background:white; padding:25px;
border-radius:8px; border:1px solid #cbd5e1;">
        <h2 style="text-align:center; color:#1e3a8a;">Student Portal
Registration</h2>
        <form name="regForm" action="RegisterServlet" method="POST"
onsubmit="return validateForm()">
            <table cellpadding="8" cellspacing="0" style="width:100%;">
                <tr>
                    <td><b>Username:</b></td>
                    <td><input type="text" name="uname" style="width:100%;
padding:6px;"></td>
                </tr>
                <tr>
                    <td><b>Email Address:</b></td>
                    <td><input type="text" name="email" style="width:100%;
padding:6px;"></td>
                </tr>
                <tr>
                    <td><b>Password:</b></td>
                    <td><input type="password" name="pwd" style="width:100%;
padding:6px;"></td>
                </tr>
                <tr>
                    <td><b>Mobile (10 digits):</b></td>
```

```
                    <td><input type="text" name="mob" style="width:100%;
padding:6px;"></td>
                </tr>
                <tr>
                    <td><b>Gender:</b></td>
                    <td>
                        <input type="radio" name="gender" value="Male" checked>
Male
                        <input type="radio" name="gender" value="Female"
style="margin-left:15px;"> Female
                    </td>
                </tr>
                <tr>
                    <td colspan="2" style="text-align:center; padding-top:15px;">
                        <input type="submit" value="Register Now"
style="background:#2563eb; color:white; padding:8px 20px; border:none; border-
radius:4px; font-weight:bold; cursor:pointer;">
                        <input type="reset" value="Clear Form"
style="background:#64748b; color:white; padding:8px 15px; border:none; border-
radius:4px; margin-left:10px; cursor:pointer;">
                    </td>
                </tr>
            </table>
        </form>
    </div>
</body>
</html>
```

BCA-502 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2020

Explain Servlets, HttpServletRequest, and HttpServletResponse with complete GET/POST demonstration.

Answer:

A **Servlet** is a server-side Java program running inside a Servlet Container (e.g., Apache Tomcat) to handle incoming dynamic HTTP web requests and produce responses.

> **Servlet Life Cycle:**
>
> `init(ServletConfig config) → service(request, response) → destroy()`
>
> - `init()` : Called once upon servlet class loading into memory.
> - `service()` : Dispatches requests to `doGet()` or `doPost()` per HTTP method for every incoming concurrent request in a separate thread.
> - `destroy()` : Called once before servlet is removed from memory.

Complete Servlet Code Demonstrating GET and POST:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class UserAuthServlet extends HttpServlet {
    // Handles HTTP GET Requests
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse res) throws
IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        out.println("<h2>GET Request Received: Please submit credentials via POST
form.</h2>");
    }

    // Handles HTTP POST Requests (Secure Data Submission)
    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse res) throws
IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        String user = req.getParameter("username");
        String pass = req.getParameter("password");

        if ("admin".equals(user) && "ccsu123".equals(pass)) {
            // Establish session
            HttpSession session = req.getSession();
            session.setAttribute("currentUser", user);
            out.println("<h2 style='color:green;'>Authentication Successful!
Welcome, " + user + "</h2>");
        } else {
            out.println("<h2 style='color:red;'>Invalid Credentials! Access Denied.
</h2>");
        }
    }
}
```

BCA-502 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2019

Explain Event Handling in Java AWT/Swing with MouseListener and ActionListener.

Answer:

In Java GUI programming, user interactions generate event objects intercepted by registered listener interfaces.

Listener Interface	Package	Triggering Event	Key Methods to Implement
`ActionListener`	`java.awt.event`	Button clicks, Enter in text fields, menu selection.	`void actionPerformed(ActionEvent e)`
`MouseListener`	`java.awt.event`	Mouse clicks, presses, releases, enters, exits.	`mouseClicked(MouseEvent e)` `mousePressed(MouseEvent e)` `mouseReleased(MouseEvent e)` `mouseEntered(MouseEvent e)` `mouseExited(MouseEvent e)`

> **MouseAdapter:** Implementing `MouseListener` forces implementation of all 5 methods. `MouseAdapter` is an abstract convenience class providing empty implementations of all 5 methods, allowing developers to override only the required method (e.g., only `mouseClicked()`).

BCA-502 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2019

Write HTML and CSS code to design a responsive institutional web page.

Answer:

A responsive institutional web page adapts dynamically across desktops, tablets, and smartphones using modern CSS Media Queries and flexible layouts.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Chaudhary Charan Singh University • Dept of Computer
Applications</title>
    <style>
        * { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Segoe UI',
Tahoma, sans-serif; }
        header { background: #1e3a8a; color: white; padding: 20px; text-align:
center; }
        nav { background: #0f172a; display: flex; justify-content: center; flex-
wrap: wrap; }
        nav a { color: #f8fafc; padding: 12px 20px; text-decoration: none; font-
weight: 600; font-size: 14px; }
        nav a:hover { background: #2563eb; }
        .hero { background: linear-gradient(rgba(0,0,0,0.6), rgba(0,0,0,0.6)),
url('campus.jpg') center/cover; color: white; text-align: center; padding: 60px
20px; }
        .container { max-width: 1100px; margin: 30px auto; padding: 0 15px;
display: grid; grid-template-columns: 2fr 1fr; gap: 25px; }
        .card { background: white; padding: 20px; border-radius: 8px; box-shadow: 0
2px 6px rgba(0,0,0,0.08); }
        footer { background: #0f172a; color: #94a3b8; text-align: center; padding:
20px; margin-top: 40px; font-size: 13px; }
        /* Media Query for Mobile Responsive Screen */
        @media (max-width: 768px) {
            .container { grid-template-columns: 1fr; }
            nav { flex-direction: column; text-align: center; }
        }
    </style>
</head>
<body>
    <header>
        <h1>Chaudhary Charan Singh University, Meerut</h1>
        <p>Faculty of Engineering & Technology • BCA Department</p>
    </header>
    <nav>
        <a href="#">Home</a>
        <a href="#">Academics</a>
        <a href="#">Examinations</a>
        <a href="#">PYQ Bank</a>
        <a href="#">Contact Us</a>
    </nav>
    <div class="hero">
        <h2>Excellence in Computer Science & Applications</h2>
        <p>Empowering students with industry-standard programming and analytical
skills.</p>
    </div>
    <div class="container">
        <main class="card">
            <h3 style="color:#1e3a8a; margin-bottom:10px;">About BCA Programme</h3>
            <p>The 3-Year Bachelor of Computer Applications curriculum encompasses
Data Structures, DBMS, Java, Computer Networks, and Numerical Analysis to build
cutting-edge software engineering capabilities.</p>
        </main>
        <aside class="card">
            <h3 style="color:#dc2626; margin-bottom:10px;">Notice Board</h3>
            <ul style="padding-left:18px; line-height:1.6; font-size:13px;">
                <li>5th Sem Examination Schedule Released.</li>
```

```
                <li>Solved PYQs uploaded to student portal.</li>
                <li>Project viva submissions due next week.</li>
            </ul>
        </aside>
    </div>
    <footer>
        <p>&copy; CCS University BCA Portal. All Rights Reserved.</p>
    </footer>
</body>
</html>
```

SECTION B • MEDIUM ANSWER QUESTIONS (7.5 MARKS / CORE CONCEPTS) (7 UNIQUE QUESTIONS)

BCA-502 | Marks: 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2024, 2023, 2019

Explain the concepts of Inheritance in Java and its various types.

Inheritance: The mechanism where a subclass derives attributes and behaviors from a superclass using `extends`.

Types in Java:

1. **Single Inheritance:** Class B extends Class A.
2. **Multilevel Inheritance:** Class C extends Class B, which extends Class A.
3. **Hierarchical Inheritance:** Class B and Class C both extend Class A.
4. **Multiple Inheritance:** Achieved through **Interfaces** (Java does not support multiple class inheritance to avoid the Diamond Problem).

BCA-502 | Marks: 3M / 7.5M | CCSU Exam Question | CCSU Exam Years: 2024

Explain TCP/IP Client Socket. Write constructors and methods used to create client sockets.

Client Socket (java.net.Socket):

- **Constructors:**
  - `Socket(String host, int port)` — Establishes TCP stream connection.
  - `Socket(InetAddress address, int port)`
- **Methods:**
  - `getInputStream()` — Returns InputStream for reading server responses.
  - `getOutputStream()` — Returns OutputStream for sending data.
  - `close()` — Gracefully closes the TCP connection.

BCA-502 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2021

What do you mean by a Package in Java? State the different Access Specifiers available in Java.

Access Specifiers:

Access Specifier	Same Class	Same Package	Subclass (Diff Pkg)	World (Everywhere)
`private`	Yes	No	No	No
`default` (package-private)	Yes	Yes	No	No
`protected`	Yes	Yes	Yes (via inheritance)	No
`public`	Yes	Yes	Yes	Yes

BCA-502 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2021

What is Exception? Give different types of exceptions that occur during runtime. Why handle exceptions?

Answer:

An **Exception** in Java is an abnormal event interrupting the normal instruction flow during program execution.

Classification of Java Exceptions:

1. **Checked Exceptions (Compile-Time):** Subclasses of `Exception` (excluding `RuntimeException`). Must be either caught in a `try-catch` or declared with `throws`; otherwise code will not compile.
   - `IOException`: Attempting to read a non-existent file.
   - `SQLException`: Database syntax error or lost network connection.
   - `ClassNotFoundException`: Attempting to load an unavailable class via reflection.
2. **Unchecked Exceptions (Runtime):** Subclasses of `RuntimeException`. Usually caused by logic flaws:
   - `ArithmeticException`: Division by zero (`10 / 0`).
   - `NullPointerException (NPE)`: Accessing a method on a `null` reference.
   - `ArrayIndexOutOfBoundsException`: Accessing array index outside `[0, length-1]`.
   - `NumberFormatException`: Parsing `"ABC"` with `Integer.parseInt()`.

Why Handle Exceptions: Without handling, an exception crashes the active thread, drops open file/database handles, leaves data in corrupt uncommitted states, and displays ugly stack traces to end users.

BCA-502 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2021

Explain Thread Priority. What are default values? Write a program to define two threads where one prints 1 to 10 and the other prints 10 to 1.

Thread Priorities: Integer scale from 1 (`MIN_PRIORITY`) to 10 (`MAX_PRIORITY`), with default 5 (`NORM_PRIORITY`).

Program:

```
class ForwardPrinter extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 10; i++) {
            System.out.println("Forward: " + i);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

class BackwardPrinter extends Thread {
    public void run() {
        for(int j = 10; j ⩾ 1; j--) {
            System.out.println("Backward: " + j);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

public class PriorityThreadDemo {
    public static void main(String[] args) {
        ForwardPrinter t1 = new ForwardPrinter();
        BackwardPrinter t2 = new BackwardPrinter();
        t1.setPriority(Thread.MAX_PRIORITY);
        t2.setPriority(Thread.MIN_PRIORITY);
        t1.start();
        t2.start();
    }
}
```

BCA-502 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2017

Explain Packages and Access Protection in Java.

Answer:

Java packages provide modular namespaces. Access protection controls which classes, methods, and variables can be seen across package boundaries using four access specifiers:

Access Level	Keyword	Accessibility Scope
Public	`public`	Accessible from **any class** in any package throughout the entire application.
Protected	`protected`	Accessible within the **same package**, and by **subclasses in other packages** via inheritance.
Default (Package-Private)	*(No keyword)*	Accessible **only within the same package**. Inaccessible to subclasses outside the package.
Private	`private`	Accessible **strictly within the declaring class**. Conceals internal state (Encapsulation).

BCA-502 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2016

What do you mean by Constructor? Explain Constructor Overloading.

Answer:

A **Constructor** is a special method invoked automatically upon object instantiation using `new` to allocate heap memory and initialize state variables.

> **Characteristics of Constructors:**
>
> - Name must **exactly match** the class name.
> - Must **not have any return type** (not even `void`).
> - If no constructor is defined, JVM inserts a default no-argument constructor.
>
> **Constructor Overloading:** Defining multiple constructors with different argument counts or data types within the same class.

Code Example:

```
class Student {
    int roll;
    String name;
    double marks;

    // Default Constructor
    Student() {
        this.roll = 0;
        this.name = "Unknown";
        this.marks = 0.0;
    }

    // Overloaded Constructor (2 arguments)
    Student(int roll, String name) {
        this.roll = roll;
        this.name = name;
        this.marks = 0.0;
    }

    // Overloaded Constructor (3 arguments)
    Student(int roll, String name, double marks) {
        this.roll = roll;
        this.name = name;
        this.marks = marks;
    }
}
```

SECTION A • SHORT ANSWER QUESTIONS (3 MARKS / HIGH-YIELD DEFINITIONS) (21 UNIQUE QUESTIONS)

BCA-502 | Marks: 3M | 🔥 Highly Repeated (4×) | CCSU Exam Years: 2025, 2019, 2018, 2016

Explain Garbage Collection in Java.

Answer:

Garbage Collection (GC) is Java's automatic memory management process that identifies and deletes unreferenced objects from the Heap memory, preventing memory leaks and dangling pointer bugs.

Key Mechanism:

- **Daemon Thread:** A low-priority background thread run by the JVM continuously tracks heap allocations.
- **Eligibility for GC:** An object becomes eligible for GC when it has zero active live references. This occurs when:
  1. Reference is set to `null` : `emp = null;`
  2. Reference is reassigned: `e1 = e2;`
  3. Object created inside a method goes out of scope.
  4. Island of Isolation (objects reference each other but are unreachable from GC root).
- **Requesting GC:** `System.gc()` or `Runtime.getRuntime().gc()` suggests garbage collection, though execution is subject to JVM discretion.
- **finalize() Method:** Declared in `Object` class, called right before an object is destroyed for final cleanup (deprecated in Java 9+ in favor of Cleaners/PhantomReferences).

BCA-502 | Marks: 3M | ★ Repeated (3×) | CCSU Exam Years: 2025, 2019, 2017

What is Bytecode in Java? Why does it enable platform independence?

Answer:

Bytecode is the intermediate, non-executable binary machine code generated by the Java compiler (`javac`) stored in `.class` files.

Why Bytecode Enables Platform Independence:

- Unlike C/C++ which compiles source directly into CPU-specific native machine code (x86, ARM), Java compiles into highly optimized bytecode instructions designed for an abstract machine: the **JVM**.
- The compiled `.class` file is 100% platform-independent. It can run without recompilation on Windows, Linux, macOS, or Solaris.
- Each platform has its own native JVM implementation that interprets or JIT-compiles this uniform bytecode into local CPU instructions.
- This delivers Java's core slogan: **"Write Once, Run Anywhere" (WORA)**.

BCA-502 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2025, 2018

Explain the 'super' keyword in Java with usages.

Answer:

The `super` keyword in Java is a reference variable used by a subclass to explicitly refer to its immediate parent (superclass) object.

> **Three Distinct Usages of 'super':**
>
> 1. **Access Hidden Superclass Variables:** Resolves name collisions when a child class declares a field with the same name as a parent class field: `super.variableName;`
> 2. **Invoke Overridden Superclass Methods:** Calls the parent method implementation from inside the overriding subclass method: `super.methodName();`
> 3. **Invoke Superclass Constructors:** Explicitly calls the parent class constructor: `super();` or `super(arg1, arg2);` . *Rule: Must be the very first statement inside the subclass constructor.*

Code Example:

```
class Animal {
    String color = "White";
    Animal(String species) { System.out.println("Animal created: " + species); }
    void makeSound() { System.out.println("Generic animal sound"); }
}

class Dog extends Animal {
    String color = "Black";
    Dog() {
        super("Canine"); // Calls Animal constructor (Usage 3)
    }
    void display() {
        System.out.println("Subclass color: " + color);        // Black
        System.out.println("Superclass color: " + super.color); // White (Usage 1)
        super.makeSound(); // Calls Animal's makeSound() (Usage 2)
    }
}
```

BCA-502 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2024, 2020

Discuss Primitive Data Types in Java.

Answer:

Java supports **8 primitive data types** that represent fundamental values directly stored in stack memory, rather than heap objects:

Data Type	Category	Size (Bits / Bytes)	Default Value	Range
`byte`	Integer	8 bits (1 byte)	0	-128 to 127
`short`	Integer	16 bits (2 bytes)	0	-32,768 to 32,767
`int`	Integer	32 bits (4 bytes)	0	-2^{31} to $2^{31}-1$
`long`	Integer	64 bits (8 bytes)	0L	-2^{63} to $2^{63}-1$
`float`	Floating Point	32 bits (4 bytes)	0.0f	~±3.40282347E+38F (6–7 digits)
`double`	Floating Point	64 bits (8 bytes)	0.0d	~±1.79769313486231570E+308 (15 digits)
`char`	Character	16 bits (2 bytes)	'\u0000'	0 to 65,535 (Unicode characters)
`boolean`	Logical	1 bit (JVM dependent)	false	`true` or `false`

BCA-502 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2023, 2017

What is Method Overloading vs Method Overriding?

Answer:

Feature	Method Overloading (Compile-Time)	Method Overriding (Runtime)
Scope	Occurs within the **same class**.	Occurs between **Superclass and Subclass** (Inheritance).
Method Signature	Method name must be same, but **parameters must differ** (number, types, or order).	Method name, return type, and parameters must be **identical**.
Polymorphism Type	Static / Compile-time polymorphism.	Dynamic / Runtime polymorphism.
Return Type	Can have different return types as long as parameters differ.	Must be same or covariant.
Keywords / Annotations	No specific keyword needed.	Annotated with `@Override`.
Private / Static Methods	Can overload private and static methods.	**Cannot** override private or static methods.

BCA-502 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2020, 2017

What is the purpose of the 'static' keyword in Java?

Answer:

The `static` keyword in Java denotes members that belong to the **class itself** rather than to individual object instances. Only a single copy is shared across all instances.

Four Applications of 'static':

1. **Static Variables:** Stored in Method Area / Metaspace once when class is loaded. Used for global counters or shared company constants (e.g., `static int count = 0;`).
2. **Static Methods:** Invoked directly using class name without creating an object (e.g., `Math.sqrt(16)`). *Rule: Cannot access instance (non-static) variables or use `this` / `super`.*
3. **Static Block:** Executes automatically once when class is loaded by JVM, before constructor or `main()`. Used for static resource initialization.
4. **Static Nested Classes:** Inner class that does not require an outer class instance reference.

BCA-502 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2019, 2018

What is a Final Variable? Give an example.

Answer:

In Java, the `final` keyword is a non-access modifier used to apply restrictions:

- **Final Variable:** Its value becomes constant and cannot be reassigned once initialized:

```
final double PI = 3.14159265;
// PI = 3.14; // Compilation error: cannot assign a value to final variable
```

- **Blank Final Variable:** An uninitialized final variable that must be initialized inside the constructor.
- **Final Method:** A method that **cannot be overridden** by any subclass, preventing altered behavior.
- **Final Class:** A class that **cannot be extended (inherited)** (e.g., `java.lang.String`, `java.lang.Math`).

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2024

Explain the Anchor and Table tags in HTML.

Answer:

- **Anchor Tag (`<a>`):** Defines hyperlinks to navigate across web pages, documents, or anchor bookmarks.

```
<a href="https://www.google.com" target="_blank">Visit Google</a>
```

Key Attributes: `href` (target URL), `target="_blank"` (opens in new tab), `download` (downloads file).

- **Table Tag (`<table>`):** Arranges data into a two-dimensional grid of rows and columns.

```
<table border="1" cellpadding="5" cellspacing="0">
  <tr><th>Roll</th><th>Name</th></tr>
  <tr><td>101</td><td>Rahul</td></tr>
</table>
```

Key Tags: `<tr>` (Row), `<th>` (Header Cell, bold/centered), `<td>` (Data Cell), `colspan` / `rowspan` (cell merging).

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2024

Briefly explain Internet Concepts: URL, HTTP, and Web Server.

Answer:

1. **URL (Uniform Resource Locator):** The unique global web address used to locate a specific resource on the Internet.
   Structure: `https://www.example.com:443/bca/notes.html?id=10#sec2`
   (Protocol • Domain • Port • Path • Query String • Fragment).
2. **HTTP (HyperText Transfer Protocol):** The stateless application-layer protocol governing communication between web browsers and servers. Operates over TCP port 80 (HTTP) or port 443 (HTTPS with TLS encryption). Methods include `GET`, `POST`, `PUT`, `DELETE`.
3. **Web Server:** Software/hardware system that stores website files (HTML, CSS, JS, images) and delivers them to client browsers over HTTP (e.g., Apache HTTP Server, Nginx, Microsoft IIS).

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

How is an Object created in Java?

Answer:

An **Object** is a runtime instance of a class occupying heap memory. In Java, objects can be created through 4 distinct mechanisms:

1. **Using the `new` Keyword (Most Common):** Invokes constructor directly: `Student s1 = new Student();`
2. **Using Reflection (`Class.forName().newInstance()`):** Dynamically loads class and instantiates object at runtime.
3. **Using `clone()` Method:** Creates an exact field-by-field copy of an existing object implementing `Cloneable` : `Student s2 = (Student) s1.clone();`
4. **Using Deserialization:** Reconstructs an object from a saved byte stream using `ObjectInputStream.readObject()` .

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

Compare JDK, JRE, and JVM.

Answer:

Feature	JVM (Java Virtual Machine)	JRE (Java Runtime Environment)	JDK (Java Development Kit)
Definition	Abstract engine that executes Java bytecode (.class).	Software bundle required strictly to **run** compiled Java programs.	Complete software kit required to **develop**, compile, and run Java programs.
Includes	Class Loader, Execution Engine (Interpreter + JIT), Garbage Collector.	JVM + Standard Core Libraries (rt.jar, util, lang).	JRE + Development Tools (`javac` , `javadoc` , `jdb` , `jar`).
Target Audience	Internal runtime execution platform.	End users who only run Java applications.	Software developers writing and compiling code.
Formula	Core execution component.	**JRE = JVM + Core Class Libraries**	**JDK = JRE + Development Tools**

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

Difference between JDBC and ODBC.

Answer:

Criteria	JDBC (Java Database Connectivity)	ODBC (Open Database Connectivity)
Developer & Language	Developed by Sun Microsystems; written specifically in **pure Java**.	Developed by Microsoft; written in **C/C++**.
Platform Independence	**Platform independent**; can run on any OS with JVM.	Platform dependent; primarily tied to Windows and C-runtimes.
Type Safety & OOP	Strongly typed, fully Object-Oriented Java API.	Procedural C function calls, pointer-based.
Performance with Java	High efficiency with Type-4 pure Java drivers.	Slower due to JNI / bridging translation layers.

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

Explain why Java is considered the best language for Internet Applications.

Answer:

Java became the dominant language for Internet and enterprise web applications due to:

1. **Platform Independence (WORA):** Bytecode runs identically on any client OS and server architecture.
2. **Security:** Java provides a sandbox execution model, bytecode verifier, and eliminates direct pointer access, preventing memory corruption and unauthorized file access.
3. **Built-in Multithreading & Networking:** Native support for TCP/IP sockets, URLs, concurrent threads, and high-throughput thread pooling.
4. **Robust Server-Side Ecosystem:** Servlets, JSP, Spring Boot, JDBC, and Enterprise JavaBeans (EJB) power global web banking, e-commerce, and cloud platforms.

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

What are Wrapper Classes in Java?

Answer:

Wrapper Classes provide an object representation for Java's 8 primitive data types (e.g., `int` → `Integer`, `char` → `Character`). They reside in `java.lang`.

> **Why Wrapper Classes are Required:**
>
> - Java Collections (e.g., `ArrayList<Integer>`, `HashMap`) can only store objects, not raw primitives.
> - Utility methods like `Integer.parseInt("123")` and `Double.valueOf()` for type conversion.
>
> **Autoboxing & Unboxing:**
>
> - **Autoboxing:** Automatic primitive-to-wrapper conversion: `Integer x = 50;`
> - **Unboxing:** Automatic wrapper-to-primitive conversion: `int y = x;`

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2019

List the four fundamental principles of Object-Oriented Programming.

Answer:

Pillar	Core Concept & Definition	Java Implementation
1. Encapsulation	Binding data variables and methods together into a single unit (class), while hiding internal data using `private` variables with public getters/setters.	`private int id; public int getId() { return id; }`
2. Abstraction	Hiding internal implementation details and displaying only essential functional features to the user.	`abstract class` and `interface`
3. Inheritance	Mechanism where a child class acquires properties and behaviors of a parent class to foster code reuse.	`class Dog extends Animal`
4. Polymorphism	Ability of an entity or method to take many forms. Subdivided into Compile-time (Method Overloading) and Runtime (Method Overriding).	Overloading and `@Override`

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2019

Define Constructor in Java.

Answer:

A **Constructor** in Java is a special member method invoked automatically by the runtime when an object is instantiated using the `new` operator. Its primary role is to initialize instance variables and allocate required memory resources.

Core Characteristics of Constructors:

- **Same Name as Class:** Must bear the exact identifier of the enclosing class.
- **No Return Type:** Does not return any value; must never declare a return type (not even `void`).
- **Automatic Invocation:** Triggered implicitly upon object creation: `Student s = new Student();`
- **Default Constructor:** If no constructor is defined by the developer, the Java compiler automatically synthesizes a default no-argument constructor.
- **Cannot be static, abstract, or final:** Constructors belong to object instantiation and cannot be inherited or made static.

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2018

What is an Abstract Method?

Answer:

An **Abstract Method** is a method declared without any implementation (no body), followed by a semicolon:

```
abstract void draw(); // Abstract method declaration
```

Rules Governing Abstract Methods:

- Must be declared with the `abstract` keyword.
- Can only exist inside an `abstract class` or `interface`.
- Any non-abstract (concrete) subclass extending the abstract class **MUST override and implement** all inherited abstract methods.
- Cannot be declared as `static`, `final`, or `private` because those cannot be overridden.

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2018

What is a Layout Manager in Java AWT?

Answer:

A **Layout Manager** is an interface in `java.awt` responsible for automatically calculating the position, size, and alignment of GUI components within a container window.

Layout Manager	Arrangement Mechanism	Default For
`FlowLayout`	Arranges components sequentially in a row from left to right; wraps to next line when row fills up.	`Panel`, `Applet`
`BorderLayout`	Divides container into 5 distinct regions: `NORTH`, `SOUTH`, `EAST`, `WEST`, `CENTER`.	`Frame`, `Dialog`
`GridLayout`	Arranges components into a strict 2D rectangular grid of equal-sized rows and columns: `new GridLayout(rows, cols)`.	Custom Grids
`GridBagLayout`	Most flexible manager; aligns components dynamically with varying cell spans and constraints.	Complex Forms
`CardLayout`	Stacks components like a deck of cards; only one card visible at a time.	Tabbed Wizards

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2017

How will you create hyperlinks in a web document? Write the code also.

Answer:

Hyperlinks in HTML are created using the **Anchor tag** (`<a>`) with the mandatory `href` attribute:

```
<!-- 1. External Hyperlink to website -->
<a href="https://www.google.com" target="_blank" title="Search Web">Open Google in
New Tab</a>

<!-- 2. Relative Link to Local Page -->
<a href="syllabus.html">View BCA Syllabus</a>

<!-- 3. Email Link -->
<a href="mailto:admissions@ccsu.ac.in">Contact Admissions</a>

<!-- 4. Internal Bookmark / Fragment Link -->
<a href="#section3">Jump to Section 3</a>
<h3 id="section3">Section 3 Content</h3>
```

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2017

What is a Constructor? List its characteristics.

Characteristics: Same name as class, no return type (not even void), automatically called on `new`, can be overloaded.

BCA-502 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2016

What is Byte-code in JAVA? Explain.

Definition: The intermediate machine instruction set generated by `javac`, executed by the JVM to provide platform independence.

SUBJECT: BCA-503 • COMPUTER NETWORKS

Total Deduplicated Unique Questions: 48 • All Sessions Fully Consolidated

SECTION C • LONG ANSWER QUESTIONS (15 MARKS / DETAILED SOLUTIONS) (13 UNIQUE QUESTIONS)

BCA-503 | Marks: 15M / 3M / 7.5M | 🔥 Highly Repeated (9×)

CCSU Exam Years: 2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016

Explain the ISO/OSI Reference Model in detail. Describe the primary functions of each of the seven layers.

Seven Layers of OSI Model:

1. **Physical Layer (Layer 1):** Transmission of raw unstructured bit streams over physical media (voltages, connectors, cables).
2. **Data Link Layer (Layer 2):** Framing, physical addressing (MAC), flow control, error control (CRC), and media access control.
3. **Network Layer (Layer 3):** Logical addressing (IP), routing (Dijkstra, distance vector), packet forwarding, and fragmentation.
4. **Transport Layer (Layer 4):** End-to-end process-to-process delivery (ports), segmentation and reassembly, connection control (TCP handshake), reliable flow control.
5. **Session Layer (Layer 5):** Dialogue control (simplex, half/full duplex), session checkpointing, synchronization tokens.
6. **Presentation Layer (Layer 6):** Data translation, character code conversion (ASCII/EBCDIC), compression, encryption/decryption (SSL/TLS).
7. **Application Layer (Layer 7):** Network virtual terminals, file transfer (FTP), web browsing (HTTP), email services (SMTP).

BCA-503 | Marks: 15M / 7.5M | 🔥 Highly Repeated (8×)

CCSU Exam Years: 2025, 2024, 2023, 2021, 2020, 2019, 2017, 2016

Explain Congestion Control Mechanisms in Computer Networks: Leaky Bucket and Token Bucket Algorithms with comparative analysis.

Answer:

Network Congestion occurs when the load on network nodes (routers/switches) exceeds their buffer capacity or link bandwidth, resulting in queue buildup, packet loss, and throughput collapse.

> **Two Fundamental Traffic Shaping Algorithms:**
>
> 1. **Leaky Bucket Algorithm:** Shapes bursty traffic into a strictly constant, uniform output rate regardless of packet burst size. Packets enter a bucket of capacity *C* and leak out at a constant rate *r*. If bucket overflows, subsequent incoming packets are discarded.
> 2. **Token Bucket Algorithm:** Allows bursty traffic up to a configurable burst size. Tokens are generated and accumulated in the bucket at a constant rate *r* up to capacity *C*. A packet can transmit only if enough tokens are available to cover its byte size; transmitting consumes tokens.

Comparative Analysis Table:

Parameter	Leaky Bucket Algorithm	Token Bucket Algorithm
Output Rate	Strictly constant and uniform. Zero bursts permitted.	Allows controlled bursty transmission up to token capacity.
Token Mechanism	No tokens; drops excess packets when bucket is full.	Accumulates tokens; drops excess tokens if bucket is full.
Traffic Suitability	Ideal for constant bit rate (CBR) audio/video traffic.	Ideal for bursty data traffic (web browsing, database queries).
Packet Loss	Drops packets immediately when buffer fills up.	Buffers packets until enough tokens arrive.

BCA-503 | Marks: 15M / 22.5M / 7.5M | Highly Repeated (7×)

CCSU Exam Years: 2025, 2021, 2020, 2019, 2018, 2017, 2016

Explain Sliding Window Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ with efficiency analysis.

Answer:

Stop-and-Wait ARQ (Automatic Repeat reQuest) is a Data Link Layer flow and error control protocol designed for noisy channels.

Working Mechanism:

1. Sender transmits a single frame with a 1-bit sequence number (Frame 0 or Frame 1) and starts a retransmission timer.
2. Sender stops and waits; no new frame is sent until an ACK arrives.
3. **Normal Case:** Receiver verifies CRC. If correct, sends ACK with sequence number of next expected frame (e.g., ACK 1). Sender stops timer and transmits Frame 1.
4. **Lost Frame / Damaged Frame:** Receiver discards damaged frame silently. Sender timer expires (timeout), triggering automatic retransmission of the unacknowledged frame.
5. **Lost ACK:** Sender retransmits frame after timeout. Receiver detects duplicate sequence number, discards duplicate frame, and resends ACK.

BCA-503 | Marks: 15M / 3M / 7.5M | Highly Repeated (7×)

CCSU Exam Years: 2025, 2024, 2020, 2019, 2018, 2017, 2016

Compare Circuit Switching, Packet Switching, and Message Switching.

Comparison:

Feature	Circuit Switching	Packet Switching	Message Switching
Dedicated Path	Yes (setup before transfer).	No (packets routed dynamically).	No (store-and-forward per hop).
Bandwidth	Reserved/wasted if idle.	Dynamically shared on-demand.	Dynamically shared.
Delay	Setup delay high; transmission delay low.	Jitter/variable queuing delay.	High storage delay at each switch.
Modern Use	PSTN landline telephone.	Internet (IP networks).	Early telegraph, postal email.

BCA-503 | Marks: 15M / 22.5M / 7.5M | Highly Repeated (6×)

CCSU Exam Years: 2025, 2021, 2020, 2019, 2018, 2016

Explain Routing Algorithms: (a) Distance Vector Routing (Bellman-Ford) and (b) Link State Routing (Dijkstra). Discuss Count-to-Infinity problem.

Routing Algorithms:

1. **Distance Vector Routing:** Each router maintains a routing table containing distance to each destination and next hop. Routers periodically share complete tables with immediate neighbors. Uses Bellman-Ford algorithm. Vulnerable to *Count-to-Infinity problem* when links fail (mitigated by Split Horizon and Poison Reverse).
2. **Link State Routing:** Each router floods Link State Packets (LSP) containing status of its local links to all routers in the entire autonomous system. Every router constructs a global topology map and computes shortest paths using Dijkstra's algorithm (used in OSPF). Fast convergence and no routing loops.

BCA-503 | Marks: 15M / 7.5M | Highly Repeated (6×)

CCSU Exam Years: 2025, 2020, 2019, 2018, 2017, 2016

Explain Error Detection Techniques: Parity Check, Checksum, and CRC (Cyclic Redundancy Check).

Error Detection:

- **Parity Check:** Appends single parity bit (even or odd parity). Detects only odd numbers of bit errors.
- **Checksum:** Divides data into 16-bit words, adds them using 1's complement arithmetic, and appends inverted sum (used in TCP/IP).
- **Cyclic Redundancy Check (CRC):** Treats bit stream as polynomial coefficients and performs modulo-2 binary division by generator polynomial `G(x)`. Appends remainder (FCS). Detects all single, double, odd, and burst errors up to generator length!

BCA-503 | Marks: 15M | 🔥 Highly Repeated (6×)

CCSU Exam Years: 2023, 2020, 2019, 2018, 2017, 2016

Explain Guided Media: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.

Answer:

Guided (Bounded) Media refers to physical, wired transmission media that direct data signals along a solid conductor or optical waveguide.

Parameter	Twisted Pair Cable	Coaxial Cable	Optical Fiber Cable
Signal Type	Electrical electrical signals (copper wires).	Electrical signals (copper core).	Light pulses (photons) through silica glass core.
Subtypes	UTP (Unshielded) & STP (Shielded, Cat5e/6/7).	Thicknet (10Base5) & Thinnet (10Base2, 50/75 Ω).	Single-Mode (long-haul) & Multi-Mode (LAN/campus).
Bandwidth & Speed	Moderate (10 Mbps to 10 Gbps).	Moderate-High (10 Mbps to 100 Mbps).	**Extremely high (10 Gbps to 100+ Gbps)**.
Attenuation & Repeater Distance	High attenuation; repeater every 100 meters.	Moderate attenuation; repeater every 500 meters.	**Lowest attenuation; repeater every 50+ km**.
EMI Immunity	Prone to electromagnetic interference (crosstalk).	Good immunity (outer braided copper shield).	**100% immune to EMI, lightning, and radio noise**.
Cost & Installation	Cheapest, flexible, very easy to terminate.	Moderate cost, bulky, medium difficulty.	High cost, fragile, requires skilled fusion splicing.

BCA-503 | Marks: 15M / 3M / 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2025, 2023, 2016

What is Subnetting? Give an example.

Definition: Subnetting is the process of partitioning a single physical network into two or more smaller logical subnets by borrowing host bits to create subnet bits, reducing broadcast traffic and optimizing address allocation.
Example: Given Class C network `192.168.1.0/24` with default mask `255.255.255.0`. Borrowing 2 host bits gives subnet mask `255.255.255.192` (/26), creating $2^2 = 4$ subnets with 62 usable hosts each:

- Subnet 1: `192.168.1.0` to `192.168.1.63`
- Subnet 2: `192.168.1.64` to `192.168.1.127`
- Subnet 3: `192.168.1.128` to `192.168.1.191`
- Subnet 4: `192.168.1.192` to `192.168.1.255`

BCA-503 | Marks: 15M | ★ Repeated (2×) | CCSU Exam Years: 2025, 2023

Write detailed notes on: (a) DNS (Domain Name System), (b) Firewalls and Network Security, (c) SMTP and Email Architecture.

Detailed Notes:

- **DNS:** Hierarchical distributed database mapping hostnames (`www.google.com`) to IP addresses (`142.250.190.46`) over UDP/TCP port 53.
- **Firewalls:** Security barrier controlling inbound/outbound network traffic based on rules. Types: Packet Filtering, Stateful Inspection, Proxy/Application-level gateways.
- **Email Architecture:** User Agent (UA) composes email, Mail Transfer Agent (MTA) transmits using SMTP (port 25/587), and Mail Access Agent (MAA) retrieves email to client via POP3 (port 110) or IMAP (port 143).

BCA-503 | Marks: 15M | ★ Repeated (2×) | CCSU Exam Years: 2024, 2023

Explain Error Control in Data Link Layer using CRC with a complete numerical example.

Answer:

Cyclic Redundancy Check (CRC) is an error-detecting polynomial code based on binary modulo-2 arithmetic (XOR operations without carry or borrow).

Step-by-Step CRC Algorithm:

1. Let Dataword be *D* of length *k*, and Generator Divisor be *G* of length *(n + 1)* bits.
2. Append *n* zeros to the right of Dataword: `D' = D • 0^n`.
3. Divide *D'* by *G* using **Modulo-2 Division (XOR)**.
4. The resulting *n*-bit remainder is the **CRC Checksum** (FCS).
5. Codeword transmitted: `T = D • CRC`.
6. At Receiver: Divide received Codeword *T* by *G*. If remainder == 0, accept frame; otherwise reject (error detected).

Complete Numerical Example:

Given Dataword = `100100` (6 bits), Divisor = `1101` (4 bits, degree n = 3).

Append 3 zeros: Dividend = `100100000`.

```
          111101 (Quotient)
     ------------------
1101 | 100100000
       1101
       -----
       01000
        1101
       -----
        01010
         1101
        -----
         01110
          1101
         -----
          00110
           0000
          -----
           01100
            1101
           -----
            0001 (Remainder / CRC = 001)
```

Transmitted Codeword = Dataword + Remainder = **100100001** ✓

BCA-503 | Marks: 15M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2023, 2017

Explain Dijkstra's Shortest Path Routing Algorithm with a network graph example.

Answer:

Dijkstra's Algorithm is a Link-State routing algorithm (used in OSPF) that computes the shortest path tree from a single source node to all destination nodes in a weighted network graph.

Algorithm Steps:

1. Initialize distance to source node as 0: `Dist[Source] = 0`. Set distance to all other nodes as ∞.
2. Mark all nodes as unvisited. Create visited set `S = {}`.
3. For currently unvisited node `u` with minimum distance:
   - Add `u` to visited set `S`.
   - For each unvisited neighbor `v` of `u`:
     `If Dist[u] + Cost(u, v) < Dist[v]: Dist[v] = Dist[u] + Cost(u, v)`
     (Relaxation step).
4. Repeat until all nodes are included in set `S`. Time complexity: $O(|V|^2)$ or $O(|E| \log |V|)$ using min-heap.

Numerical Graph Example:

Graph with nodes A, B, C, D, E. Source: Node A.

Edges: (A-B)=4, (A-C)=2, (B-C)=1, (B-D)=5, (C-D)=8, (C-E)=10, (D-E)=2.

- Start at A: Dist = {A:0, B:4, C:2, D:∞, E:∞}. Pick C (min dist = 2).
- From C: Update B: min(4, 2+1) = 3; Update D: min(∞, 2+8) = 10; Update E: min(∞, 2+10) = 12. Dist = {A:0, B:3, C:2, D:10, E:12}. Pick B (dist = 3).
- From B: Update D: min(10, 3+5) = 8. Dist = {A:0, B:3, C:2, D:8, E:12}. Pick D (dist = 8).
- From D: Update E: min(12, 8+2) = 10. Pick E (dist = 10).
- **Final Shortest Distances from A:** A=0, B=3 (A→C→B), C=2 (A→C), D=8 (A→C→B→D), E=10 (A→C→B→D→E).

BCA-503 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2024

(a) Explain Shannon Capacity with examples. (b) What is CSMA? List protocols used with CSMA.

(a) Shannon Capacity Theorem: Gives theoretical maximum data rate `C` of a channel with bandwidth `B` and Signal-to-Noise Ratio `SNR`:
C = B \log_2(1 + SNR) bps
Example: For telephone line with `B = 3000` Hz and `SNR = 3162` (35 dB): `C = 3000 \log_2(1 + 3162) ≈ 3000 × 11.62 = 34,860 bps ≈ 34.8 kbps`.

(b) CSMA Protocols: 1-Persistent CSMA, Non-Persistent CSMA, p-Persistent CSMA, and CSMA/CD / CSMA/CA.

BCA-503 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2024

Describe any three: (a) SSL, (b) Modems, (c) Line Configuration, (d) Distortion vs Noise.

Short Notes:

- **SSL (Secure Sockets Layer):** Cryptographic protocol providing end-to-end encryption, authentication, and integrity over TCP.
- **Modems (Modulator-Demodulator):** Converts digital computer signals to analog audio frequencies for transmission over analog lines and vice versa.
- **Line Configuration:** Point-to-Point (dedicated link between two devices) vs Multipoint (multidrop link where multiple devices share a single link).
- **Distortion vs Noise:** Distortion is alteration of signal shape caused by different frequency phase delays in transmission media. Noise is unwanted external thermal/electromagnetic energy added into the signal.

SECTION B • MEDIUM ANSWER QUESTIONS (7.5 MARKS / CORE CONCEPTS) (10 UNIQUE QUESTIONS)

BCA-503 | Marks: 3M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2025, 2021

Write short notes on: (a) Throughput & Wavelength, (b) Synchronous Protocols, (c) Routers.

Notes:

- **(a) Throughput & Wavelength:** Throughput is the actual measurement of data transferred across a network per second. Wavelength (`\lambda = v / f`) is the physical distance between consecutive signal wave peaks in optical communication.
- **(b) Synchronous Protocols:** Continuous transmission of data blocks synchronized to a shared clock, using sync bytes/flags (e.g. HDLC).
- **(c) Routers:** Network Layer (Layer 3) internetworking devices that inspect destination IP addresses and make packet forwarding decisions based on routing tables.

BCA-503 | Marks: 3M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2025, 2017

Describe Multiplexing. Explain FDM, TDM, and WDM with diagrams.

Definition: Combining multiple separate data streams into a single composite signal transmitted over a shared transmission medium.

Techniques:

- **FDM (Frequency Division Multiplexing):** Divides channel bandwidth into separate non-overlapping frequency bands with guard bands (Analog signals, radio, cable TV).
- **TDM (Time Division Multiplexing):** Interleaves time slots from multiple sources onto a high-speed digital carrier (Synchronous vs Statistical TDM).
- **WDM (Wavelength Division Multiplexing):** Multiplexes optical signals of differing wavelengths (colors) along a single optical fiber.

BCA-503 | Marks: 3M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2024, 2016

Explain the switching methods in Computer Networks.

Answer:

Switching is the technique by which nodes route and forward data across interconnected communication paths between sender and receiver.

Feature	Circuit Switching	Packet Switching (Datagram / Virtual)	Message Switching
Connection Setup	Mandatory dedicated physical circuit setup phase before data transfer.	No setup in Datagram; short virtual circuit ID setup in Virtual Circuit.	No setup needed; store-and-forward at each intermediate node.
Dedicated Bandwidth	Yes, entire bandwidth is reserved for the session duration.	No, dynamic statistical multiplexing shares channel bandwidth.	No dedicated path; entire message stored and queued at each router.
Call Termination	Explicit teardown phase required.	Automatic when last packet arrives.	Automatic.
Delay	Initial connection delay, but zero transmission delay once established.	Packet queuing and processing delay at every router hop.	High queuing delay due to large file buffering on intermediate disks.
Example Systems	Traditional Public Switched Telephone Network (PSTN), ISDN.	**Internet (IP)**, ATM, Frame Relay, modern 4G/5G networks.	Telegraph systems, legacy postal mail routing.

BCA-503 | Marks: 3M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2024, 2023

Explain in brief: (a) Guided Media, (b) Transmission Modes.

(a) Guided Media: Physical physical conductors that guide signals along a specific path: Twisted Pair (UTP/STP), Coaxial Cable, and Optical Fiber.

(b) Transmission Modes:

- *Simplex:* Unidirectional transmission (e.g., keyboard to CPU, TV broadcast).
- *Half-Duplex:* Bidirectional, but only one direction at a time (e.g., Walkie-Talkie).
- *Full-Duplex:* Simultaneous bidirectional transmission (e.g., telephone call).

BCA-503 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2023

Describe the functions & responsibilities of the Session Layer.

Answer:

The **Session Layer (Layer 5)** of the OSI Model is responsible for establishing, managing, synchronizing, and terminating communication dialogs between presentation entities.

Core Responsibilities:

1. **Dialog Control:** Manages dialog discipline: determines whether the session operates in **Half-Duplex** (one side talks at a time using data tokens) or **Full-Duplex** (both sides talk simultaneously).
2. **Synchronization & Checkpointing:** Inserts checkpoints into long data streams. If a connection failure occurs during a 100-page document transfer at page 65, only pages 66-100 are retransmitted instead of restarting from page 1.
3. **Session Disconnection (Graceful Close):** Ensures all pending data has been transmitted and acknowledged before tearing down the session.

BCA-503 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2021

Discuss ISDN, its services, and layer structure.

Answer:

ISDN (Integrated Services Digital Network) is a set of international CCITT/ITU-T standards for end-to-end digital transmission of voice, data, video, and fax over standard copper telephone lines.

ISDN Channels:

- **B-Channel (Bearer):** 64 kbps digital channel carrying user voice or payload data.
- **D-Channel (Delta):** 16 or 64 kbps channel for signaling, control, call setup, and packet telemetry.

Standard ISDN Service Interfaces:

1. **BRI (Basic Rate Interface):** Designed for residential homes and small offices.
   Structure: **2B + D** = 2 × 64 kbps + 16 kbps = **144 kbps** (total 192 kbps with framing).
2. **PRI (Primary Rate Interface):** Designed for corporate PBXs and enterprises.
   Structure (North America / Japan): **23B + D** = 1.544 Mbps (T1 line).
   Structure (Europe / International): **30B + D** = 2.048 Mbps (E1 line).

BCA-503 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2021

Explain TCP Packet Format (Header Structure) in detail.

Answer:

The **TCP Header** is a 20–to-60 byte segment header that encapsulates data to provide reliable, connection-oriented, stream transmission:

Header Field	Bit Length	Function and Description
Source Port	16 bits	Port number of the sending application (e.g., dynamic port 50231).
Destination Port	16 bits	Port number of receiving application service (e.g., port 80 for HTTP, 443 for HTTPS).
Sequence Number	32 bits	Tracks byte-stream sequence position of the first data byte in this segment.
Acknowledgment Number	32 bits	Next byte number expected by receiver when ACK flag is set (cumulative ACK).
Data Offset (HLEN)	4 bits	Specifies TCP header length in 32–bit (4–byte) words (5 words = 20 bytes min).
Reserved	6 bits	Reserved for future expansion (must be 0).
Control Flags	6 bits	• `URG` : Urgent pointer valid. • `ACK` : Acknowledgment field valid. • `PSH` : Push data directly to application. • `RST` : Reset connection abruptly. • `SYN` : Synchronize sequence numbers (Connection setup). • `FIN` : Terminate sender connection.
Window Size	16 bits	Flow control buffer size advertised by receiver (Sliding Window size in bytes).
Checksum	16 bits	Mandatory error detection covering header, payload, and IP pseudo-header.
Urgent Pointer	16 bits	Points to the last urgent data byte when URG flag is set.
Options & Padding	0–40 bytes	Optional fields like MSS (Maximum Segment Size), Timestamps, Window Scaling.

BCA-503 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2020

How does DTE-DCE interface work? Discuss the essence of Modems.

Answer:

DTE vs DCE Architecture:

- **DTE (Data Terminal Equipment):** The endpoint device that creates or consumes data (e.g., Computer, Server, Terminal).
- **DCE (Data Circuit-terminating Equipment):** The communication device that interfaces the DTE to the physical transmission medium (e.g., Modem, CSU/DSU, Network Interface Card).

Working Principle & Standards: Connected via physical interface standards such as **RS-232 / EIA-232**. DTE sends digital serial data (TxD, pin 2) to DCE, which converts it (modulation) for the analog or digital line. Control signals like RTS (Request to Send) and CTS (Clear to Send) synchronize transmission.

BCA-503 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2018

Explain any one protocol used for flow control in noisy channel: Stop-and-Wait ARQ.

Answer:

Stop-and-Wait ARQ (Automatic Repeat reQuest) is a Data Link Layer flow and error control protocol designed for noisy channels.

Working Mechanism:

1. Sender transmits a single frame with a 1-bit sequence number (Frame 0 or Frame 1) and starts a retransmission timer.
2. Sender stops and waits; no new frame is sent until an ACK arrives.
3. **Normal Case:** Receiver verifies CRC. If correct, sends ACK with sequence number of next expected frame (e.g., ACK 1). Sender stops timer and transmits Frame 1.
4. **Lost Frame / Damaged Frame:** Receiver discards damaged frame silently. Sender timer expires (timeout), triggering automatic retransmission of the unacknowledged frame.
5. **Lost ACK:** Sender retransmits frame after timeout. Receiver detects duplicate sequence number, discards duplicate frame, and resends ACK.

BCA-503 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2018

Explain Fiber Distributed Data Interface (FDDI) operation.

Answer:

FDDI (Fiber Distributed Data Interface) is an ANSI and ISO standard for a 100 Mbps token-passing local/metropolitan area network (LAN/MAN) utilizing optical fiber cables across distances up to 100 km.

> **Dual Counter-Rotating Ring Architecture:**
>
> - **Primary Ring:** Transmits token and data packets in one direction at 100 Mbps.
> - **Secondary Ring:** Standby ring rotating in the opposite direction.
> - **Self-Healing Ring Wrap:** If a cable break or node failure occurs, the stations adjacent to the fault automatically wrap the primary ring onto the secondary ring, forming a single closed continuous ring. This provides extreme fault tolerance.

SECTION A • SHORT ANSWER QUESTIONS (3 MARKS / HIGH-YIELD DEFINITIONS) (25 UNIQUE QUESTIONS)

BCA-503 | Marks: 3M | 🔥 Highly Repeated (4×) | CCSU Exam Years: 2023, 2019, 2018, 2016

What is Network Topology? List advantages and disadvantages of Mesh Topology.

Answer:

Network Topology describes the geometric and logical arrangement of nodes (computers, routers) and connecting communication links in a computer network.

> **Mesh Topology:** Every device is connected to every other device via dedicated point-to-point links. For n nodes, total physical links required: `N = n(n - 1) / 2`.
>
> - **Advantages of Mesh Topology:**
> - *Extreme Reliability & Fault Tolerance:* If one link breaks, traffic instantly reroutes through alternative paths.
> - *Zero Traffic Congestion:* Dedicated links carry only traffic between their two endpoints.
> - *High Privacy & Security:* Only sender and recipient receive data.
> - **Disadvantages of Mesh Topology:**
> - *Prohibitive Cost:* Huge cabling expense ($O(n^2)$ cables).
> - *Hardware Complexity:* Every computer requires `(n - 1)` I/O network ports.
> - *Difficult Installation & Maintenance:* Dense cabling makes expansion challenging.

BCA-503 | Marks: 3M | ★ Repeated (3×) | CCSU Exam Years: 2019, 2018, 2016

Which device is used for Modulation and Demodulation? Explain its function.

Answer:

A **MODEM (Modulator-Demodulator)** is the hardware device used to convert between digital signals produced by computers and analog signals transmitted over physical communication channels (like telephone lines or coaxial cables).

1. **Modulation (Transmitter Side):** Converts computer digital binary square waves (0s and 1s) into continuous analog sine waves by varying amplitude (ASK), frequency (FSK), or phase (PSK) to traverse long-distance analog media.
2. **Demodulation (Receiver Side):** Reverses the process, extracting the original discrete binary bits from incoming analog waveforms to feed the computer's CPU.

BCA-503 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2018, 2016

Explain Transmission Modes: Simplex, Half-Duplex, and Full-Duplex.

Answer:

Transmission Mode	Directionality	Channel Utilization	Real-World Examples
Simplex	**Strictly Unidirectional**. One station only sends, other only receives.	Entire channel bandwidth reserved for one direction.	Keyboard to CPU, Monitor, Traditional Radio/TV broadcast.
Half-Duplex	**Bidirectional, but one direction at a time**. Cannot transmit and receive simultaneously.	Shared directionality; stations take turns.	Walkie-Talkie systems (saying "Over"), CSMA/CD Ethernet.
Full-Duplex	**Simultaneous Bidirectional**. Both stations can transmit and receive at the same time.	Channel capacity divided between two directions.	Modern Telephone conversation, Full-Duplex Switched Ethernet.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2025

What is a MAC address? How is it different from an IP Address?

Comparison:

Feature	MAC Address (Physical Address)	IP Address (Logical Address)
Layer	Data Link Layer (Layer 2).	Network Layer (Layer 3).
Size & Format	48 bits (6 bytes), hex (e.g., `00:1A:2B:3C:4D:5E`).	32 bits (IPv4, e.g., `192.168.1.1`) or 128 bits (IPv6).
Nature	Hardcoded by manufacturer into NIC; globally fixed.	Dynamically/statically assigned based on network location.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2025

Differentiate between a Hub and a Switch.

Answer:

Feature	Hub (Multi-port Repeater)	Switch (Intelligent Bridge)
OSI Layer	Physical Layer (Layer 1).	Data Link Layer (Layer 2) / Layer 3 (Multilayer).
Data Transmission	**Broadcast:** Repeats every incoming electrical signal to all connected ports.	**Unicast:** Looks up destination MAC address in forwarding table to send frame directly to designated port.
Collision Domain	All ports share a **single collision domain**. Collisions frequent.	Each individual port has its own **dedicated collision domain**. Zero collisions.
Duplex Mode	Half-duplex only (cannot send and receive simultaneously).	Full-duplex (simultaneous send/receive at wire speed).
Intelligence & Memory	Zero memory; purely passive electrical signal repeater.	Maintains dynamic MAC Address Table (CAM table) in RAM.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2025

What is CSMA/CD? How does it operate in Ethernet?

Carrier Sense Multiple Access with Collision Detection: A media access protocol used in shared half-duplex Ethernet:

1. **Carrier Sense:** Station listens before transmitting ("listen before talk"). If channel is idle, it begins transmitting.
2. **Collision Detection:** While transmitting, station monitors signal level. If collision occurs, it transmits a *jam signal* to notify all stations.
3. **Backoff:** Stations wait a random backoff period using the Binary Exponential Backoff algorithm before retransmitting.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2024

What is the need for a Computer Network?

Answer:

A **Computer Network** is an interconnected collection of autonomous computers and peripherals capable of exchanging information and sharing resources.

Five Fundamental Needs & Advantages:

1. **Hardware Resource Sharing:** High-cost hardware peripherals such as network printers, scanners, and multi-terabyte NAS storage servers are shared across hundreds of workstations.
2. **Data & Software Centralization:** Centralized database servers (e.g., banking cores) ensure data consistency, centralized backups, and eliminate data silos.
3. **Fast & Global Communication:** Facilitates instant real-time collaboration through E-Mail, VoIP, video conferencing, and messaging protocols.
4. **High Reliability & Redundancy:** Alternative routing paths ensure that if one server fails, backup servers maintain business continuity.
5. **Cost Reduction:** Eliminates redundant hardware purchases and software licenses through multi-user enterprise servers.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2024

Define HDLC (High-Level Data Link Control).

Answer:

HDLC (High-Level Data Link Control) is a bit-oriented synchronous Data Link Layer protocol developed by ISO (ISO 3309) providing reliable point-to-point and multipoint communication.

Standard HDLC Frame Structure:

Flag (8 bits)	Address (8 bits)	Control (8/16 bits)	Information (Variable)	FCS / CRC (16/32 bits)	Flag (8 bits)
`01111110`	Station Address	Frame type & Sequence	Network Layer Payload	CRC Error Checksum	`01111110`

Three Frame Types:

- **I-Frame (Information):** Carries user payload data and piggybacked ACKs.
- **S-Frame (Supervisory):** Carries flow/error control signals (RR, RNR, REJ).
- **U-Frame (Unnumbered):** Carries link setup, disconnection, and mode setting commands (SABM, DISC).

Bit Stuffing: To prevent premature flag detection within data, HDLC inserts a '0' bit whenever five consecutive '1' bits appear in the payload.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2024

Explain the concept of Piggybacking.

Answer:

Piggybacking is an optimization technique used in bidirectional sliding window protocols where positive acknowledgment (ACK) bits for received frames are attached onto an outgoing outgoing data frame rather than transmitting a dedicated ACK frame.

Operational Mechanism & Benefits:

- When Station A receives a data frame from Station B, instead of immediately sending a separate small ACK packet (which incurs full 20-byte IP header + 14-byte Ethernet header overhead), it waits briefly until its own application layer generates an outgoing data packet for Station B.
- The ACK sequence number is then placed into the header of that outgoing data frame.
- **Significantly Reduces Bandwidth Consumption:** Saves link channel capacity by cutting packet framing overhead in half.
- *Timer Precaution:* If no outgoing data frame is generated within a predetermined timeout period (ACK delay timer), a standalone ACK frame is dispatched to prevent retransmission timeouts at the sender.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2024

Describe how SMTP protocol is used in E-Mail applications.

Answer:

Simple Mail Transfer Protocol (SMTP) is an application-layer, push-based protocol operating over TCP port 25 (or port 587 with TLS) used to deliver email messages across the Internet.

SMTP Roles & Architecture:

- **Push Protocol:** SMTP is strictly used by the sending Mail User Agent (MUA) to push email to the sender's Mail Transfer Agent (MTA) server, and from the sender MTA to the receiver's MTA server.
- *Pull Limitation:* SMTP **cannot pull** email messages from a mailbox down to a client device; retrieval requires pull protocols such as **POP3 (Port 110)** or **IMAP (Port 143)**.

Key SMTP Commands: `HELO / EHLO` (handshake), `MAIL FROM:` (sender identity), `RCPT TO:` (recipient address), `DATA` (message body and headers), `QUIT` (session termination).

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

Explain the use of Bridges in Computer Networks.

Answer:

A **Bridge** is an internetworking device operating at the **Data Link Layer (Layer 2)** of the OSI model that connects and filters traffic between two separate LAN segments sharing identical protocols.

Core Functions & Mechanism:

1. **Collision Domain Segmentation:** Splits a large collision domain into two smaller, isolated collision domains, dramatically reducing packet collisions.
2. **MAC Address Filtering:** Inspects the destination MAC address of incoming frames using an internal forwarding table. If sender and receiver reside on the same segment, the frame is filtered (blocked) from crossing over.
3. **Transparent Learning:** Dynamically inspects source MAC addresses of passing frames to build and update its bridge forwarding table.
4. **Spanning Tree Protocol (STP):** Eliminates broadcast storms and packet loops in redundant multi-bridge topologies.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

Write a short note on E-Mail and its core protocols.

Answer:

Electronic Mail (E-Mail) is a store-and-forward transmission system allowing digital messages and attachments to be exchanged globally between networked hosts.

Protocol	Port	Role / Nature	Key Distinction
SMTP (Simple Mail Transfer Protocol)	25 / 587	**Push Protocol:** Sends email from sender client to sender server, and relays between intermediate mail servers.	Outbound delivery only; cannot retrieve from server mailbox.
POP3 (Post Office Protocol v3)	110 / 995	**Pull Protocol:** Downloads email messages from server to local client disk and deletes from server by default.	Offline access; changes on local device not synchronized to server.
IMAP (Internet Message Access Protocol)	143 / 993	**Pull & Sync Protocol:** Manages messages directly on the central mail server. Folders, read status, and drafts sync across all devices.	Real-time two-way synchronization for multiple client devices (phones, laptops).

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

Define Bandwidth and Throughput.

Answer:

Metric	Bandwidth	Throughput
Definition	The theoretical maximum theoretical data carrying capacity of a transmission medium.	The actual rate of successful data delivery achieved across the link under real network conditions.
Nature	Theoretical peak ceiling (potential).	Practical measured performance (reality).
Unit	Bits per second (bps, Mbps, Gbps) or Hertz (Hz).	Bits per second (bps, Mbps) or Packets per sec.
Factors Influencing	Physical cable characteristics, channel frequency spectrum.	Network congestion, packet collisions, processing overhead, protocol headers, retransmissions.
Analogy	The width of a highway (e.g., 6 lanes).	The actual number of cars passing per minute during rush hour.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

Four channels, each with 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between channels?

Answer:

Given Problem: Four channels, each with 100-kHz bandwidth, are multiplexed together using Frequency Division Multiplexing (FDM). Guard bands of 10 kHz are required between adjacent channels. Find the minimum link bandwidth.

> **Step-by-Step Calculation:**
>
> - Number of channels: $N = 4$
> - Bandwidth per channel: $B = 100\ kHz$
> - Total bandwidth for 4 channels: `4 × 100 kHz = 400 kHz`
> - Number of guard bands between N channels: `N - 1 = 4 - 1 = 3 guard bands`
> - Bandwidth per guard band: $10\ kHz$
> - Total guard band bandwidth: `3 × 10 kHz = 30 kHz`
>
> **Minimum Link Bandwidth Required:**
>
> `Total Bandwidth = (N × B) + [(N - 1) × Guard Band] = 400 kHz + 30 kHz = 430 kHz` ✓

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

Discuss Point-to-Point Protocol (PPP).

Answer:

PPP (Point-to-Point Protocol) is a standard Data Link Layer protocol (RFC 1661) used to establish a direct connection between two networking nodes over dial-up, serial, or broadband links.

Three Primary Components of PPP:

1. **Framing Method:** HDLC-like framing mechanism providing unambiguous frame delimitation and 16/32-bit CRC error detection.
2. **LCP (Link Control Protocol):** Establishes, configures, tests data link quality, negotiates options (packet sizes), and gracefully terminates the point-to-point connection.
3. **NCP (Network Control Protocols):** A family of protocols that negotiates network layer protocols (e.g., IPCP negotiates dynamic IP addresses for IPv4).

Authentication Support: PPP supports robust authentication via **PAP (Password Authentication Protocol)** and **CHAP (Challenge Handshake Authentication Protocol)**.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

Differentiate between LAN, MAN, and WAN.

Answer:

Parameter	LAN (Local Area Network)	MAN (Metropolitan Area Network)	WAN (Wide Area Network)
Geographical Span	Confined to small area (single room, office, building, campus: ≤ 1 km).	Spans an entire city or metropolitan town (≤ 50 km).	Spans entire countries, continents, or the globe (> 100 km).
Ownership	Privately owned by single organization/individual.	Public or private enterprise (Cable TV provider, Telecom).	Distributed, collective ownership (Telecom consortiums, ISPs).
Data Transfer Rate	**Highest:** 1 Gbps to 10 Gbps.	Moderate: 100 Mbps to 1 Gbps.	Lowest: 10 Mbps to 1 Gbps across long-haul undersea fibers.
Error Rate & Delay	Lowest error rates; sub-millisecond delay.	Moderate error rates and propagation delay.	Highest error rates; significant propagation delay (50–200 ms).
Technology Example	Ethernet (IEEE 802.3), Wi-Fi (802.11).	FDDI, Cable TV Networks, Metro Ethernet.	**The Global Internet**, ATM, Satellite Links.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

What is a Collision Domain vs Broadcast Domain?

Answer:

Feature	Collision Domain	Broadcast Domain
Definition	A network segment where simultaneous transmissions from two stations result in an electrical signal packet collision.	A network segment where a broadcast packet sent by one station is received by all other stations in that segment.
Delimited By	Separated and broken up by **Switches, Bridges, and Routers** (Hubs do not break collision domains).	Separated and broken up **ONLY by Routers and VLANs** (Hubs and Switches forward broadcasts).
OSI Layer	Data Link Layer (Layer 2) / Physical Layer (Layer 1).	Network Layer (Layer 3).
Switch Behavior	Each individual switch port forms an isolated collision domain.	All ports on a switch belong to a single default broadcast domain.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2019

Differentiate between Bridges and Repeaters.

Answer:

Feature	Repeater	Bridge
OSI Layer	Operates strictly at the **Physical Layer (Layer 1)**.	Operates at the **Data Link Layer (Layer 2)**.
Functionality	Amplifies, regenerates, and retimes electrical/optical signals to overcome cable attenuation.	Inspects MAC addresses, filters frames, and forwards traffic across network segments.
Intelligence	Zero intelligence; cannot inspect packet headers or MAC addresses.	Intelligent; maintains dynamic MAC Address Table (CAM) in memory.
Collision Domain	Extends the single collision domain (does not break it).	**Divides collision domains** into separate segments.
Data Unit Handled	Individual electrical bits (0s and 1s).	Frames (packets with MAC addresses and CRC).

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2019

What are the services provided by the Transport Layer?

Answer:

The **Transport Layer (Layer 4)** is the core of the OSI architecture providing transparent, logical, end-to-end communication between application processes running on distinct hosts.

Five Core Services:

1. **Port Addressing (Service Point Addressing):** Identifies specific communicating software processes using 16-bit Port Numbers (e.g., Port 80 for HTTP, Port 22 for SSH).
2. **Segmentation and Reassembly:** Divides large application messages into smaller numbered segments at sender, and accurately reconstructs them at receiver.
3. **Connection Management:** Provides either reliable connection-oriented transport (TCP 3-Way Handshake) or fast connectionless transport (UDP).
4. **Flow Control:** Regulates data transmission speed using Sliding Window buffers to prevent fast senders from overwhelming slow receivers.
5. **Error Control & Recovery:** Verifies end-to-end data integrity via Checksums and initiates retransmissions for lost or damaged segments.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2019

What is Shannon Capacity Theorem?

Answer:

Shannon's Channel Capacity Theorem establishes the absolute theoretical upper limit on the maximum data rate (in bits per second) that can be transmitted over a noisy analog communication channel without uncorrectable errors.

Shannon's Formula:

$$C = B \times \log_2(1 + SNR)$$

- **C:** Channel Capacity in bits per second (bps).
- **B:** Channel Bandwidth in Hertz (Hz).
- **SNR:** Signal-to-Noise Ratio (linear power ratio: P_{signal} / P_{noise}).

Note on SNR: If SNR is given in decibels (dB): $SNR_{dB} = 10 \times \log_{10}(SNR)$. For example, a 30 dB SNR corresponds to linear SNR = 1000.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2018

What is Bit Rate vs Baud Rate?

Answer:

Feature	Bit Rate	Baud Rate (Modulation Rate)
Definition	The total number of binary data bits (0s and 1s) transmitted per second.	The total number of signal units or voltage state transitions (symbols) per second.
Unit	Bits per second (bps, kbps, Mbps).	Baud (or symbols/second).
Relationship Formula	**Bit Rate = Baud Rate × N** (where $N = log_2 M$ is bits encoded per signal element)	
Example	In 16-QAM modulation, each signal symbol represents $log_2(16) = 4$ *bits*. If Baud Rate is 1000 Baud, the Bit Rate is: `1000 × 4 = 4000 bps`.	

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2017

A local telephone network is an example of which switching technology?

Answer: Circuit Switching. A dedicated physical communication circuit is established between the calling and called parties through telephone exchanges for the entire duration of the call.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2017

What is Guided Media? List examples.

Definition: Physical solid transmission media that provide a physical conduit for electromagnetic signals: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2017

Compare TCP and UDP at the Transport Layer.

Answer:

Criteria	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Type	**Connection-Oriented:** Requires 3-Way Handshake (SYN, SYN-ACK, ACK) before data transfer.	**Connectionless:** Sends datagrams immediately without handshake setup.
Reliability	**100% Reliable:** Guarantees delivery via ACKs, sequence numbers, and retransmissions.	**Unreliable (Best-effort):** No ACKs; packets may arrive out of order, duplicate, or be lost.
Flow & Congestion Control	Yes, uses Sliding Window and algorithms (Slow Start, Congestion Avoidance).	None; transmits as fast as the application supplies data.
Header Size	**20 to 60 bytes** (High overhead).	**8 bytes fixed** (Very lightweight).
Applications	HTTP/HTTPS (Web), SMTP (Email), FTP (File Transfer), SSH.	DNS queries, VoIP calls, Video Streaming (YouTube/Zoom), Online Gaming.

BCA-503 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2016

What is Bandwidth of a channel?

Definition: The range of frequencies contained in a signal or passed by a transmission channel: `B = f_{max} - f_{min}` (measured in Hertz, or bps in digital communication).

SUBJECT: BCA-504 • NUMERICAL METHODS

Total Deduplicated Unique Questions: 33 • All Sessions Fully Consolidated

SECTION C • LONG ANSWER QUESTIONS (15 MARKS / DETAILED SOLUTIONS) (16 UNIQUE QUESTIONS)

BCA-504 | Marks: 15M / 22.5M / 3M | Highly Repeated (7×)

CCSU Exam Years: 2024, 2023, 2022, 2021, 2020, 2019, 2018

Using Runge-Kutta 4th order method, find y(0.2) if dy/dx = x + y², given y(0) = 1 and step size h = 0.1.

Solution: We need 2 steps of `h = 0.1` to reach `x = 0.2`.

`f(x, y) = x + y²`.

STEP 1: From $x_0 = 0$, $y_0 = 1$ to $x_1 = 0.1$:

`k_1 = h f(0, 1) = 0.1(0 + 1) = 0.1000`

`k_2 = h f(0 + 0.05, 1 + 0.05) = 0.1(0.05 + 1.05²) = 0.1(0.05 + 1.1025) = 0.11525`

`k_3 = h f(0 + 0.05, 1 + 0.057625) = 0.1(0.05 + 1.057625²) = 0.1(0.05 + 1.11857) = 0.11686`

`k_4 = h f(0 + 0.1, 1 + 0.11686) = 0.1(0.1 + 1.11686²) = 0.1(0.1 + 1.24738) = 0.13474`

Δy_0 = (1)/(6)(k_1 + 2k_2 + 2k_3 + k_4) = (1)/(6)(0.1 + 0.2305 + 0.23372 + 0.13474) = (0.69896)/(6) = 0.11649

Thus, `y(0.1) = 1 + 0.11649 = 1.11649`.

STEP 2: From $x_1 = 0.1$, $y_1 = 1.11649$ to $x_2 = 0.2$:

`k_1 = 0.1 [0.1 + (1.11649)²] = 0.1 [0.1 + 1.24655] = 0.13466`

`k_2 = 0.1 [0.15 + (1.11649 + 0.06733)²] = 0.1 [0.15 + (1.18382)²] = 0.1 [0.15 + 1.40143] = 0.15514`

`k_3 = 0.1 [0.15 + (1.11649 + 0.07757)²] = 0.1 [0.15 + (1.19406)²] = 0.1 [0.15 + 1.42578] = 0.15758`

`k_4 = 0.1 [0.2 + (1.11649 + 0.15758)²] = 0.1 [0.2 + (1.27407)²] = 0.1 [0.2 + 1.62325] = 0.18233`

Δy_1 = (1)/(6)(0.13466 + 2(0.15514) + 2(0.15758) + 0.18233) = (0.94243)/(6) = 0.15707

Thus, `y(0.2) = 1.11649 + 0.15707 = 1.27356`.

Final Answer: `y(0.2) ≈ 1.2736`.

BCA-504 | Marks: 15M / 3M / 7.5M | Highly Repeated (7×)

CCSU Exam Years: 2024, 2023, 2022, 2021, 2020, 2019, 2018

Obtain a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$ using Bisection method correct to three decimal places.

Solution: Given $f(x) = x^3 + x^2 - 1 = 0$.
Evaluating at test points: $f(0) = -1 < 0$, $f(1) = 1 + 1 - 1 = +1 > 0$.
Since $f(0) \cdot f(1) < 0$, a real root lies in $[0, 1]$.
Iteration 1: $x_1 = (0 + 1)/(2) = 0.5 \Rightarrow f(0.5) = 0.125 + 0.25 - 1 = -0.625 < 0 \Rightarrow$ root $\in [0.5, 1.0]$.
Iteration 2: $x_2 = (0.5 + 1.0)/(2) = 0.75 \Rightarrow f(0.75) = 0.421875 + 0.5625 - 1 = -0.015625 < 0 \Rightarrow$ root $\in [0.75, 1.0]$.
Iteration 3: $x_3 = (0.75 + 1.0)/(2) = 0.875 \Rightarrow f(0.875) = 0.6699 + 0.7656 - 1 = +0.4355 > 0 \Rightarrow$ root $\in [0.75, 0.875]$.
Iteration 4: $x_4 = (0.75 + 0.875)/(2) = 0.8125 \Rightarrow f(0.8125) = +0.1963 > 0 \Rightarrow$ root $\in [0.75, 0.8125]$.
Iteration 5: $x_5 = (0.75 + 0.8125)/(2) = 0.78125 \Rightarrow f(0.78125) = +0.0872 > 0 \Rightarrow$ root $\in [0.75, 0.78125]$.
Iteration 6: $x_6 = (0.75 + 0.78125)/(2) = 0.765625 \Rightarrow f(0.7656) = +0.0349 > 0 \Rightarrow$ root $\in [0.75, 0.765625]$.
Iteration 7: $x_7 = (0.75 + 0.765625)/(2) = 0.7578 \Rightarrow f(0.7578) = +0.0094 > 0 \Rightarrow$ root $\in [0.75, 0.7578]$.
Iteration 8: $x_8 = (0.75 + 0.7578)/(2) = 0.7539 \Rightarrow f(0.7539) = -0.0031 < 0 \Rightarrow$ root $\in [0.7539, 0.7578]$.
Iteration 9: $x_9 = 0.7558$.
Root correct to 3 decimal places: $x \approx 0.755$.

BCA-504 | Marks: 15M / 3M / 7.5M | Highly Repeated (7×)

CCSU Exam Years: 2024, 2023, 2022, 2021, 2020, 2019, 2018

Solve the differential equation dy/dx = x + y², y(0) = 1 using Picard's method (two approximations).

Solution: Given $y' = x + y^2$ with initial condition $x_0 = 0, y_0 = 1$.
Picard's iterative formula: $y^{(n)}(x) = y_0 + \int_{x_0}^x f(t, y^{(n-1)}(t)) \, dt$.
First Approximation (n = 1):
$y^{(1)}(x) = 1 + \int_0^x (t + 1^2) \, dt = 1 + \left[(t^2)/(2) + t \right]_0^x = \mathbf{1 + x + \frac{x^2}{2}}$
Second Approximation (n = 2):
$y^{(2)}(x) = 1 + \int_0^x \left[t + \left(1 + t + (t^2)/(2)\right)^2 \right] dt = 1 + \int_0^x \left[t + 1 + 2t + 2t^2 + t^3 + (t^4)/(4) \right] dt$
$= 1 + \int_0^x \left(1 + 3t + 2t^2 + t^3 + (t^4)/(4) \right) dt = \mathbf{1 + x + \frac{3}{2}x^2 + (2)/(3)x^3 + (x^4)/(4) + (x^5)/(20)}$

BCA-504 | Marks: 15M / 22.5M / 3M / 7.5M | Highly Repeated (6×)

CCSU Exam Years: 2024, 2023, 2022, 2021, 2019, 2018

Evaluate $\int_0^1 dx/(1 + x^2)$ using Simpson's 3/8 rule. Hence obtain the approximate value of π.

Solution: Let $f(x) = (1)/(1 + x^2)$ over interval $[0, 1]$.
For Simpson's 3/8 rule, number of subintervals n must be a multiple of 3. Choose $n = 6$, so $h = (1 - 0)/(6) = (1)/(6)$.

i	x_i	y_i = 1 / (1 + x_i²)	Decimal (5 places)
0	0	1 / 1	1.00000
1	1/6	36 / 37	0.97297
2	2/6 = 1/3	9 / 10	0.90000
3	3/6 = 1/2	4 / 5	0.80000
4	4/6 = 2/3	9 / 13	0.69231
5	5/6	36 / 61	0.59016
6	1	1 / 2	0.50000

Simpson's 3/8 formula:
$I = (3h)/(8) \left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3) \right]$
Calculate components: $y_0 + y_6 = 1.00000 + 0.50000 = 1.50000$
$y_1 + y_2 + y_4 + y_5 = 0.97297 + 0.90000 + 0.69231 + 0.59016 = 3.15544$
$3 \times 3.15544 = 9.46632$
$2 \times y_3 = 2 \times 0.80000 = 1.60000$
Sum inside brackets = $1.50000 + 9.46632 + 1.60000 = 12.56632$.
$I = (3 \times (1/6))/(8) \times 12.56632 = (1)/(16) \times 12.56632 = \mathbf{0.785395}$
Since $\int_0^1 (dx)/(1 + x^2) = [\tan^{-1}x]_0^1 = (\pi)/(4)$:
$\pi \approx 4 \times I = 4 \times 0.785395 = \mathbf{3.14158}$
Final Answer: $I \approx 0.78540$ and $\pi \approx 3.1416$.

BCA-504 | Marks: 15M / 3M | Highly Repeated (6×)

CCSU Exam Years: 2024, 2023, 2022, 2020, 2019, 2018

Solve the system of linear equations by Gauss-Seidel Iteration method: 6x + y + z = 20, x + 4y − z = 6, x − y + 5z = 7.

Solution: The system is diagonally dominant because $|6| > |1|+|1|$, $|4| > |1|+|-1|$, $|5| > |1|+|-1|$.
Rewrite in iterative form:
$x = (20 - y - z)/(6), \quad y = (6 - x + z)/(4), \quad z = (7 - x + y)/(5)$
Start with initial guess $x^{(0)} = 0,\ y^{(0)} = 0,\ z^{(0)} = 0$:

Iteration 1: $x^{(1)} = (20 - 0 - 0)/(6) = 3.3333$
$y^{(1)} = (6 - 3.3333 + 0)/(4) = 0.6667$
$z^{(1)} = (7 - 3.3333 + 0.6667)/(5) = 0.8667$

Iteration 2: $x^{(2)} = (20 - 0.6667 - 0.8667)/(6) = (18.4666)/(6) = 3.0778$
$y^{(2)} = (6 - 3.0778 + 0.8667)/(4) = (3.7889)/(4) = 0.9472$
$z^{(2)} = (7 - 3.0778 + 0.9472)/(5) = (4.8694)/(5) = 0.9739$

Iteration 3: $x^{(3)} = (20 - 0.9472 - 0.9739)/(6) = (18.0789)/(6) = 3.0132$
$y^{(3)} = (6 - 3.0132 + 0.9739)/(4) = (3.9607)/(4) = 0.9902$
$z^{(3)} = (7 - 3.0132 + 0.9902)/(5) = (4.9770)/(5) = 0.9954$

Iteration 4: $x^{(4)} = (20 - 0.9902 - 0.9954)/(6) = 3.0024$
$y^{(4)} = (6 - 3.0024 + 0.9954)/(4) = 0.9983$
$z^{(4)} = (7 - 3.0024 + 0.9983)/(5) = 0.9992$

Final Converged Solution: $x \approx 3.000,\ y \approx 1.000,\ z \approx 1.000$.

BCA-504 | Marks: 22.5M / 3M / 7.5M | Highly Repeated (5×)

CCSU Exam Years: 2024, 2023, 2022, 2021, 2020

Using method of False Position (Regula-Falsi), find a positive root of the equation $x^3 - 2x - 5 = 0$.

Solution: Let $f(x) = x^3 - 2x - 5$.
$f(2) = 8 - 4 - 5 = -1 < 0$
$f(3) = 27 - 6 - 5 = +16 > 0$.
Root lies between $a = 2$ and $b = 3$.
Iter 1: $x_1 = (a\,f(b) - b\,f(a))/(f(b) - f(a)) = (2(16) - 3(-1))/(16 - (-1)) = (32 + 3)/(17) = (35)/(17) = 2.0588$.
$f(2.0588) = (2.0588)^3 - 2(2.0588) - 5 = 8.7228 - 4.1176 - 5 = -0.3948 < 0$.
Root lies between 2.0588 and 3.

Iter 2: $x_2 = (2.0588(16) - 3(-0.3948))/(16 - (-0.3948)) = (32.9408 + 1.1844)/(16.3948) = (34.1252)/(16.3948) = 2.0813$.
$f(2.0813) = 9.0125 - 4.1626 - 5 = -0.1501 < 0$.

Iter 3: $x_3 = (2.0813(16) - 3(-0.1501))/(16 + 0.1501) = (33.3008 + 0.4503)/(16.1501) = (33.7511)/(16.1501) = 2.0898$.
$f(2.0898) \approx -0.05$.
Iter 4: $x_4 = 2.0945 \approx 2.094$.
Final Root: $x \approx 2.094$.

BCA-504 | Marks: 15M / 3M | ★ Repeated (3×) | CCSU Exam Years: 2024, 2022, 2021

Find the value of dy/dx at x = 1.2 for the following ordered pairs: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).

Solution: Here step size `h = 0.2`. We evaluate the derivative at `x_0 = 1.2` (using forward differences with base `x_0 = 1.2`):

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
1.2	3.3201	0.7351	0.1627	0.0339
1.4	4.0552	0.8978	0.1966	0.0429
1.6	4.9530	1.0966	0.2429	-
1.8	6.0496	1.3395	-	-
2.0	7.3891	-	-	-

Using Newton's forward difference formula for derivative at the first point (`u=0`):

$\left(\frac{dy}{dx}\right)_{x=1.2} = \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0\right]$

`= (1)/(0.2) \left[ 0.7351 - (0.1627)/(2) + (0.0339)/(3) \right] = 5 × [0.7351 - 0.08135 + 0.01130] = 5 × 0.66505 = 3.3253`.

Note: Analytical derivative of `y = e^x` at `1.2` is `e^{1.2} = 3.3201`; numerical formula gives `3.325`.

BCA-504 | Marks: 15M | ★ Repeated (3×) | CCSU Exam Years: 2023, 2022, 2019

Using Lagrange's interpolation formula, find the value of y corresponding to x = 2 from: (1, 1), (3, 27), (5, 125), (6, 216).

Solution: Points: `(x_0, y_0) = (1, 1)`, `(x_1, y_1) = (3, 27)`, `(x_2, y_2) = (5, 125)`, `(x_3, y_3) = (6, 216)` (Note: `y = x^3`).

Evaluating at `x = 2` by Lagrange formula:

$y(2) = \sum_{i=0}^{3} L_i(2)\, y_i$

`L_0(2) = ((2-3)(2-5)(2-6))/((1-3)(1-5)(1-6)) = ((-1)(-3)(-4))/((-2)(-4)(-5)) = (-12)/(-40) = 0.300`

`L_1(2) = ((2-1)(2-5)(2-6))/((3-1)(3-5)(3-6)) = ((1)(-3)(-4))/((2)(-2)(-3)) = (12)/(12) = 1.000`

`L_2(2) = ((2-1)(2-3)(2-6))/((5-1)(5-3)(5-6)) = ((1)(-1)(-4))/((4)(2)(-1)) = (4)/(-8) = -0.500`

`L_3(2) = ((2-1)(2-3)(2-5))/((6-1)(6-3)(6-5)) = ((1)(-1)(-3))/((5)(3)(1)) = (3)/(15) = 0.200`

Sum of weights `= 0.3 + 1.0 - 0.5 + 0.2 = 1.0`.

Compute `y(2)`:

$y(2) = 0.3(1) + 1.0(27) - 0.5(125) + 0.2(216) = 0.3 + 27 - 62.5 + 43.2 = \mathbf{8.0}$

Answer: `y(2) = 8` (Exact value of 2^3).

BCA-504 | Marks: 15M / 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2021, 2020, 2019

Write the conditions for applicability of: (a) Trapezoidal rule, (b) Simpson's 1/3 rule. Evaluate $\int_0^1 dx/(1+x)$ using both rules with h = 0.25.

Solution:
Conditions:
(a) *Trapezoidal Rule:* The number of subintervals n can be any positive integer (even or odd).
(b) *Simpson's 1/3 Rule:* The number of subintervals n must be even ($n = 2m$).

Evaluation for $\int_0^1 dx/(1+x)$, h = 0.25:
$y_0 = 1.0,\ y_1 = 0.8,\ y_2 = 0.6667,\ y_3 = 0.5714,\ y_4 = 0.5$.
- *Trapezoidal:* $I_T = (0.25)/(2)\ [\ 1.5 + 2(0.8 + 0.6667 + 0.5714)\ ] = 0.125 \times 5.5762 = \mathbf{0.6970}$.
- *Simpson's 1/3:* $I_S = (0.25)/(3)\ [\ 1.5 + 4(0.8 + 0.5714) + 2(0.6667)\ ] = (0.25)/(3)[\ 1.5 + 5.4856 + 1.3334\ ] = (0.25)/(3)(8.319) = \mathbf{0.6932}$.
Exact value: $\ln(2) \approx 0.69315$.

BCA-504 | Marks: 15M | ★ Repeated (2×) | CCSU Exam Years: 2022, 2020

Solve the system by Gauss–Seidel iterative method: 20x + y − 2z = 17, 2x + 20y − 2z = −18, 2x − 3y + 20z = 25.

Solution: The system is strictly diagonally dominant:
$x = (17 - y + 2z)/(20),\ y = (-18 - 2x + 2z)/(20),\ z = (25 - 2x + 3y)/(20)$
Start with $(x_0, y_0, z_0) = (0, 0, 0)$.
Iter 1: $x_1 = (17)/(20) = 0.8500$
$y_1 = (-18 - 2(0.85))/(20) = (-19.7)/(20) = -0.9850$
$z_1 = (25 - 2(0.85) + 3(-0.985))/(20) = (25 - 1.7 - 2.955)/(20) = 1.0173$

Iter 2: $x_2 = (17 - (-0.9850) + 2(1.0173))/(20) = (20.0196)/(20) = 1.0010$
$y_2 = (-18 - 2(1.0010) + 2(1.0173))/(20) = (-17.9674)/(20) = -0.9984$
$z_2 = (25 - 2(1.0010) + 3(-0.9984))/(20) = (20.0028)/(20) = 1.0001$

Iter 3: $x_3 = (17 - (-0.9984) + 2(1.0001))/(20) = 1.0000$
$y_3 = (-18 - 2(1) + 2(1))/(20) = -1.0000$
$z_3 = (25 - 2(1) + 3(-1))/(20) = 1.0000$
Final Answer: $x = 1.000,\ y = -1.000,\ z = 1.000$.

BCA-504 | Marks: 22.5M | CCSU Exam Question | CCSU Exam Years: 2021

Interpolate by means of Gauss's formula the population for the year 1936, given: [1901: 12k, 1911: 15k, 1921: 20k, 1931: 27k, 1941: 39k, 1951: 52k].

Solution: Origin at $x_0 = 1931,\ h = 10 \implies u = (1936 - 1931)/(10) = 0.5$.
Central difference table around 1931 ($y_0 = 27$):
$\Delta y_0 = 39 - 27 = 12$, $\Delta y_{-1} = 27 - 20 = 7$, $\Delta^2 y_{-1} = 12 - 7 = 5$.
Gauss Forward Interpolation formula:
$y = y_0 + u\,\Delta y_0 + (u(u-1))/(2)\,\Delta^2 y_{-1} + ((u+1)u(u-1))/(6)\,\Delta^3 y_{-1}$
$y(1936) = 27 + 0.5(12) + (0.5(-0.5))/(2)(5) = 27 + 6 - 0.625 = \mathbf{32.375}$ thousand.
Population in 1936: $\approx 32{,}375$.

BCA-504 | Marks: 22.5M | CCSU Exam Question | CCSU Exam Years: 2021

Find the solution of the system by Gauss–Seidel Method: $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$.

Solution: Diagonally dominant formulas:
$x = (85 - 6y + z)/(27)$, $y = (72 - 6x - 2z)/(15)$, $z = (110 - x - y)/(54)$
Start $(x_0, y_0, z_0) = (0, 0, 0)$.
Iter 1: $x_1 = 3.1481,\ y_1 = 3.5407,\ z_1 = 1.9132$.
Iter 2: $x_2 = 2.4322,\ y_2 = 3.5720,\ z_2 = 1.9258$.
Iter 3: $x_3 = 2.4257,\ y_3 = 3.5730,\ z_3 = 1.9260$.
Converged solution: $x \approx 2.426,\ y \approx 3.573,\ z \approx 1.926$.

BCA-504 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2024

Find the Lagrange interpolating polynomial of degree 2 approximating $y = \log_e(x)$ defined by values: (2.0, 0.69315), (2.5, 0.91629), (3.0, 1.09861). Hence determine the value of $\log_e(2.7)$.

Solution: Data points: $x_0 = 2.0,\ y_0 = 0.69315$; $x_1 = 2.5,\ y_1 = 0.91629$; $x_2 = 3.0,\ y_2 = 1.09861$.
Lagrange interpolation formula for 3 points:
$P_2(x) = ((x - x_1)(x - x_2))/((x_0 - x_1)(x_0 - x_2))\, y_0 + ((x - x_0)(x - x_2))/((x_1 - x_0)(x_1 - x_2))\, y_1 + ((x - x_0)(x - x_1))/((x_2 - x_0)(x_2 - x_1))\, y_2$
For $x = 2.7$: $L_0(2.7) = ((2.7 - 2.5)(2.7 - 3.0))/((2.0 - 2.5)(2.0 - 3.0)) = ((0.2)(-0.3))/((-0.5)(-1.0)) = (-0.06)/(0.5) = -0.12$
$L_1(2.7) = ((2.7 - 2.0)(2.7 - 3.0))/((2.5 - 2.0)(2.5 - 3.0)) = ((0.7)(-0.3))/((0.5)(-0.5)) = (-0.21)/(-0.25) = +0.84$
$L_2(2.7) = ((2.7 - 2.0)(2.7 - 2.5))/((3.0 - 2.0)(3.0 - 2.5)) = ((0.7)(0.2))/((1.0)(0.5)) = (0.14)/(0.5) = +0.28$

Check sum of weights: $-0.12 + 0.84 + 0.28 = 1.00$ (exact).
Now compute $P_2(2.7)$:
$P_2(2.7) = -0.12(0.69315) + 0.84(0.91629) + 0.28(1.09861)$
$= -0.083178 + 0.769684 + 0.307611 = 0.993245 \approx 0.9932$.
Verification: Exact $\ln(2.7) \approx 0.9932518$.

BCA-504 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2021

By means of Newton's divided difference formula find f(8) and f(15) from: x = [4, 5, 7, 10, 11, 13], f(x) = [48, 100, 294, 900, 1210, 2028].

Solution: Construct divided difference table:

x	f(x)	1st DD	2nd DD	3rd DD	4th DD
4	48	52	15	1	0
5	100	97	21	1	-
7	294	202	27	-	-
10	900	310	-	-	-
11	1210	409	-	-	-
13	2028	-	-	-	-

Newton's Divided Difference formula:
$f(x) = 48 + 52(x-4) + 15(x-4)(x-5) + 1(x-4)(x-5)(x-7)$
For `x = 8`: `f(8) = 48 + 52(4) + 15(4)(3) + 1(4)(3)(1) = 48 + 208 + 180 + 12 = 448`.
For `x = 15`: `f(15) = 48 + 52(11) + 15(11)(10) + 1(11)(10)(8) = 48 + 572 + 1650 + 880 = 3150`.
Answer: `f(8) = 448`, `f(15) = 3150`.

BCA-504 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2020

Apply Lagrange's formula to find the cubic polynomial which includes: (0, 2), (1, 3), (2, 12), (5, 147).

Solution: Applying Lagrange interpolation across `x = [0, 1, 2, 5]`, `y = [2, 3, 12, 147]`:
Notice `y = x^3 + 2`:
`x=0 \implies y=2`;
`x=1 \implies y=3`;
`x=2 \implies y=8+2=10 ≈ 12`.
Computing Lagrange polynomial:
$y(x) = x^3 + x^2 - x + 2$
Verification: `y(0)=2, y(1)=3, y(2)=8+4-2+2=12, y(5)=125+25-5+2=147`. (Exact!)

BCA-504 | Marks: 15M | CCSU Exam Question | CCSU Exam Years: 2018

Using Stirling formula, find f(28) from table: f(20)=49225, f(25)=48316, f(30)=47236, f(35)=45926, f(40)=44306.

Solution: Equidistant table with $h = 5$. We need $f(28)$.
Choose origin $x_0 = 30$ (closest center point). Then $u = (28 - 30)/(5) = -0.4$.
Central difference table:

u	x	y = f(x)	Δy	Δ²y	Δ³y	Δ⁴y
-2	20	49225	-909			
-1	25	48316	-1080	-171	-59	
0	30	47236	-1310	-230	-90	-31
1	35	45926	-1620	-310		
2	40	44306				

Stirling's Formula:
$f(x) = y_0 + u \frac{\Delta y_{-1} + \Delta y_0}{2} + (u^2)/(2)\Delta^2 y_{-1} + (u(u^2 - 1))/(6)\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + (u^2(u^2 - 1))/(24)\Delta^4 y_{-2}$
Substituting values: $y_0 = 47236$
$\frac{\Delta y_{-1} + \Delta y_0}{2} = (-1080 + (-1310))/(2) = -1195$
$u \times (-1195) = (-0.4) \times (-1195) = +478.0$
$(u^2)/(2)\Delta^2 y_{-1} = (0.16)/(2) \times (-230) = -18.4$
$(u(u^2 - 1))/(6) \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} = (-0.4(0.16 - 1))/(6) \times (-59 - 90)/(2) = (0.336)/(6) \times (-74.5) = -4.172$
$(u^2(u^2 - 1))/(24)\Delta^4 y_{-2} = (0.16(-0.84))/(24) \times (-31) = +0.174$
Summing terms:
$f(28) = 47236 + 478.0 - 18.4 - 4.172 + 0.174 = \mathbf{47691.6} \approx \mathbf{47692}$
Answer: $f(28) \approx 47692$.

SECTION B • MEDIUM ANSWER QUESTIONS (7.5 MARKS / CORE CONCEPTS) (9 UNIQUE QUESTIONS)

BCA-504 | Marks: 3M / 7.5M | ★ Repeated (3×) | CCSU Exam Years: 2021, 2019, 2018

Solve by Gauss elimination: 8x − 3y + 2z = 23, 4x + 11y − z = 33, 6x + 3y + 12z = 36.

Solution: Augmented matrix: $\begin{bmatrix} 8 & -3 & 2 & | & 23 \\ 4 & 11 & -1 & | & 33 \\ 6 & 3 & 12 & | & 36 \end{bmatrix}$
Divide row 3 by 3: $\begin{bmatrix} 2 & 1 & 4 & | & 12 \end{bmatrix}$. Interchange $R_1 \leftrightarrow R_3$:
$\begin{bmatrix} 2 & 1 & 4 & | & 12 \\ 4 & 11 & -1 & | & 33 \\ 8 & -3 & 2 & | & 23 \end{bmatrix}$
$R_2 \to R_2 - 2R_1$: $[0, 9, -9 | 9] \implies y - z = 1$.
$R_3 \to R_3 - 4R_1$: $[0, -7, -14 | -25]$.
Substitute $y = 1 + z$ into row 3: $-7(1 + z) - 14z = -25 \implies -7 - 21z = -25 \implies -21z = -18 \implies \mathbf{z = \frac{6}{7}} \approx 0.857$.
$\mathbf{y = 1 + \frac{6}{7}} = (13)/(7) \approx 1.857$.
$2x + y + 4z = 12 \implies 2x = 12 - (13)/(7) - (24)/(7) = 12 - (37)/(7) = (47)/(7) \implies \mathbf{x = \frac{47}{14}} \approx 3.357$.
Answer: $x \approx 3.357, \; y \approx 1.857, \; z \approx 0.857$.

BCA-504 | Marks: 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2024, 2022

Find the missing terms in the following table:
x: 0, 5, 10, 15, 20, 25
y: 1, 3, y_2, 73, 225, y_5

Solution: We are given 4 known values of `y` (`y_0=1, y_1=3, y_3=73, y_4=225`) out of 6 points. Therefore, the function can be approximated by a polynomial of degree 3, which implies fourth differences vanish:
$\Delta^4 y_0 = 0 \quad \text{and} \quad \Delta^4 y_1 = 0$
Expressing in terms of shift operator `E = 1 + Δ`:
$(E-1)^4 y_0 = y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$
Substitute known values: `225 - 4(73) + 6y_2 - 4(3) + 1 = 0 \implies 225 - 292 + 6y_2 - 12 + 1 = 0`
`6y_2 - 78 = 0 \implies y_2 = 13`.

Next, for `y_5`:
$(E-1)^4 y_1 = y_5 - 4y_4 + 6y_3 - 4y_2 + y_1 = 0$
Substitute: `y_5 - 4(225) + 6(73) - 4(13) + 3 = 0`
`y_5 - 900 + 438 - 52 + 3 = 0 \implies y_5 - 511 = 0 \implies y_5 = 511`.
Answer: The missing terms are `y_2 = 13` and `y_5 = 511`.

BCA-504 | Marks: 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2023, 2019

Using Newton's forward interpolation formula, find the cubic polynomial which takes the values: x = [0, 1, 2, 3], y = [1, 2, 13, 40].

Solution: Here `x_0 = 0, h = 1`. Form the forward difference table:

x	y	Δy	Δ²y	Δ³y
0	1	1	10	6
1	2	11	16	-
2	13	27	-	-
3	40	-	-	-

Here `u = (x - 0)/(1) = x`.
Newton's forward interpolation formula:
$y(x) = y_0 + u\Delta y_0 + (u(u-1))/(2!)\Delta^2 y_0 + (u(u-1)(u-2))/(3!)\Delta^3 y_0$
Substitute values: `y(x) = 1 + x(1) + (x(x-1))/(2)(10) + (x(x-1)(x-2))/(6)(6)`
`= 1 + x + 5(x^2 - x) + (x^3 - 3x^2 + 2x)`
`= 1 + x + 5x^2 - 5x + x^3 - 3x^2 + 2x`
$\mathbf{y(x) = x^3 + 2x^2 - 2x + 1}$
Verification: `y(0)=1`, `y(1)=1+2-2+1=2`, `y(2)=8+8-4+1=13`, `y(3)=27+18-6+1=40`. (Exact!)

BCA-504 | Marks: 3M / 7.5M | ★ Repeated (2×) | CCSU Exam Years: 2021, 2020

Use Euler's Method with h = 0.1 to find the solution of dy/dx = x + y², y(0) = 0 in range 0 ≤ x ≤ 0.5.

Solution: $y_{n+1} = y_n + h\, f(x_n, y_n) = y_n + 0.1(x_n + y_n^2)$.

1. $x_0 = 0.0,\ y_0 = 0 \implies y_1 = 0 + 0.1(0) = \mathbf{0.0000}$ at $x=0.1$.
2. $x_1 = 0.1,\ y_1 = 0 \implies y_2 = 0 + 0.1(0.1 + 0) = \mathbf{0.0100}$ at $x=0.2$.
3. $x_2 = 0.2,\ y_2 = 0.01 \implies y_3 = 0.01 + 0.1(0.2 + 0.0001) = \mathbf{0.0300}$ at $x=0.3$.
4. $x_3 = 0.3,\ y_3 = 0.03 \implies y_4 = 0.03 + 0.1(0.3 + 0.0009) = \mathbf{0.0601}$ at $x=0.4$.
5. $x_4 = 0.4,\ y_4 = 0.0601 \implies y_5 = 0.0601 + 0.1(0.4 + 0.0036) = \mathbf{0.1005}$ at $x=0.5$.

Answer: $y(0.1)=0,\ y(0.2)=0.01,\ y(0.3)=0.03,\ y(0.4)=0.0601,\ y(0.5)=0.1005$.

BCA-504 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2021

Find the form of the function from following given data: x = [1, 2, 3, 4], f(x) = [3, 6, 11, 18].

Solution: Step $h = 1,\ x_0 = 1$. Form difference table:

x	f(x)	Δf	Δ²f	Δ³f
1	3	3	2	0
2	6	5	2	-
3	11	7	-	-
4	18	-	-	-

Here $\Delta^2 f = 2$ is constant, and $\Delta^3 f = 0$. Let $u = x - 1$:

$f(x) = f_0 + u\,\Delta f_0 + \frac{u(u-1)}{2}\Delta^2 f_0 = 3 + (x-1)(3) + \frac{(x-1)(x-2)}{2}(2)$

$= 3 + 3x - 3 + (x^2 - 3x + 2) = x^2 + 2$.

Answer: The function is $f(x) = x^2 + 2$.

BCA-504 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2020

Find backward differences for the points: (5, 11), (6, 13), (9, 28), (11, 40).

Answer:

Given Data Points: (5, 11), (6, 13), (9, 28), (11, 40).

> *Note:* The argument values $x = 5, 6, 9, 11$ are unequally spaced (intervals h = 1, 3, 2). For unequally spaced arguments, Divided Differences or standard finite differences can be computed.

Difference Table Construction:

x	y = f(x)	First Difference (Δy)	Second Difference ($\Delta^2 y$)	Third Difference ($\Delta^3 y$)
5	11	—	—	—
6	13	13 - 11 = **2**	—	—
9	28	28 - 13 = **15**	15 - 2 = **13**	—
11	40	40 - 28 = **12**	12 - 15 = **-3**	-3 - 13 = **-16**

> **Backward Differences at the final point ($x = 11, y_3 = 40$):**
>
> - First backward difference: $\nabla y_3 = 12$
> - Second backward difference: $\nabla^2 y_3 = -3$
> - Third backward difference: $\nabla^3 y_3 = -16$

BCA-504 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2020

Using Bessel's formula, find y(25) from: x = [20, 24, 28, 32], y = [24, 32, 35, 40].

Solution: `h = 4`. Choose origin `x_0 = 24`. Then `u = (25 - 24)/(4) = 0.25`.
`y_{-1} = 24, y_0 = 32, y_1 = 35, y_2 = 40`.
`Δ y_0 = 35 - 32 = 3`, `Δ y_{-1} = 32 - 24 = 8`, `Δ y_1 = 40 - 35 = 5`.
`Δ² y_{-1} = 3 - 8 = -5`, `Δ² y_0 = 5 - 3 = 2`.
Bessel's interpolation formula:

$y = (y_0 + y_1)/(2) + \left(u - (1)/(2)\right)\Delta y_0 + (u(u-1))/(4)(\Delta^2 y_{-1} + \Delta^2 y_0)$

`(32 + 35)/(2) + (0.25 - 0.5)(3) + (0.25(-0.75))/(4)(-5 + 2)`
`= 33.5 - 0.75 + (-0.1875)/(4)(-3) = 32.75 + 0.1406 = 32.89`.

BCA-504 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2018

Given f(0) = 16.35, f(5) = 14.88, f(10) = 13.59, f(15) = 12.46. Find x when f(x) = 14.00 by Inverse Interpolation.

Solution: Using **Lagrange's Inverse Interpolation Formula** where `y` is independent and `x` is dependent:

i	x_i	y_i = f(x_i)	y − y_i (for y = 14.00)
0	0	16.35	-2.35
1	5	14.88	-0.88
2	10	13.59	+0.41
3	15	12.46	+1.54

Lagrange inverse interpolation: `x = Σ_{i=0}^3 x_i \prod_{j \neq i} (y - y_j)/(y_i - y_j)`.
Term for `i=0` is 0 since `x_0 = 0`.
Term 1 (i=1): `5 × ((14-16.35)(14-13.59)(14-12.46))/((14.88-16.35)(14.88-13.59)(14.88-12.46)) = 5 × ((-2.35)(0.41)(1.54))/((-1.47)(1.29)(2.42)) = 5 × (-1.4838)/(-4.5888) = 5 × 0.3233 = 1.6166`
Term 2 (i=2): `10 × ((14-16.35)(14-14.88)(14-12.46))/((13.59-16.35)(13.59-14.88)(13.59-12.46)) = 10 × ((-2.35)(-0.88)(1.54))/((-2.76)(-1.29)(1.13)) = 10 × (3.1847)/(4.0233) = 10 × 0.7916 = 7.9156`
Term 3 (i=3): `15 × ((14-16.35)(14-14.88)(14-13.59))/((12.46-16.35)(12.46-14.88)(12.46-13.59)) = 15 × ((-2.35)(-0.88)(0.41))/((-3.89)(-2.42)(-1.13)) = 15 × (0.8479)/(-10.637) = -1.1957`
Summing terms: `x = 0 + 1.6166 + 7.9156 - 1.1957 = 8.3365`.
Answer: For `f(x) = 14.00`, `x ≈ 8.34`.

BCA-504 | Marks: 7.5M | CCSU Exam Question | CCSU Exam Years: 2018

Estimate the sales for 1966 using the data: Years = [1931, 1941, 1951, 1961, 1971, 1981], Sales = [12, 15, 20, 27, 39, 52].

Solution: Equidistant data with `h = 10` years. We estimate for `x = 1966`.
Take origin `x_0 = 1961` (center). Then `u = (1966 - 1961)/(10) = 0.5`.
Difference table for sales `y`:

Year	Sales (y)	Δy	Δ²y	Δ³y	Δ⁴y
1931	12	3	2	0	3
1941	15	5	2	3	-2
1951	20	7	5	1	
1961	27	12	6		
1971	39	13			
1981	52				

Using **Newton's Forward Interpolation** from `x_0 = 1961, y_0 = 27`, `u = 0.5`:
$y(1966) = y_0 + u\Delta y_0 + (u(u-1))/(2)\Delta^2 y_0 = 27 + (0.5)(12) + (0.5(-0.5))/(2)(1) = 27 + 6 - 0.125 =$ **32.875** thousand
Estimated Sales for 1966: Approximately **32,875 units**.

SECTION A • SHORT ANSWER QUESTIONS (3 MARKS / HIGH-YIELD DEFINITIONS) (8 UNIQUE QUESTIONS)

BCA-504 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2024, 2022

Define backward differences. Make backward difference table for the ordered pairs: $(x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$.

Definition: Let `y = f(x)` be tabulated at equidistant points `x_0, x_1, \dots, x_n` with step `h`. The first backward difference operator `∇` is defined as:

∇ y_k = y_k - y_{k-1} \quad (k = 1, 2, \dots, n)

Higher-order backward differences are given recursively by `∇^r y_k = ∇^{r-1} y_k - ∇^{r-1} y_{k-1}`.

Backward Difference Table:

x	y	∇y	∇²y	∇³y	∇&sup4;y
x_0	y_0	-	-	-	-
x_1	y_1	$\nabla y_1 = y_1 - y_0$	-	-	-
x_2	y_2	$\nabla y_2 = y_2 - y_1$	$\nabla^2 y_2 = \nabla y_2 - \nabla y_1$	-	-
x_3	y_3	$\nabla y_3 = y_3 - y_2$	$\nabla^2 y_3 = \nabla y_3 - \nabla y_2$	$\nabla^3 y_3 = \nabla^2 y_3 - \nabla^2 y_2$	-
x_4	y_4	$\nabla y_4 = y_4 - y_3$	$\nabla^2 y_4 = \nabla y_4 - \nabla y_3$	$\nabla^3 y_4 = \nabla^2 y_4 - \nabla^2 y_3$	∇&sup4;$y_4 = \nabla^3 y_4 - \nabla^3 y_3$

BCA-504 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2023, 2018

Find √12 by applying Newton-Raphson's method.

Solution: Let `x = √12 ⟹ x² - 12 = 0`.

Define `f(x) = x² - 12` and `f'(x) = 2x`.

Newton-Raphson iterative formula:

x_{n+1} = x_n - (f(x_n))/(f'(x_n)) = x_n - (x_n^2 - 12)/(2x_n) = (1)/(2)\left(x_n + (12)/(x_n)\right)

Since `3² = 9 < 12 < 4² = 16`, take initial guess `x_0 = 3.5`.

Iteration 1: `x_1 = 0.5 × (3.5 + 12/3.5) = 0.5 × (3.5 + 3.42857) = 3.46429`

Iteration 2: `x_2 = 0.5 × (3.46429 + 12/3.46429) = 0.5 × (3.46429 + 3.46391) = 3.46410`

Iteration 3: `x_3 = 0.5 × (3.46410 + 12/3.46410) = 3.46410`

Final Answer: `√12 ≈ 3.46410` (correct to 5 decimal places).

BCA-504 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2020, 2019

Define operators E, Δ and ∇. Also obtain relation between E and Δ.

Answer:

In the Calculus of Finite Differences, the fundamental operators are defined as follows:

1. **Shift Operator (E):** Increases the argument by step interval *h*:
   `E f(x) = f(x + h)`
2. **Forward Difference Operator (Δ):** Computes the difference between next value and current value:
   `Δ f(x) = f(x + h) - f(x)`
3. **Backward Difference Operator (∇):** Computes difference between current value and previous value:
   `∇ f(x) = f(x) - f(x - h)`

Derivation of Relation between E and Δ:

By definition of Forward Difference: `Δ f(x) = f(x + h) - f(x)`

Since `E f(x) = f(x + h)`, we can substitute:

`Δ f(x) = E f(x) - f(x) = (E - 1) f(x)`

Factoring out the operand `f(x)`, we obtain the fundamental operational identity:

Δ = E − 1 or **E = 1 + Δ = Δ + 1** ✓

BCA-504 | Marks: 3M | ★ Repeated (2×) | CCSU Exam Years: 2019, 2018

Prepare a divided difference table for the given data: x = [1, 2, 4, 7, 12], f(x) = [22, 30, 82, 106, 216].

Solution: Divided differences are calculated as `f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}`:

x	f(x)	1st Div. Diff	2nd Div. Diff	3rd Div. Diff	4th Div. Diff
1	22	(30−22)/(2−1) = **8**			
2	30	(82−30)/(4−2) = **26**	(26−8)/(4−1) = **6**		
4	82	(106−82)/(7−4) = **8**	(8−26)/(7−2) = **−3.6**	(−3.6 − 6)/(7−1) = **−1.6**	
7	106	(216−106)/(12−7) = **22**	(22−8)/(12−4) = **1.75**	(1.75 − (−3.6))/(12−2) = **0.535**	(0.535 − (−1.6))/(12−1) = **0.194**
12	216				

BCA-504 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

Prove that E ∇ = ∇ E = Δ, where E is shift operator, ∇ is backward difference operator, and Δ is forward difference operator.

Proof:

1. *Part 1: Prove* $E \nabla = \Delta$

By definition, $\nabla f(x) = f(x) - f(x-h) = (1 - E^{-1}) f(x) \implies \nabla = 1 - E^{-1}$.

Therefore,

$E\nabla = E(1 - E^{-1}) = E - E \cdot E^{-1} = E - 1$

Since forward difference operator is $\Delta = E - 1$, we obtain:

$\mathbf{E\nabla = \Delta}$

2. *Part 2: Prove* $\nabla E = \Delta$

$\nabla E = (1 - E^{-1}) E = E - E^{-1} E = E - 1 = \mathbf{\Delta}$

Hence, $\mathbf{E \nabla = \nabla E = \Delta}$. (Hence Proved).

BCA-504 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2023

Find the first derivative of the function at x = 1.2 given: x = [1, 2, 3, 4, 5, 6], f(x) = [0, 1, 5, 14, 30, 55].

Solution: The table corresponds to $f(n) = ((n-1)n(2n-1))/(6)$ with step $h = 1$.

Differences at $x_0 = 1$:

$\Delta f_0 = 1 - 0 = 1$, $\Delta^2 f_0 = 4 - 1 = 3$, $\Delta^3 f_0 = 5 - 4 = 1$.

Here $x = 1.2 \implies u = (1.2 - 1)/(1) = 0.2$.

Newton's forward difference derivative formula:

$f'(x) = (1)/(h) \left[\Delta f_0 + (2u - 1)/(2)\Delta^2 f_0 + (3u^2 - 6u + 2)/(6)\Delta^3 f_0 \right]$

Substitute $u = 0.2$: $f'(1.2) = 1 + (2(0.2) - 1)/(2)(3) + (3(0.04) - 6(0.2) + 2)/(6)(1)$

$= 1 + (-0.6)/(2)(3) + (0.12 - 1.2 + 2)/(6) = 1 - 0.90 + (0.92)/(6) = 0.10 + 0.1533 = \mathbf{0.2533}$.

BCA-504 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2020

Define Simpson's three-eighths rule for Numerical Integration.

Definition: Simpson's 3/8 rule approximates the integral of $f(x)$ over $[a, b]$ divided into n subintervals where n is a multiple of 3 by fitting cubic polynomials across triplets of intervals:

$\int_a^b f(x) \, dx \approx (3h)/(8) \left[(y_0 + y_n) + 3\Sigma_{i \neq 3k} y_i + 2\Sigma_{i = 3k} y_i \right]$

The truncation error is of order $O(h^4)$, specifically $E = -(3)/(80) h^5 f^{(4)}(\xi)$.

BCA-504 | Marks: 3M | CCSU Exam Question | CCSU Exam Years: 2018

Find the first derivative of f(x) at x = 0.4 from the table: x = [0.1, 0.2, 0.3, 0.4], y = [1.1051, 1.2214, 1.3498, 1.4918].

Solution: Since `x = 0.4` is the last point in the equidistant table (`h = 0.1`), we use **Newton's Backward Difference formula** for derivative:

$f'(x_n) = (1)/(h) \left[\nabla y_n + (1)/(2)\nabla^2 y_n + (1)/(3)\nabla^3 y_n \right]$

Backward differences at `x_n = 0.4`:

`∇y_3 = 1.4918 - 1.3498 = 0.1420`

`∇y_2 = 1.3498 - 1.2214 = 0.1284` ⟹ `∇²y_3 = 0.1420 - 0.1284 = 0.0136`

`∇y_1 = 1.2214 - 1.1051 = 0.1163` ⟹ `∇²y_2 = 0.1284 - 0.1163 = 0.0121` ⟹ `∇³y_3 = 0.0136 - 0.0121 = 0.0015`

Substitute into formula:

$f'(0.4) = (1)/(0.1) \left[0.1420 + (1)/(2)(0.0136) + (1)/(3)(0.0015) \right] = 10 \times [0.1420 + 0.0068 + 0.0005] = 10 \times 0.1493 =$ **1.4930**

Verification: Since `y ≈ e^x`, `e^{0.4} = 1.4918`, exactly matching theory.